VASTRANGAM

# Vastrangam BOS

**Business Operating System — one business, one brain.**

22 modules · 113 apps · 16 working today · compiled 2026-08-24

## What this document is

Two documents in one, because they answer two different questions and people ask both.

**Part One — The System** is the reader's tour: what Vastrangam BOS is, how one garment moves through it, every module and every app, and the rules that hold everywhere.

**Part Two — The Plan of Action** is the builder's document: the one law, the data core, the module-to-module wiring, five end-to-end flows, all 22 modules in build order with a diagram each, and what "done" means.

Both are generated from `brand/site/modules.js`, the one canonical list. Neither this page nor either part contains a module count, an app name or an app order typed by hand — which is why they cannot disagree with each other or with the software.

## Where the build actually stands

**16 of 113 apps are working today.** Each one is a real single-file application that carries its own self-tests and passes a click-through audit in both editions. Every other app in this document is marked **designed, not yet built**, and is described as a specification rather than as something you can open. Nothing here is described as finished that is not.

## Companies and channels — the short answer

**The system is not limited to three companies, and never was.**

A company is a row. A channel — a marketplace account, the D2C site, a POS counter, a B2B desk, an export buyer — is also a row. Every business record carries the company it belongs to; every sale also carries the channel it came through. **Ten companies with ten channels each is the same tables and the same code as three companies and seven marketplaces.**

| | Today's data | The design |
|---|---|---|
| Companies | 3 | a table, no ceiling |
| Channels per company | 7 marketplaces + D2C, B2B, export, POS | a table, no ceiling |
| Stock | one number per SKU | one number per SKU — never per channel |
| Books | one ledger per company | one ledger per company |
| Group | sum minus inter-company trade | sum minus inter-company trade |

This is checked rather than claimed. `core/tests/core.test.js` builds ten companies with ten channels each — a hundred channels — posts an order down every one plus ten inter-company sales, and asserts that every company's books balance, that no journal line anywhere points at another company's account, and that the group figure is the plain sum **minus** inter-company trade: ₹2,10,500 gross, ₹50,000 eliminated, ₹1,60,500 group. The same builder is then called for eleven companies and eleven channels with nothing in the code changed.

The reporting side behaves the same way. **Vastrangam_BOS_Data_Studio.html** reads your own sale, return and karigar workbooks in the browser — no upload, no account, no internet — and emits one pair of quantity columns per company **found in the sheets**. A fourth company is a new sheet in the workbook, not a new version of the software.

---

## The honesty rules this document is written under

1. Nothing is described as finished that is not. Every app is marked working today or designed.
2. No count is typed from memory. Modules, apps and build state are read from the canonical list.
3. No figure is invented. Where a rate or a price is missing, the tool posts zero and names the item rather than guessing — a guessed rate is a wrong payment to a real person.
4. Where something could not be verified, it says so instead of implying it was.

---

# PART ONE — THE SYSTEM

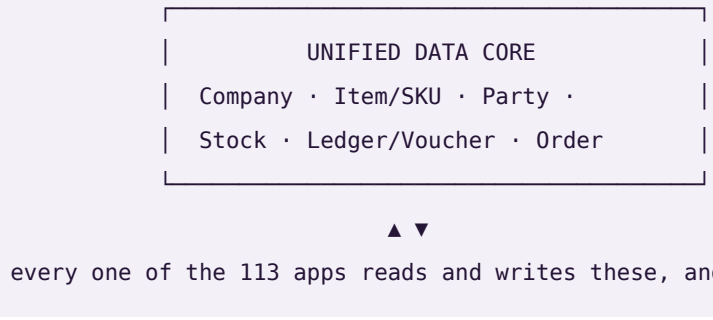
**The Business Operating System for Vastrangam Group: 22 modules and 113 apps over one shared data core.**

This file is the whole system in plain text — every module, every app, and what each one reads and writes. It is generated from `brand/site/modules.js`, the same file the website and every PDF read, so nothing here can disagree with them. The counts below are not typed in; they are counted from that file each time this page is built.

| | |
|---|---|
| **Modules** | 22, built in dependency order — a module is only built once everything it needs exists |
| **Apps** | 113 |
| **Working today** | 16 — each opens in a browser, carries its own self-tests and passes the click-through audit in both editions |
| **Engine working, screen to come** | 2 — the arithmetic is written and passing its own tests on the command line; there is no screen on it yet, so it is not counted above |
| **Still to build** | 95 |
| **Companies** | Vastrangam (invoices VS) · Ethnic Fashion trading as Go4Fashion (invoices EF, SKUs GF) · Adini Couture (invoices AC) |
| **Shared data core** | Company · Item/SKU · Party · Stock · Ledger/Voucher · Order |
| **Key difference** | Not a suite of integrated apps. One application over one database, so there is no sync step and no second copy of any master record |
| **Compliance** | Double-entry books with CGST/SGST/IGST, TDS, TCS, input credit on **accepted** goods, GSTR-1 and GSTR-3B, filed per registration |
| **Channels** | Amazon, Flipkart, Myntra, Meesho, Ajio, Nykaa, JioMart, plus your own storefront, the Surat counter, boutique wholesale and export |
| **Security** | Row-level isolation per company; outside services connect with scoped, revocable keys — **never account passwords** |
| **Deployment** | Hosted, or single-file apps that run by double-clicking with no install and no internet |

## The one idea

Every module reads and writes the **same six records**. That is the physical reason a single goods receipt can touch stock, the books, quality and sourcing in the same instant.



**One stock number, not one per channel.** The last piece sold at the Surat counter disappears from Myntra and Flipkart in the same instant — not three hours later as a cancellation, because cancellations are what a seller rating is lost to.

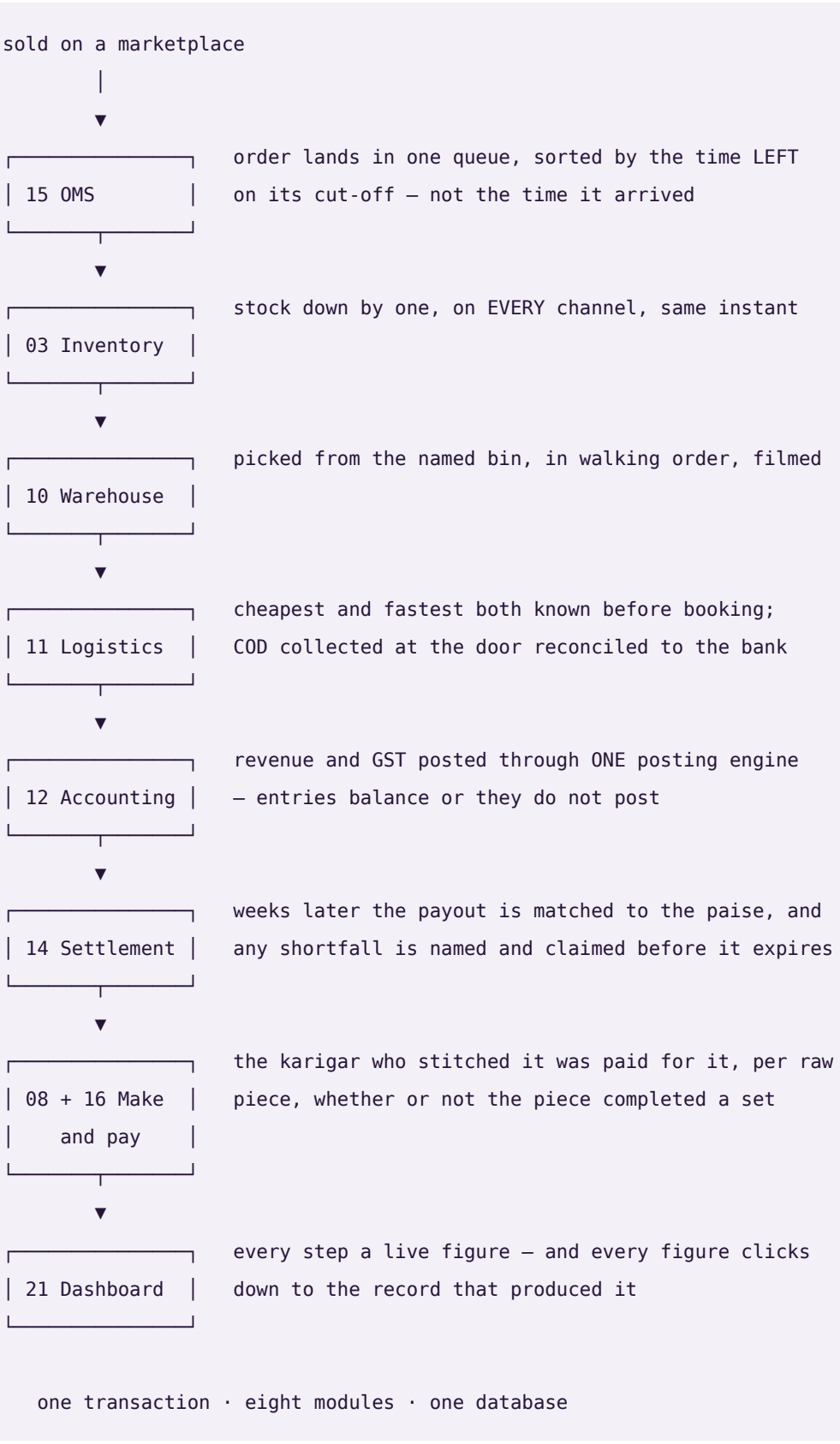
**Accepted — not ordered — is what counts.** You order 100 metres. 100 arrive. Quality accepts 96. Most systems increase stock by 100 and claim tax credit on 100. This one increases stock by **96**, claims input credit on **96**, raises a debit note for the 4 rejected, and lowers that mill's accept rate — automatically.

**Nothing derived is ever stored.** Outstanding, ageing, risk, promise dates, cost per piece and profit per design are recomputed on read. A stored total is a number that can drift away from the documents underneath it; a computed one cannot.

---

## How one garment moves through it

The test of whether this is one system or 113 programs sharing a login: sell a single garment and follow it.




## Every module and every app

Listed in build order. Each app is marked **working today** or **designed, not yet built** — nothing is described as finished that is not.

### Module 01 · Platform

*The spine the whole house runs on.*

Not a module you open — the layer underneath all 22. Who can see what, how Vastrangam is configured, and a record of everything that ever happened.

**Reads from:** Every module **Writes to:** Every module

| App | What it does | State |
|---|---|---|
| Identity, Settings & Audit | Users, roles and permissions, company switching, tax and numbering setup — and an immutable record of everything that ever happened. | designed, not yet built |
| Industry Packs | What your trade calls things, the stages your work moves through, the extra fields your records need, the documents you issue and the reference data you start with — all of it loaded as a row of configuration, never as a separate version of the software. Pick the pack that matches your trade and the whole system changes its words: the same order screen reads as a job, a matter, a consignment or an appointment, with the same columns underneath. A pack may rename what exists, add fields to tables that exist, and switch discretionary rules off; it may never invent a concept, add a field to a table that does not exist, contain any executable code, or switch off the guarantees — company scoping, the audit trail, money never being a float — because a trade may change its vocabulary and may not opt out of the things the books are trusted for. Adding a trade nobody anticipated is therefore a file somebody writes, not a release somebody ships. | designed, not yet built |
| **Ask & Print** | At an exhibition in Hyderabad, send one line from your phone: “ledger Kalamandir”, “print slips”. It comes back as a PDF, or it prints at the Surat office — with nothing plugged into your phone and nothing at the office open to the internet. | working today |
| Communications | WhatsApp commands and broadcasts, email and SMS, and the handful of scheduled jobs that carry a nudge to the right person without anyone remembering to send it. Every capability here has more than one interchangeable provider — none of them is ever the source of a figure the business reports. | designed, not yet built |
| WhatsApp Command Console | The shop floor does not open a laptop. A karigar or a staff member sends a short message — in, out, today’s pieces, leave, an advance — and it becomes a real record: attendance with the time and place it was marked, a production report against the design, a request in the approvals queue. The end-of-day prompt asks what is still open and how long it needs, so tomorrow starts from what is actually pending rather than from memory. Marking attendance checks the phone is at the unit within the radius set for it, with a grace window; a person outside it is not refused, they are flagged for the manager, because a system that locks someone out of being paid for being early at the wrong gate has failed at its job. Every override is recorded with who made it. The words a worker types are in the language they speak, and nothing here ever asks for a password, a bank detail or a document number. | designed, not yet built |
| Data Privacy & Consent | What a person’s data may be used for, captured as consent at the point it is given and honoured everywhere downstream — including the right to have it corrected or removed. Retention (how long a record is kept) and deletion (a person’s right to have their own data removed) are two different policies, tracked separately, because a rule that keeps records for the law and a request from a person to be forgotten do not resolve the same way. | designed, not yet built |
| **Provider Router & Cost Guard** | The rule that no capability depends on one outside service, enforced at the moment it matters instead of merely promised. Every capability has an ordered fallback list ending on an option that needs nothing connected, so a courier API that stops answering at 9pm, or an AI key that hits its quota mid-catalogue, drops to the next option instead of stopping the work. A provider that keeps failing is tripped out of the list entirely and retried once after a cooldown rather than hammered; retries inside one provider wait twice as long each time. And every paid call is counted in paise against a ceiling you set — over the ceiling the paid provider is refused, not warned about, and the work completes on a free one. Because | engine working, screen to come |

| App | What it does | State |
|---|---|---|
| | every capability is guaranteed a built-in or by-hand option, a spent budget can stop the spending without ever stopping the business. | |
| Payment Data Scope | A written statement of exactly which systems ever see a card or bank credential, and which never do — because every card-capable screen in this system hands that moment to a payment provider's own secured field and never stores or even passes the number through application code. The statement is what an auditor or a partner asks for before they will connect to this system. | designed, not yet built |

### Module 02 · Design & Sampling

*A style exists on paper before it exists as stock.*

A product moves from a first idea to something the business can actually make and sell — specification, sample rounds, costed trials and sign-off — before it is ever entered as a catalog record. Building this first means Inventory & Catalog never has to invent a style it has no real specification for.

**Reads from:** CRM **Writes to:** Inventory & Catalog · Manufacturing

| App | What it does | State |
|---|---|---|
| PLM & Development | Concept to a design that can actually be made: fabric and trim specification, sample rounds with the mill, costed trials against a target price, and sign-off — every version kept, so last season's costing is still there. | designed, not yet built |
| Design / IP Register | What protects a design once it exists — trademark or copyright status, the date it was first shown, and a flag the moment a near-identical listing turns up elsewhere. Every version already kept by PLM & Development gets a filed status here instead of just a date stamp, because a design with no ownership record on file is a design nobody can defend. | designed, not yet built |

### Module 03 · Inventory & Catalog

*One stock number everyone trusts.*

The most important number in the house: one quantity per design and size, per godown, per stage — greige, dyed, in stitching, finished, listed. Read and written by every other module. And one product record every marketplace lists from.

**Reads from:** Design & Sampling · Every module **Writes to:** Every module

| App | What it does | State |
|---|---|---|
| Stock | Live quantity by design, size and location, fabric in metres and pieces in numbers, with reorder alerts, lot tracking, set kits and dead-stock ageing. | designed, not yet built |
| Catalog / PIM | One record per design — fabric, work, length, colour, size chart, images, HSN, MRP and what each panel actually sells it at — scored for Myntra and Amazon readiness before it lists. It also carries the two things everything downstream needs: the code each panel knows the design by, mapped to yours, and the packed size and weight that decide the courier rate and settle every weight dispute. Every listing's state on every panel is here too — live, waiting for approval, blocked, archived — with the quality score that decides whether anyone sees it. | designed, not yet built |
| Kit & Combo SKU | A sellable SKU made of component SKUs — a three-piece set sold as one listing. Selling the kit decrements each component at order time, so stock is right for every piece in the set, not just the set itself. | designed, not yet built |
| Master-Data Hygiene | Duplicate detection and merge across customers, vendors and designs, and a dead-stock register for what has not moved in months. Protects every downstream report from the same record existing twice under two names. | designed, not yet built |

### Module 04 · CRM

*Know every boutique, chain and customer completely.*

One record per party — a Kalamandir or a Rajmandir, a Surat walk-in or a Myntra buyer — carrying every enquiry, order, return, agreement and conversation, whichever channel it arrived on.

**Reads from:** Every module **Writes to:** Sales · E-commerce / OMS · Marketing

| App | What it does | State |
|---|---|---|
| **CRM & Customer 360** | Enquiry to confirmed order, then the full lifetime: what they bought, what came back, what they are worth and which new range to show them first. | working today |
| **Documents & eSign** | Mill agreements, job-work contracts, signed delivery challans, export documents and boutique credit terms filed against the party or order they belong to — found by that record, not by hunting through a folder. | working today |
| **Helpdesk & Live Chat** | A boutique asking where its parcel is, or a customer asking about a size — the question becomes a ticket tied to the order, with the whole history already open. | working today |
| Forms & Feedback (NPS) | A short form after delivery, and the score it produces attached to the design or item it is actually about — not just the buyer — so a complaint-prone item surfaces as a pattern instead of a scatter of individual gripes. | designed, not yet built |

### Module 05 · Sales

*Counter, wholesale, website and export — one order book.*

The Surat counter, the boutique wholesale book, the website and the export shipment all write to the same order and draw on the same stock number. And the parcel is followed to the door, because a sale is not done until the COD money is in.

**Reads from:** Inventory & Catalog · CRM · Warehouse · Logistics **Writes to:** Inventory & Catalog · Accounting & GST · Warehouse · Logistics

| App | What it does | State |
|---|---|---|
| **D2C Sales** | Orders from your own storefront, cart to dispatch, with loyalty and partial COD on a ₹4,400 anarkali. | working today |
| **B2B & Credit** | Boutique and chain orders on credit limits and tier pricing, with outstanding aged against each party's own agreed terms. | working today |
| **Export** | Commercial invoice, packing list, LUT bond and IGST-refund tracking for the Gulf and UK buyers. | working today |
| **POS** | Counter billing at Udhna that draws on the same stock as the website — no second stock register. | working today |
| **Quotes & Proforma** | Send a quote, convert it to a confirmed order in one click. | working today |
| Couriers & AWB | Book the parcel on the order, compare couriers for that pin code, print the label with the design code on it, and follow the AWB to the door. | designed, not yet built |
| Subscriptions | A schedule that raises its own invoice on its cycle and follows up on its own when a payment fails — for anything sold as a standing order rather than a one-off. | designed, not yet built |
| Customisation & Made-to-Measure | The order that does not exist in the catalogue: a buyer sends reference pictures and their own measurements, a price is agreed over several messages, and the piece is made for them. All of it is one record — the references, the measurement set, every quote in the negotiation and what was finally agreed, the advance taken to start work and the balance taken before dispatch. When the order is accepted it opens a production order like any other, so a bespoke piece is costed, stitched, checked and posted exactly as a catalogue piece is. Two legs of money on one order is the part most systems get wrong: the advance is earned when the work starts, the balance is owed until the piece ships, and the ledger shows both separately rather than one payment appearing when the whole thing is over. | designed, not yet built |

## Module 06 · Planning & Requirements (MRP)

*Turn what is selling into what to buy and make.*

Confirmed orders and demand history have to become a plan before Purchase can buy anything or Manufacturing can start anything — otherwise buying and making are both just guessing. This module sits between the two: it reads what is actually selling and what is already committed, and turns that into requirement, not the other way around.

**Reads from:** Sales · E-commerce / OMS · Inventory & Catalog **Writes to:** Purchase · Manufacturing

| App | What it does | State |
|---|---|---|
| Demand Forecast & Signal | What sold, by SKU and by period, turned into a short-term forecast — the number every requisition and every production order downstream is measured against instead of a guess. | designed, not yet built |
| Requirement Explosion (MRP run) | Confirmed demand exploded through the bill of materials into what raw material to buy and what to produce, and by when — so a purchase requisition or a production order always traces back to real demand, never to a feeling that stock is getting low. | designed, not yet built |
| Open-to-Buy / Budget Ceiling | A spending ceiling set per period regardless of what the demand signal suggests, so a real signal cannot on its own commit more money than the business has decided to risk this season. Every requisition the MRP run drafts is checked against what is left of the ceiling before it goes to Purchase. | designed, not yet built |

## Module 07 · Purchase

*Nothing over-billed by a mill gets paid.*

The buy side end to end — mills, dyers, job workers and packing suppliers — with the control that stops you paying for metres you rejected.

**Reads from:** Inventory & Catalog · Planning & Requirements (MRP) · Manufacturing **Writes to:** Inventory & Catalog · Accounting & GST · Quality & Compliance

| App | What it does | State |
|---|---|---|
| **Procurement** | Enquiry to purchase order to goods receipt, with a strict three-way match: you ordered 100 metres, 100 arrived, quality accepted 96, and the bill is only cleared for 96. | working today |
| **Vendor Management** | Mill 360 — payables, ageing, a real risk score from accept rate and spend concentration, and sourcing that follows performance rather than habit. | working today |
| Insurance Register | What cover exists over stock in transit, stock in the warehouse and product liability, matched against what is actually moving through Purchase and Warehouse right now — so real value in transit is never quietly running with no cover tracked anywhere. | designed, not yet built |

## Module 08 · Manufacturing

*Know what a piece really costs to make.*

From the cut plan to the finished piece — what each karigar earned, what the dyer charged, what the zari cost, and what that design actually cost before you priced it.

**Reads from:** Purchase · Planning & Requirements (MRP) · Design & Sampling **Writes to:** Inventory & Catalog · HR & Payroll · Accounting & GST · Quality & Compliance

| App | What it does | State |
|---|---|---|
| Production Orders | Cutting, stitching, embroidery, washing, finishing and checking — your own stages, with work-in-progress visible at each and nothing lost at the dyer. | designed, not yet built |
| Piece-rate & Contractors | Karigars paid by the piece: pooled set completion, per-garment rates, alterations, rework and advances resolved into one payout. | designed, not yet built |
| BOM & Consumption | What each design consumes — metres of fabric, zari, lining, buttons, packing — costed at today's mill rates. | designed, not yet built |
| Maintenance | Machines and the building: what is due for service, when it was last done, what it cost, and what stopped while it was down. | designed, not yet built |

### Module 09 · Quality & Compliance

*Certify what was received and what was made.*

Quality inspection used to live buried as one step inside Manufacturing; it stands on its own here because a rejection at goods-receipt and a rejection on the production floor are the same discipline, and because the certificates a business holds — the proof it follows a standard — belong next to the inspections that back them up, not scattered across email.

**Reads from:** Purchase · Manufacturing **Writes to:** Purchase · Manufacturing · Inventory & Catalog

| App | What it does | State |
|---|---|---|
| Quality Control | Accept, reject or send for rework, with reasons that feed the mill's accept rate and the karigar's record. | designed, not yet built |
| Certificate & Compliance Register | Every standard the business or a vendor holds — a safety, labour or environmental certification — with its issue date, its expiry, and the audit that backs it, so a certificate about to lapse is visible before a buyer asks for it and finds it already expired. What this register tracks is what a sustainability report downstream is actually built from. | designed, not yet built |

### Module 10 · Warehouse

*Pick the right design first time — and prove what you sent.*

Bin-level instructions and barcode scanning so the right piece leaves the godown and stock stays honest — and a recording of each parcel being packed, because a wrong-return claim is settled by footage, not by argument.

**Reads from:** Sales · E-commerce / OMS · Inventory & Catalog **Writes to:** Inventory & Catalog · Sales · E-commerce / OMS

| App | What it does | State |
|---|---|---|
| Picking & Bins | Pick lists in walking order through the godown, by design and size, so nobody crosses the floor twice. | designed, not yet built |
| Barcode Operations | Scan to pick, pack, dispatch and run a physical stock count from a phone — the same scan whether the order came from a marketplace, your Shopify site or the counter. | designed, not yet built |
| Packing Video | Every parcel filmed as it is packed and indexed by its order number, so when a panel says the wrong piece was sent, the clip goes into the claim. | designed, not yet built |

### Module 11 · Logistics

*The courier network — rates, failed deliveries and the COD money.*

Booking one parcel happens on the order. This module is the network behind it: what Delhivery, Blue Dart and the rest charge to that pin code before you pick one, what happens to a delivery that fails in a small town, and whether the cash collected at the door reached your bank.

**Reads from:** Sales · E-commerce / OMS · Warehouse **Writes to:** Accounting & GST · Sales · E-commerce / OMS

| App | What it does | State |
|---|---|---|
| Rates & Zones | Every courier’s rate card by zone, weight slab and service — so the cheapest and the fastest option for this parcel are both known before it is booked. | designed, not yet built |
| NDR & RTO Rescue | A failed delivery worked while it can still be saved — reattempt, call, correct the address — before it becomes a return you pay for twice. | designed, not yet built |
| COD Remittance | What the courier collected at the door against what reached the Surat account, parcel by parcel, with every shortfall named and aged. | designed, not yet built |
| Handover & Manifest | What is expected out today against what the pickup boy actually took, per courier and per service. The manifest to hand him, the one-time code to confirm it, and a signed note of what was left behind — so a parcel lost between the packing table and the van has an owner. | designed, not yet built |
| Fleet | Your own delivery vehicles, if you run any — what each trip cost, what is due for service, and what that adds to freight. Optional; most businesses run couriers only. | designed, not yet built |

### Module 12 · Accounting & GST

*Books that always balance — and no BUSY needed.*

A full double-entry ledger built for Indian compliance, keeping the books itself. B2B sales, returns, mill purchases, payments and receipts are entered by hand because a person decides them; every website, marketplace and counter sale posts itself.

**Reads from:** Every module **Writes to:** Finance Reports · Treasury & Financial Planning

| App | What it does | State |
|---|---|---|
| Accounting | Double-entry books where every voucher balances and the trial balance always ties. | designed, not yet built |
| Invoicing | GST tax invoices and receipts, worked out from the lines to the paise. Where a panel raises its own invoice you keep both numbers on the order — theirs and your own series — so the panel's paperwork and your books point at the same sale. | designed, not yet built |
| Expenses | Spend captured by category with approvals, and bill OCR to save typing. | designed, not yet built |
| GST & Tax | CGST, SGST, IGST, TDS, TCS, input credit, GSTR-1 and GSTR-3B — filed per registration, so a group where one company is registered and another is not files exactly what each one owes and nothing it does not. | designed, not yet built |
| ITC Reconciliation | Every input credit matched against the government's own GSTR-2A/2B before a return is filed, so a credit claimed is a credit that is actually on record. | designed, not yet built |
| Receivables, Payables & PDC | Payments and receipts allocated against named open invoices, and a register for post-dated cheques that posts only on the date they are realised, not the date they were written. | designed, not yet built |
| Fixed Assets & Depreciation | The asset register with both Straight-Line and Written-Down-Value depreciation tracked side by side, and disposal posting its gain or loss straight to the books. | designed, not yet built |
| Year-End Close & Period Lock | Profit-and-loss accounts reset and balance-sheet accounts carry forward at year end, and a reviewed period locks against backdated edits until an admin unlocks it — an act that is itself logged. | designed, not yet built |
| Finance Reports | P&L, balance sheet, and profit by channel, design and SKU — so you know which anarkali actually earned money after commission, shipping and returns. | designed, not yet built |

### Module 13 · Treasury & Financial Planning

*Know what cash is coming, not just what already arrived.*

Accounting records what happened; this module is concerned with what happens next — how much cash is actually expected, when, and whether spend against a budget is on track before the month closes and turns the answer into history.

**Reads from:** Accounting & GST · Sales · Purchase **Writes to:** Accounting & GST

| App | What it does | State |
|---|---|---|
| Cash Flow Forecast | Expected receipts and expected payments laid out by week, drawn from open invoices and open bills rather than typed in by hand — so a cash shortfall is visible weeks before the date it would actually bite. | designed, not yet built |
| Banking & Reconciliation | Bank statement lines matched against the ledger's own record of what should have moved, with anything unmatched surfaced instead of silently carried forward. | designed, not yet built |
| Budget vs Actual | A budget set per category and period, and actual spend from Accounting tracked against it as the period runs — not only after it closes, when the only thing left to do is explain the variance. | designed, not yet built |

### Module 14 · Settlement

*Get paid what the panels owe you — cycle by cycle.*

Matching one payout to one order line happens in OMS. This is the level above: the settlement cycle each panel runs, the commission it actually charged against the rate card it published, and the TCS it deducted in your name.

**Reads from:** E-commerce / OMS · Accounting & GST **Writes to:** Accounting & GST

| App | What it does | State |
|---|---|---|
| Payout Cycles | Every settlement cycle each panel runs — what it should pay, what actually landed in the bank, and on which day — so a late payout is visible the day it is late, not at month end. | designed, not yet built |
| Fee & Commission Audit | The commission a panel publishes for a category against what it actually took, style by style. A quiet rate change is caught the first time it is applied, not at year end — and your seller tier sits on the same screen, because the tier is what the rate card hangs off, and slipping out of one quietly costs more than any single deduction. | designed, not yet built |
| TCS & TDS Register | Every rupee the panels deducted as TCS, and TDS on job work, matched against the portal's own figures — so the credit you claim is the credit you are owed. | designed, not yet built |

### Module 15 · E-commerce / OMS

*Seven panels, one queue — and every rupee accounted for.*

Stop logging into Myntra, then Flipkart, then Ajio. Every marketplace order lands in one pipeline and your stock goes out to all of them — then the money side closes out in the same module: what the panel paid, what it kept as commission, what came back, and what it still owes you.

**Reads from:** Inventory & Catalog · CRM · Sales · Accounting & GST · Logistics · Settlement **Writes to:** Inventory & Catalog · Accounting & GST · Warehouse · Logistics · Settlement

| App | What it does | State |
|---|---|---|
| **Marketplace OMS** | Myntra, Flipkart, Ajio, Amazon, Meesho, Nykaa and JioMart in a single queue — processed all together, channel-wise, or design-wise. The stages the panels really use, with the right cut-off counting down on each order — a quick-commerce or air-shipped order is not due at the same hour as a standard one. Priority orders at the top, and the day grouped by design so a Muskan Purple is picked once for eleven parcels instead of eleven times. | working today |
| **Order Management** | One pipeline from new to delivered, whether the order came from a seller panel, your Shopify or WooCommerce site, a dealer or the counter. | working today |
| Manual Data Check | The order and return sheets you already download from the panels, and the offline registers from the three shops — one file or a whole ZIP — read back as ten cross-checks: net sale after commission and fees, month, design, state, wrong returns, SPF claims, ads, payouts and GST. Every figure clicks through to the transactions behind it. | designed, not yet built |
| Reconciliation | Match every marketplace payout to the order line that earned it, and expose the gap. | designed, not yet built |
| Claims & Disputes | Weight disputes, SPF shortfalls, parcels lost in transit and returns that came back with a different piece inside — filed as claims with the packing footage attached, and answered before they close. A claim awaiting your reply is money; one closed for no response is nothing, so the days left sit beside the amount. | designed, not yet built |
| Returns / RMA | Customer returns, courier returns and wrong returns kept apart — because only one of the three is really your fault, and only one of them turns into dead stock. | designed, not yet built |
| Channels & Storefronts | Connect a channel once and it stays in step: catalogue out, price out, stock out, orders in. Seven marketplaces and any website platform — Shopify, WooCommerce, Magento, BigCommerce, Wix or a custom site over its own API — each switchable without touching your data. Shopsy and any other storefront a channel runs alongside its main one counts as its own channel here. Where a channel has no open interface, its own downloaded report is a first-class way in. A channel may also know you by a different trading name — that is a label on the channel, not a second company, so it tags the order and the payout without ever splitting your books. | designed, not yet built |
| Labels & Documents | The panel gives you a PDF; this hands the packing table something it can work from. Cropped to 4×6 for every channel, your design code printed large where the panel left it off, the invoice and slip merged behind it, and the whole batch to the label printer in one job. Reprint one parcel without redoing the lot — and no customer’s name and address is ever uploaded to an outside website to be cropped. | designed, not yet built |
| Listing & Catalog Manager | Bulk-create and bulk-edit listings across every channel from the one product record in Inventory & Catalog, and catch the mismatches that quietly cost sales: listed but out of stock, or in stock but never listed. | designed, not yet built |
| Size / Fit Recommendation AI | A fit suggestion at the point of purchase, built from the item’s own measurements and the return history of buyers who picked each size — aimed straight at the return reason that costs the most: the right item in the wrong size. | designed, not yet built |

| App | What it does | State |
|---|---|---|
| AR / Virtual Try-On | A way to see drape, fit and colour on a screen before buying, for items where a flat photo alone leaves too much to guess. | designed, not yet built |

### Module 16 · HR & Payroll

*Pay everyone right, on time.*

Office staff on a monthly salary and karigars paid by the piece, in one register, with attendance driving both and the festival advance already deducted.

**Reads from:** Manufacturing **Writes to:** Accounting & GST

| App | What it does | State |
|---|---|---|
| Staff & Contractors | Attendance marked by tap, effective-dated salary, and karigar piece-rate earnings in a single register. | designed, not yet built |
| Time-off & Advances | Leave, Diwali advances, and exactly how they change this month’s payout before you approve it. | designed, not yet built |
| Appraisal & Hiring | Performance reviews and a hiring pipeline that ends in an employee record. | designed, not yet built |
| Recruitment | The pipeline before someone becomes an employee — an opening, the people who applied for it, a trial piece where the work itself is the interview, and the decision with its reason kept. It matters more here than in most trades: a karigar is taken on for skill on a particular garment, and the trial output is the evidence, so it is recorded against the design and the rate that would apply rather than remembered as an impression. A candidate who is not taken on now stays findable when the same skill is needed in a busy month, and their personal documents are held under the same consent and retention rules as anyone else’s, not in a folder on somebody’s phone. | designed, not yet built |
| Payout Execution | Where the calculation in the earnings register actually turns into money leaving the business — bank batch, UPI, cash against a signed receipt — with the method and the reference recorded against every payout, so the register’s total and the money that actually moved can always be checked against each other. | designed, not yet built |

### Module 17 · Marketing

*Sell more without cutting the price.*

Plan the festive calendar, run the campaigns, and let rules keep you competitive on the panels without giving the margin away.

**Reads from:** Inventory & Catalog · CRM **Writes to:** Sales · E-commerce / OMS

| App | What it does | State |
|---|---|---|
| Social Calendar | Plan and publish across every channel from one calendar. | designed, not yet built |
| Campaigns | Email, SMS and WhatsApp campaigns measured on real revenue, not opens. | designed, not yet built |
| Repricing Engine | Rules per panel and per design — floor, ceiling, match-lowest and a festive override — so a Diwali sale does not quietly go below cost. And what each change actually did: a design whose orders fell after a price rise shows as exactly that, next to the rule that raised it. | designed, not yet built |
| Automation | If this happens, do that — across any module, without writing code. | designed, not yet built |
| Blog & Pages | How to drape it, what to wear it to, which fabric for which season — written, scheduled and published to your own site with the meta and internal links already set. | designed, not yet built |
| Events | Trade shows and exhibitions worked as a channel of their own — booth, budget and every lead captured on the floor landing straight in CRM instead of on a stack of business cards. | designed, not yet built |
| Website & Page Builder | The storefront itself, built by dragging sections into place rather than by editing a theme file — hero, product grid, size guide, lookbook, contact form — each block reading live from the catalogue, so a price or a stock state on a landing page is the same number the order screen uses instead of a figure someone pasted in and forgot. Blog & Pages above writes articles into a site that already exists; this is for the businesses that do not have one, and it is the gap that shows up plainly when this module list is set beside a mature open-source ERP: they ship a full site builder next to the blog, and until now this did not. | designed, not yet built |
| Markdown / Clearance Optimization | The same rule engine that reprices for competitiveness, aimed at ageing stock instead: when to start discounting it and by how much, before it becomes a warehouse write-off rather than a sale at a lower margin. | designed, not yet built |

### Module 18 · AI Content Engine

*Write it, shoot it, cut it — from your own catalogue.*

Listings, ads, reels and product photography generated from your own designs, in a voice that sounds like one person from Surat rather than a template — so the words match the piece and the picture is the size Myntra actually wants.

**Reads from:** Inventory & Catalog **Writes to:** Marketing · E-commerce / OMS

| App | What it does | State |
|---|---|---|
| Content Engine | Fourteen stages in your own voice — buyer psychology, competitor reading, hooks, the product description, marketplace copy for Amazon and Myntra, ad variations, reel scripts, song lyrics for the reel, the calendar, size chart and alt text. | designed, not yet built |
| Image Studio | A phone photo becomes a listing image: layers, free transform, background removal, Myntra 1080×1440 and every other channel preset, watermark and SEO alt text. | designed, not yet built |
| Video Studio | Text and image to video, reels and ad cuts sized for every channel. | designed, not yet built |
| Design Studio | Banners, festive creatives and thumbnails — templates, layers, undo and redo, any colour, exact sizing and stock elements, exporting PNG, JPG or PDF at whatever size the panel or the printer asks for. | designed, not yet built |
| **Motion Renderer** | A reel rendered from a page of HTML and CSS — the same layers, fonts and brand colours the Design Studio already uses — into a real MP4, on this machine, with nothing uploaded. It is not a screen recording: the clock is faked, the animation is seeked to the exact instant of each frame, and only then is that frame captured, so the render does not care whether the machine was busy. Rendering the same festival banner twice produces the same file to the byte, which is what makes a reel something you can check and re-cut rather than something you have to watch all the way through and hope about. | engine working, screen to come |
| Narration Studio | A voice over the reel, in the language the buyer actually speaks — the same script the Content Engine wrote, spoken. The default needs nothing installed and no key, because every modern browser can already speak; a self-hosted or cloud voice sits behind it as an interchangeable provider for anyone who wants a cloned or branded one. Long scripts are split at sentence boundaries and rejoined, so a two-minute description is not cut off at whatever limit a service imposes. A voice cloned from a real person is only ever used with that person's recorded consent, filed against them in Data Privacy & Consent like any other permission. | designed, not yet built |
| Image Generation Slot | Generated imagery — a model on a background you do not have to shoot, a festival backdrop, a lifestyle scene — as a capability with interchangeable providers rather than a bet on one service: a queue of jobs, a preview while it works, inpainting to fix one region, upscaling and face correction. This one needs a graphics card. Image models cannot run on an ordinary office machine or on the container this system is built in, so what ships is the queue, the review screen and the provider slot, with the generating itself done by whichever engine you point it at — your own GPU box, or a cloud service, swapped without touching anything else. Said plainly here because the alternative is a screen that looks finished and produces nothing. | designed, not yet built |
| Publisher | One push sends the listing, images and copy to the website and every panel, and reports back what went live and what a panel rejected, with the reason. | designed, not yet built |

## Module 19 · SEO, AEO & AIO

*Be found by a search box, an answer box and an AI.*

Content already exists once this module is reached; here it is made findable — by a traditional search engine, by the answer box above the results, and by the AI assistants now answering shopping questions directly instead of sending someone to a results page.

**Reads from:** Inventory & Catalog · AI Content Engine **Writes to:** Marketing

| App | What it does | State |
|---|---|---|
| Technical SEO & Schema | Structured data, sitemaps and page-level technical checks against your own storefront and content pages, so a search engine can read what a page is actually about instead of guessing from the text alone. | designed, not yet built |
| Answer-Engine Optimization | Content shaped to be quoted directly by an answer box or a voice assistant — a clear, citable answer near the top of the page — rather than written only for a person to scroll through. | designed, not yet built |
| AI-Engine Visibility Tracking | Whether and how this business is actually cited when someone asks an AI assistant a shopping question in this category, tracked over time — the same discipline as rank tracking, aimed at a newer kind of result page. | designed, not yet built |

### Module 20 · Projects & Collaboration

*The work that is not an order — and the talking around it.*

An exhibition in Hyderabad, a boutique’s custom order, a new godown fit-out, a legal matter with a supplier. Work that is not a sales order still has a deadline, a cost and documents — and it belongs on the same records as everything else.

**Reads from:** CRM · Sales · HR & Payroll · Inventory & Catalog **Writes to:** Accounting & GST · HR & Payroll · CRM

| App | What it does | State |
|---|---|---|
| Projects & Cases | An exhibition, a custom order for a chain, a fit-out or a dispute — stages you define, owners, deadlines, documents, hours and real cost, all on one record the ledger can see. | designed, not yet built |
| Timesheets & Planning | Who is on what this week and the hours that actually went in — against a project, an exhibition or a machine — with billable and non-billable kept apart. | designed, not yet built |
| Approvals | One queue for everything waiting on a yes: a mill purchase order, a boutique discount, a leave day, a credit note, a payment. The rule that sent it there is next to it, and the decision goes on the record. | designed, not yet built |
| Forum | Questions and answers that outlive a chat — for customers, dealers or staff — with the useful ones kept where the next person will actually find them. | designed, not yet built |
| Automation Studio | The place a person builds “when this happens, do that” by dragging it out and watching it run — a trigger, the steps after it, a branch where the answer decides which way to go — over the same event stream every module already writes to. Marketing’s Automation aims that idea at campaigns; this is the general one, reaching any module: a payment marked short holds the next dispatch and opens a claim, a karigar’s pooled sets crossing a threshold raises the payout for approval, a return marked damaged writes off the piece and messages the buyer. Every run is kept — what fired it, each step, what each step returned — because an automation nobody can inspect afterwards is a rule the business cannot trust with its money. | designed, not yet built |
| Discuss | The conversation attached to the record it is about — this order, this mill bill, this dispute — so a year later the reason for the decision is still sitting beside it. | designed, not yet built |
| Knowledge Base | A searchable internal wiki of standard operating procedures, scoped to the role it applies to, so how a task is meant to be done is written down once instead of carried in one person’s head. | designed, not yet built |

### Module 21 · Dashboard & BI

*See the whole house without asking anyone.*

Every number rolls up here as work happens — the day’s marketplace orders, what the karigars finished, what is still lying at the dyer, what the mills are owed. No exports, no waiting for month-end.

**Reads from:** Every module **Writes to:** —

| App | What it does | State |
|---|---|---|
| **CEO Dashboard** | Cash, sales by channel, stock by design, profit per piece and the alerts that matter — one screen, refreshed as the day runs. | working today |
| **Report Builder** | Drag the fields you want into a report and save it for the whole team. | working today |
| **Group Consolidation** | Ethnic Fashion, Vastrangam and Adini Couture as one set of figures, inter-company transfers removed, so the group position is real rather than three spreadsheets added together. Adini Couture has no registration of its own and mainly does job work — it still counts in the group, without being pulled into a return it does not belong in. Add the fourth company the day you open it. | working today |
| Excel Dashboard Builder | A full workbook — financial summary, HR, purchase, sales, inventory and production, GST, expenses — generated from the live records behind every other screen, with each company shown as its own row and a consolidated row that is a formula over them, never a separately typed total. | designed, not yet built |
| ESG / Sustainability Reporting | Water usage, chemical compliance, waste and packaging, reported from the same certificate and audit records Quality & Compliance already keeps — so a sustainability report is a query over evidence already on file, not a separate exercise assembled once a year from scratch. | designed, not yet built |

### Module 22 · AI Assistant, Agents & Automation

*Ask the house a question — and let the routine work run itself.*

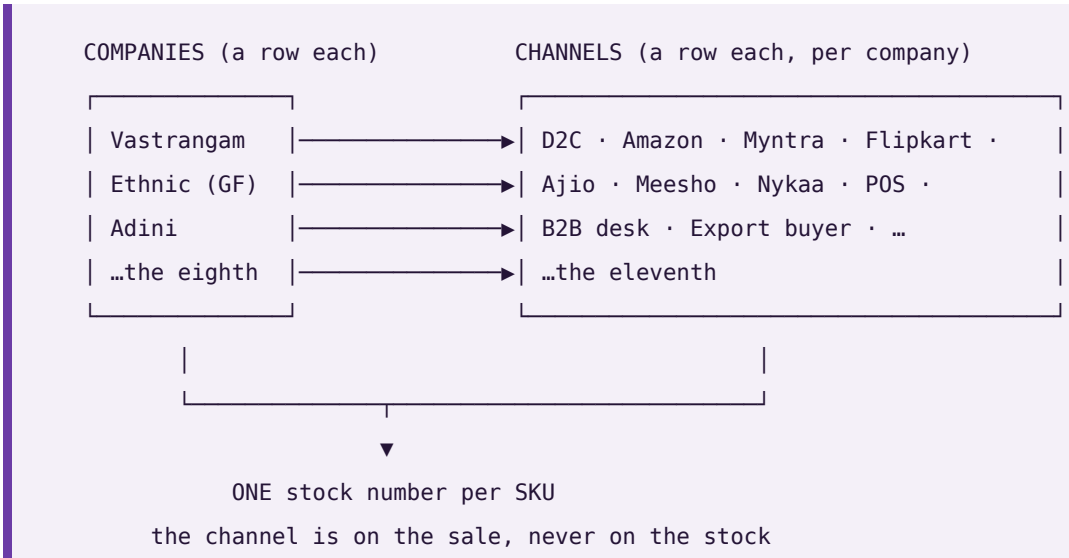
Last for the same reason Dashboard & BI is late: something that answers questions about the whole business can only be built once the whole business is in one place. Three different things live here and the difference between them matters. An ASSISTANT answers a question you asked, from the records, with the records attached. A CHATBOT holds the same conversation with your customer instead of you. An AGENT is given a job rather than a question and works out the steps itself. That last one is what separates this module from Automation Studio in Module 20, where a person draws the steps in advance and the rule runs the same way every time; and from Automation in Module 17, which fires marketing campaigns and nothing else. Both of those stay exactly as they are — this module sits above them and calls them, rather than replacing either. Module 18 writes content; this module answers and acts.

**Reads from:** Every module **Writes to:** Projects & Collaboration · CRM · Marketing

| App | What it does | State |
|---|---|---|
| AI Assistant | Ask in your own words — “what did Myntra actually pay us last week, and what is still short?”, “which designs did Kalamandir return most” — and get the answer with the rows it came from underneath it, each clicking through to the record. It reads the same ledger and the same settlement lines the accounts screen reads, so its figure and the books agree. When it cannot find the answer it says so; it never puts a plausible number in place of a real one. | designed, not yet built |
| AI Chatbot | The same engine facing your buyer, on the website and on WhatsApp: where is my order, will the L fit me, I want to return this saree. It reads the real order and the real size chart, not a script written last season, and it says “let me get someone” for anything about money or a complaint — handing over into the Surat helpdesk queue with the whole chat already attached. It never asks a buyer for a card number, a bank detail or a password. | designed, not yet built |
| AI Agents | A job rather than a question: “chase every unmatched payout line from last week and draft the claim for each.” It works out the steps and stops where a person has to decide. Filing the claim, moving money, changing a price or messaging a boutique all wait for your yes. | designed, not yet built |
| Agent Guardrails & Run Log | What each agent is allowed to touch, written down as a scope rather than trusted to a prompt, and every run recorded step by step: what started it, what it read, what it proposed, what a person approved, what it actually changed. Kept in the same audit trail as everything else, with the same absence of an off switch. An agent whose working nobody can inspect afterwards is not a colleague, it is an unexplained entry in the books. | designed, not yet built |
| Knowledge & Retrieval | The index that makes the answers grounded: your own designs, rate cards, settlement files, standard procedures and past decisions, searchable so a reply quotes what is actually on file instead of what a model remembers about the trade in general. Permission-scoped at the row, so two people asking the same question get answers drawn only from what each of them may already see. | designed, not yet built |

## Companies and channels — how many is up to you

You have three companies and sell on seven marketplaces. Neither of those is a setting this system was built around, and neither is a ceiling. A company is a **row**. A channel is a **row**. Every business record carries the company it belongs to, and every sale carries the channel it came through. Ten companies selling on ten channels each is the same three tables and the same code as three and seven.




**Three things this buys you, and one it deliberately refuses.**

**Each company's books are its own.** Its trial balance balances on its own. No report can reach across into another company's rows — not by convention, but because a journal line can only point at an account that belongs to the same company, and a test checks that no line anywhere ever does.

**The group is the sum minus what you sold yourselves.** When Vastrangam sells to Ethnic, that is revenue in one set of books and cost in another. Adding the companies up would report a group turnover the group never earned from the outside world. Every entry that names a sister company is eliminated at group level, and the consolidation returns all three numbers — gross, eliminated, group — so you can see the elimination rather than take it on trust.

**The channel is a dimension of the sale, never of the stock.** You can read this month by channel, by company, or by both. What you cannot do is keep a separate stock number per channel, and that is on purpose: the last piece sold on one marketplace has to vanish from the other ten at that instant, which per-channel inventory cannot do.

**And the tool follows your sheets.** Drop a workbook into the Data Studio and the report's columns come from the sheets that are actually in it. Two companies today gives two pairs of columns; a fourth company is a new sheet in the workbook, not a new version of the software.

> **This is checked, not claimed.** `core/tests/core.test.js` *builds ten companies with ten channels each — a hundred channels — posts an order down every one plus ten inter-company sales, and asserts every company's books balance, that no journal line points at another company's account, and that the group figure is the plain sum* **minus** *inter-company trade: ₹2,10,500 gross, ₹50,000 eliminated, ₹1,60,500 group. It then calls the same builder for eleven companies and eleven channels with no code changed. The Data Studio's own tests do the matching thing on the reporting side: ten companies in one workbook produce ten pairs of columns.*

## The rules that hold everywhere

1. **No app depends on any single outside company.** Every capability — books, marketplaces, AI writing, couriers, payments, messaging, storage, GST, printing, barcode — has several interchangeable providers and a by-hand option, so the system works with nothing connected at all. **A provider named as the source of a figure is a bug**; providers move messages and money, the ledger originates numbers.
2. **The books are this system's own.** No other accounting package is required, ever. Tally, BUSY and Zoho remain available for anyone already running one — nothing assumes them, and no figure is ever sourced from one.
3. **Nothing ever asks for an account password.** Outside services connect with a scoped, revocable key. *This system will never ask you for a marketplace, bank or account password. If any screen ever does, it is not this system.*
4. **Money is integer paise, never a floating-point number**, because float arithmetic accumulates the error that eventually shows up as a trial balance that will not tie.
5. **Values that change over time are effective-dated** — a salary, a price, a tax rate, a commission. March payroll resolves the salary in force *in March*. A missing value is an error, never silently treated as zero.
6. **Nothing is deleted, only deactivated.** A person who leaves keeps their name attached to years of earnings, approvals and audit rows that must still resolve.
7. **The audit trail has no off switch.** Eight years, before-and-after values, as the MCA rule requires — because an audit trail that can be switched off is one that gets silenced exactly when it matters.
8. **Gates, not warnings.** Each app refuses one thing outright, because a warning gets clicked through on a busy afternoon. A cash-on-delivery order cannot be packed below its advance. A bill for more than was accepted cannot be paid. Every gate is also a self-test.
9. **It is not trained. It is built.** No model learns from your data. Every rule is written down, visible on the Wiring screen, and checked by a self-test — so it is right on day one.
10. **You can reach it from anywhere, but it cannot be reached into.** Ask & Print takes a plain line from your phone and sends a PDF back. The office reaches out; the internet never reaches in.

---

## How it is verified

Nothing ships because it looked right on a screen.

1. **The arithmetic, with no screen involved.** Each engine runs in isolation and its self-tests execute against seeded data.
2. **Every screen and every control, in a real browser.** Each build opens in headless Chromium; every screen is visited and every interactive control on it is clicked. Any console error fails the build.
3. **The real job, with the result asserted.** Not "does the button click" but "did the thing happen". A control that looks alive but changes nothing fails the build.
4. **Against the owner's own figures.** Where the business already knows the answer, the software has to reproduce it — and where it cannot, the reason is named rather than the number quietly adjusted. The reference report the business produced by hand covers April 2025 to June 2027 and totals **25,307 sets, 59,110 pieces and ₹26,90,062** across 143 designs and 29 karigar units. Run today against the workbooks as

they now stand, the engine returns **16,662 sets, 36,229 pieces and ₹17,45,911** across 128 designs and 20 karigars — because the FY2026-27 workbook has since been restructured into one payment sheet per team and no longer carries a design grid at all, so that year's rows cannot be read from it. The verification does not paper over this: it places **every** design in the reference report into a bucket with a named cause — matched exactly, changed at source, rate added since, incomplete-set rule, or only present in the FY2026-27 grid — and fails on any design whose difference has no explanation. There are currently none. A mismatch is a bug, not a rounding difference; an unreadable input is a stated limitation, not a passing test.

5. **A structural audit.** Every "comes from" on every Wiring screen must name a module that actually exists, no vendor name may ever be the source of a figure, and the app count in every file must match this one.
6. **The shipped copy, not the working copy.** The packaged archive is extracted and re-tested in the folder a customer would open it in, because a packaging step that quietly renames a file breaks nothing until it is in somebody's Downloads folder.

---

### The honesty charter

This is the standard the build is held to, and it is written down because a standard nobody wrote down is a standard nobody can be held to.

1. **Nothing is called finished that is not.** Every deliverable is labelled tool, stub, mockup or spec. A mockup stays labelled a mockup even when a working version would be more impressive to show. Generation features stay badged as mockups until a real paid API is actually wired.
2. **Counts are counted, never claimed.** Every module and app figure on this page is read from the canonical module list when the page is built. No number here was typed by hand.
3. **Progress is reported as it is.** If tests fail, the failure is shown with its output. If a step was skipped, it is named as skipped. "Done" means implemented, tested and checked against the original request — not "the code has been written".
4. **The gap is stated, not buried.** 16 of 113 apps work today. A further 2 have a working, tested engine but no screen on it yet, and are counted separately rather than folded in to make the first number look larger. The remaining 95 are designed and specified. Those 16 still run on their own storage, and rewiring them onto the shared core is the first job of Module 01 — until that is done they are good tools, not yet one system.
5. **Uncertainty is surfaced, not smoothed over.** Where something cannot be verified, it is reported as unverified rather than presented as fact.

---

*Vastrangam BOS · one business, one brain · 22 modules · 113 apps · one shared data core · 16 working today*

---

# PART TWO — THE PLAN OF ACTION

## One platform. Every function. Wired together.

The build plan for a multi-company operating system covering a Surat ethnic & western fashion manufacturer running three sister companies — D2C, seven marketplaces, B2B and export on one order book, one stock number and one ledger.

**22 modules · 109 apps · 16 apps working today · 2 more with their engine running · 91 to build.** Built in dependency order, Module 01 first through Module 22 last. Every module is finished — every app on the shared database, verified in a browser — before the next begins.

**How to read this.** Part I is the integration layer: the law the whole system obeys, the data core that makes it one system rather than 21 programs, and the wiring between modules. Part II is the 21 modules one at a time, each with a diagram of how it actually functions. Every section carries a diagram because the point of this document is that you can see the machine working, not just read a list of features.

**Honest framing.** Nothing in this document is described as finished unless it is. Each app is marked **BUILT** (a working file you can open today, carrying its own self-tests), **ENGINE WORKING** (the arithmetic written and passing its own tests on the command line, with no screen on it yet) or **SPEC** (designed to this behaviour, not yet written). The count above is the true one: 16 built and 2 engines, of 109. Section A7 gives the two commands that check the engines in about a minute.

---

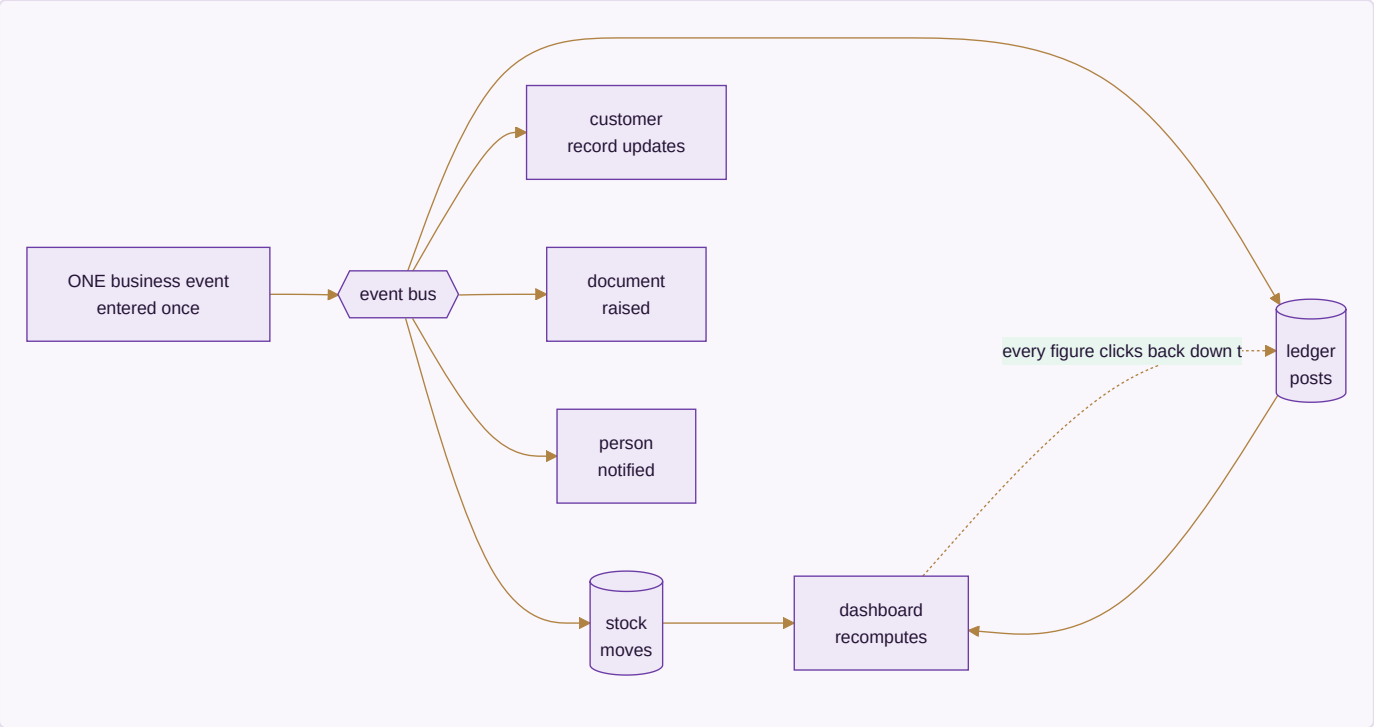
# PART I — HOW THE SYSTEM WORKS

---

### A0 · THE ONE LAW — INTEGRATION

**A business event is entered once and flows everywhere it belongs.** No module re-enters data another module already has. No number is maintained separately from the record that created it.

This is the whole design. Everything else in this document serves it.




**The test of the law.** Any figure on any screen can be clicked down to the record that produced it, and that record down to the document that created it. A dashboard number is a live query over the ledger — never a total someone maintained on the side. If a figure cannot be traced, it is a defect, not a rounding difference.

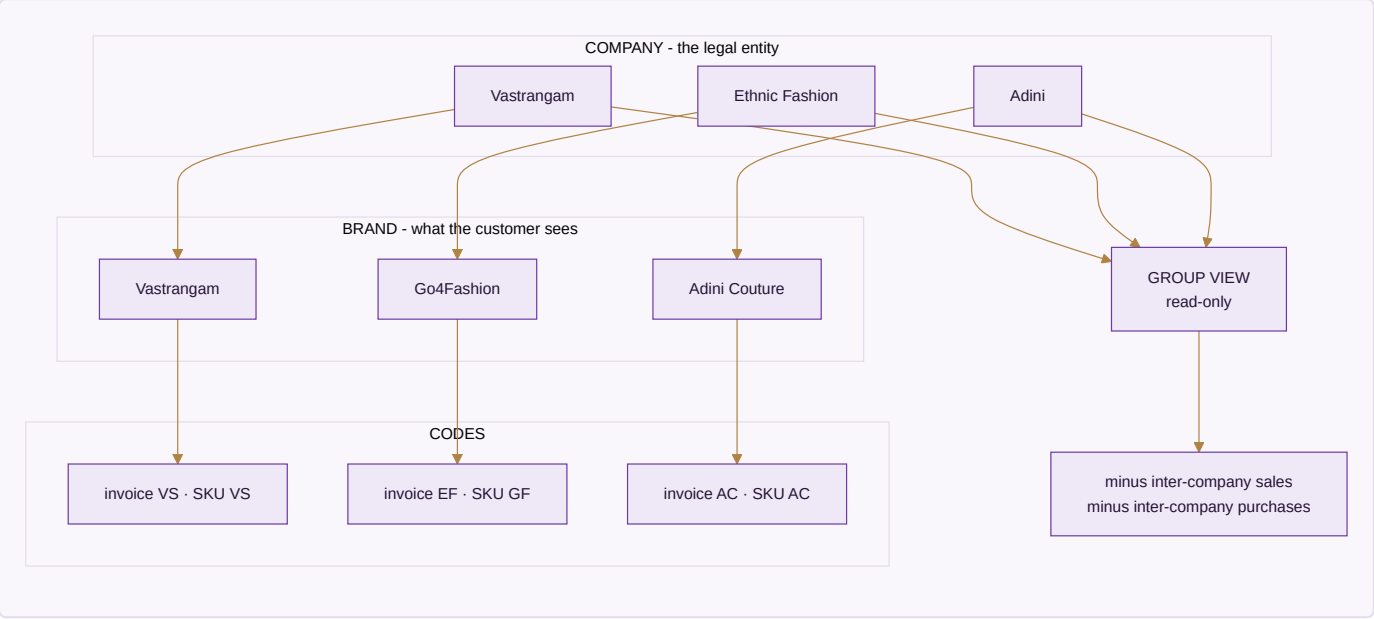
### The eight cascades that must fire by themselves

These are the specific chains the platform has to make automatic. If one of them needs a human to re-key something, the system has failed at the only thing that makes it a system.

| A single action… | …updates every one of these, in one transaction |
|---|---|
| **A sale**, any channel | stock deducted → invoice with GST → ledger (Dr Debtor/Bank, Cr Sales + Output GST) → customer outstanding and lifecycle → settlement expectation → dashboard |
| **A marketplace order pull** | sales order created → stock reserved → pick list → fulfilment → on payout, reconciliation and variance check → books |
| **A settlement import** | each line matched to its order → commission, fees, TCS, TDS to books → variance → claim raised → bank reconciliation → SKU profit → dashboard |
| **A return** | credit note → refund → a wrong return becomes dead stock, never restocked → return cost to P&L → return rate feeds design analytics |
| **A karigar production report** | pooled set completion and piece-rate earnings → finished stock in → payout in HR → Karigar Wages posted → cost per piece → design profit |
| **A material purchase** | PO → GRN → three-way match → raw stock in → vendor payable → input tax credit → landed cost into COGS |
| **A staff attendance mark** | effective-dated salary resolved → payroll → Salaries posted → productivity cost → design cost |
| **A generated image or listing** | asset library → product listing on the storefront and each marketplace → marketing calendar → published → campaign return tracked back |

## A1 · THE THREE COMPANIES

The single most common way a multi-company system goes quietly wrong is collapsing three different ideas into one field. They are kept apart at the root of the database.




**Company ≠ brand ≠ prefix.** Ethnic Fashion the company trades as Go4Fashion the brand; its invoices read EF and its SKUs read GF. Reports group by *company*. SKUs use the *brand code*. Invoices use the *company prefix*.

Three separate fields, because they genuinely are three different things.

**Group is a lens, not an entity.** The consolidated view is read-only. Group profit is the sum of the three companies minus inter-company sales and purchases — a real elimination, so moving stock from one pocket to another never inflates group turnover. Editing happens inside a company; the group only reports.

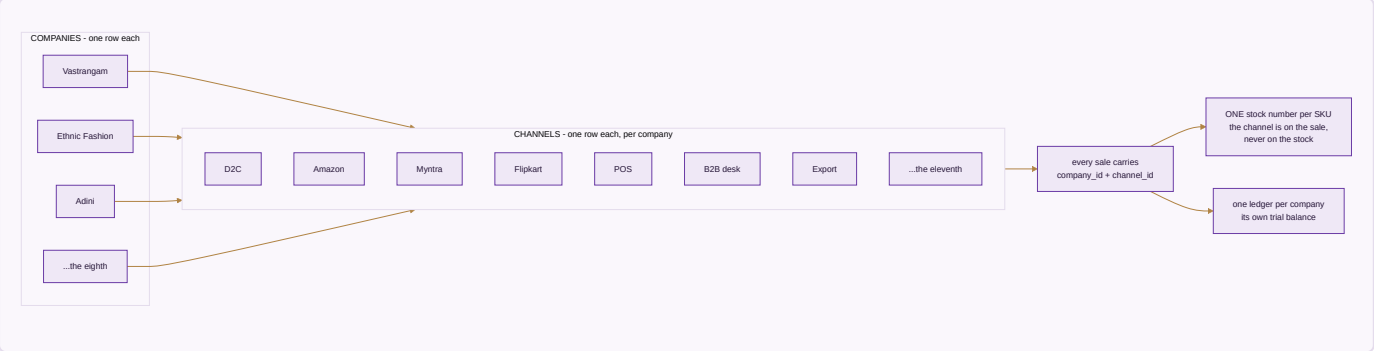
**A company without its own registration is still a company.** A job-work arm that holds no GSTIN counts in the group figures without being dragged into a return it does not belong in.

**Three is today's data, not the design.** Nothing above is built for the number three. `companies` is an ordinary table; adding an eighth is inserting a row. The next section is about the other half of that — the channels those companies sell through.

---

## A1b · COMPANIES × CHANNELS — WHY NEITHER NUMBER IS FIXED

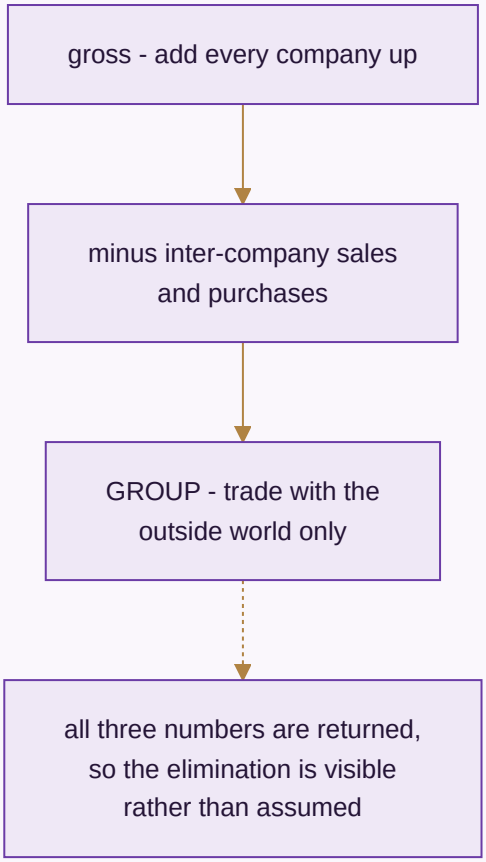
A **channel** is a way a company sells: its own site, a marketplace seller account, a POS counter, a B2B desk, an export buyer. Like a company it is a row, not a column and not a list in the code. Three companies on seven marketplaces is the data this business has now; ten on ten is the same three tables.




**Every business row carries its company.** A journal line can only point at an account belonging to the same company, so one company cannot read another's figures — checked by a test rather than left to discipline.

**The channel is a dimension of the sale.** `channel_id` sits on the stock movement and on the journal entry, so any figure can be read by channel, by company, or by both. It deliberately does *not* sit on the stock row: inventory is not per channel, because the last piece sold on one marketplace must vanish from the other ten in the same instant.

**The group eliminates what the companies sold each other.** An entry whose other side is a sister company carries `counterparty_company_id`, and consolidation removes exactly those.



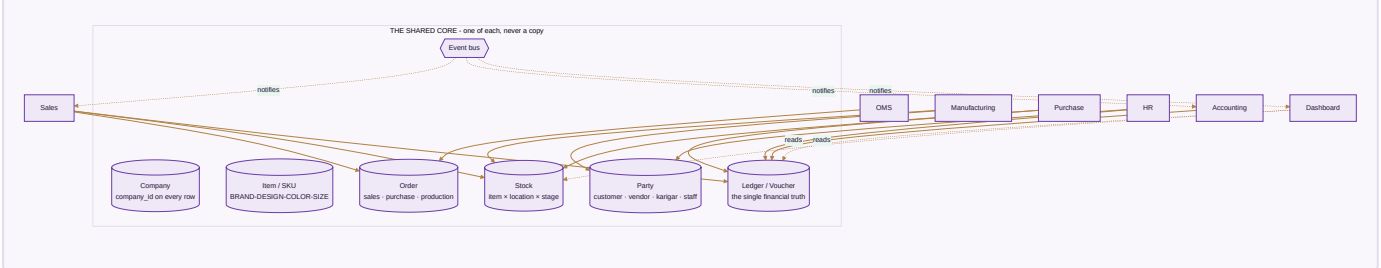

**This is proved, not asserted.** `core/tests/core.test.js` builds **10 companies × 10 channels = 100 channels**, posts one order down every channel plus ten inter-company sales, and checks each company's figures channel by channel, that every trial balance still balances, that no journal line anywhere points at another company's account, and that the group figure is the sum minus inter-company trade — **₹2,10,500 gross, ₹50,000 eliminated, ₹1,60,500 group**. The same builder is then called for 11 × 11 with nothing in the code changed.

The reporting side matches: the Data Studio detects every `<Company> Sale` / `<Company> Return` sheet pair present in an uploaded workbook and emits one pair of quantity columns per company. A fourth company is a new sheet, not a new release.

---

## A2 · THE UNIFIED DATA CORE

This is the physical reason a sale can touch stock and the books at the same instant: there is one set of master records underneath every module, not one silo per module.



**The seven things there is exactly one of.** Company · Item/SKU · Party · Stock · Ledger · Order · the event bus that lets modules notify each other. Every business table carries `company_id`, and row-level security enforces it in the database — so even a bug in application code cannot leak one company's data into another's.

**Why this matters more than any feature.** The predecessor to this system was a set of separate browser tools, each with its own private storage. A sale recorded in the sales tool never reached the accounting tool. Every integration promise in this document rests on that being fixed: one core, one event bus, and the order the sales module writes is the identical row the accounting module reads.

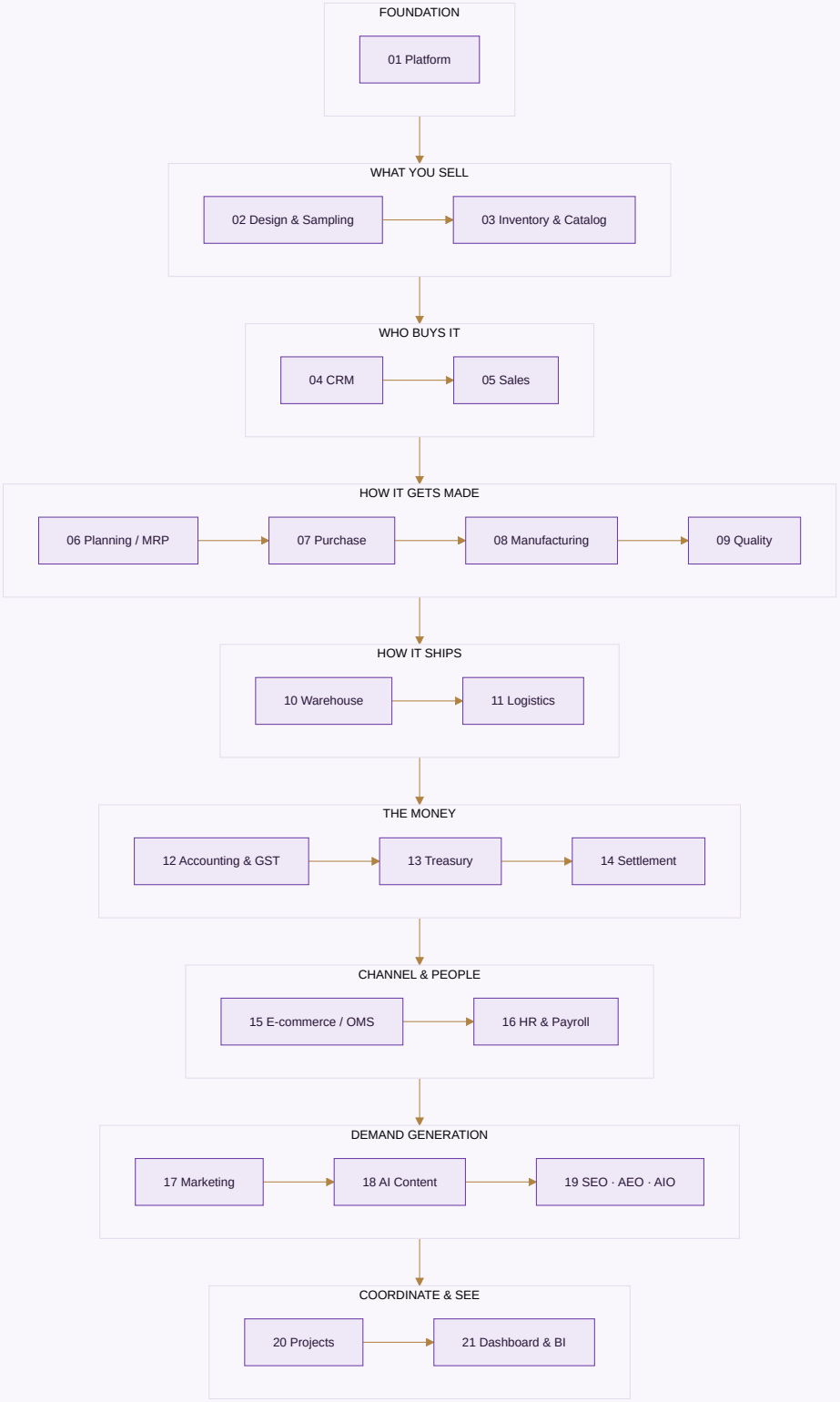
### Four rules the core enforces before any module is built

1. **Money is integer paise, never a floating-point number.** Currency arithmetic in floats accumulates error that eventually shows up as a trial balance that will not tie.
2. **Values that change over time are effective-dated** — a salary, a price, a tax rate, a commission. A payroll run for March resolves the salary in force *in March*, not today's. Zero rows in force is an error, never treated as zero.
3. **Nothing is deleted, only deactivated.** A person who leaves is marked inactive; their name is attached to years of earnings, approvals and audit rows that must still resolve.
4. **The audit trail cannot be switched off.** Every edit records who, when, and the value before and after, kept for eight years as the MCA rule requires. It is not a setting that defaults to on — there is no switch, because an audit trail with an off switch is one that gets silenced exactly when it matters.

---

## A3 · THE MODULE MAP — 22 MODULES

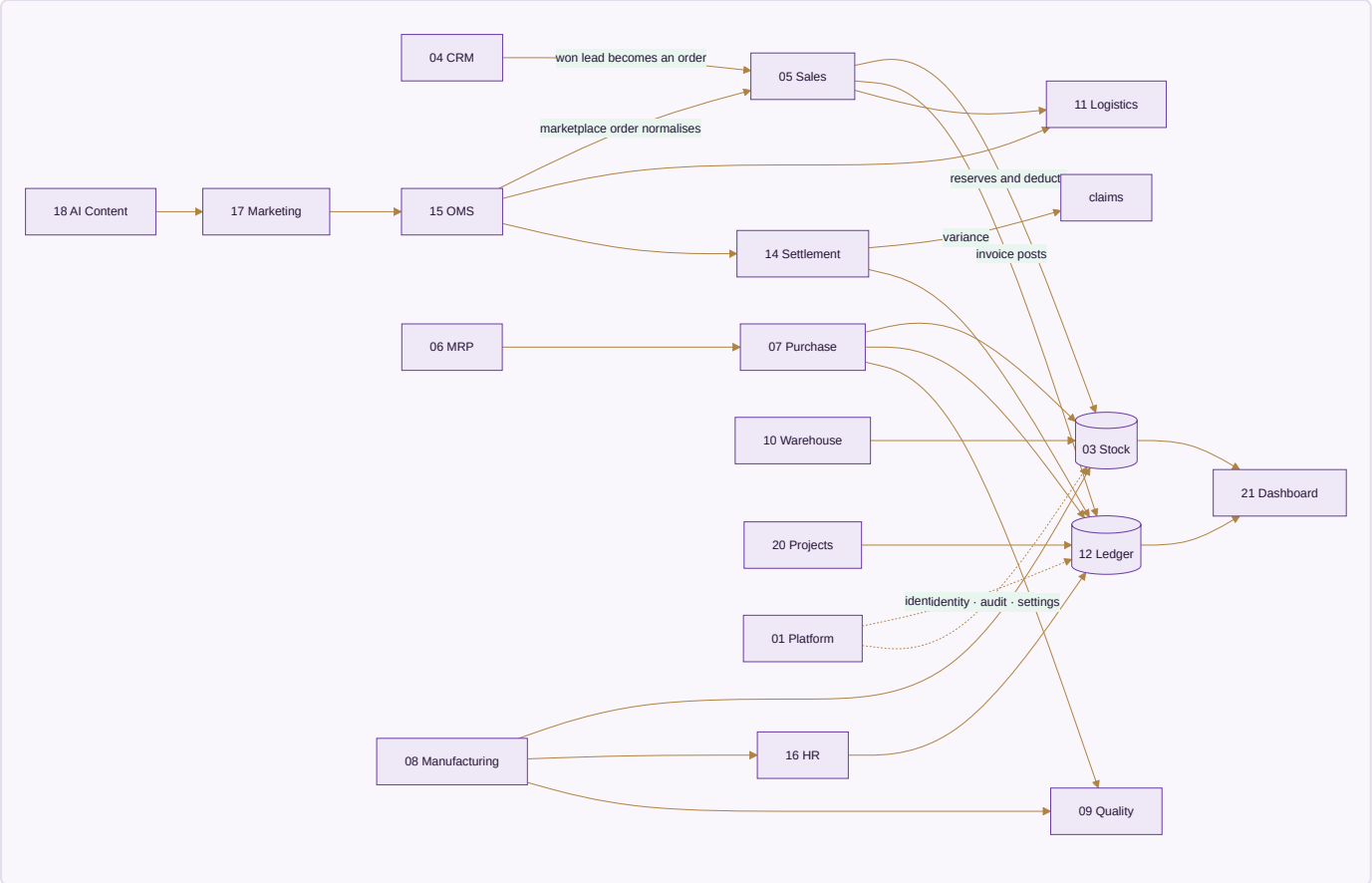
Every module lives inside one application: one login, one company switcher, one data core. The numbering below is the build order, and the build order is dependency order — a module is only built once everything it needs already exists.


FOUNDATION
01 Platform
WHAT YOU SELL
02 Design & Sampling
03 Inventory & Catalog
WHO BUYS IT
04 CRM
05 Sales
HOW IT GETS MADE
06 Planning / MRP
07 Purchase
08 Manufacturing
09 Quality
HOW IT SHIPS
10 Warehouse
11 Logistics
THE MONEY
12 Accounting & GST
13 Treasury
14 Settlement
CHANNEL & PEOPLE
15 E-commerce / OMS
16 HR & Payroll
DEMAND GENERATION
17 Marketing
18 AI Content
19 SEO · AEO · AIO
COORDINATE & SEE
20 Projects
21 Dashboard & BI

| # | Module | Apps | Built | Engine | To build |
|---|---|---|---|---|---|
| 01 | Platform | 8 | 1 | 1 | 6 |
| 02 | Design & Sampling | 2 | 0 | 0 | 2 |
| 03 | Inventory & Catalog | 4 | 0 | 0 | 4 |
| 04 | CRM | 4 | 3 | 0 | 1 |
| 05 | Sales | 8 | 5 | 0 | 3 |
| 06 | Planning & Requirements (MRP) | 3 | 0 | 0 | 3 |
| 07 | Purchase | 3 | 2 | 0 | 1 |
| 08 | Manufacturing | 4 | 0 | 0 | 4 |
| 09 | Quality & Compliance | 2 | 0 | 0 | 2 |
| 10 | Warehouse | 3 | 0 | 0 | 3 |
| 11 | Logistics | 5 | 0 | 0 | 5 |
| 12 | Accounting & GST | 9 | 0 | 0 | 9 |
| 13 | Treasury & Financial Planning | 3 | 0 | 0 | 3 |
| 14 | Settlement | 3 | 0 | 0 | 3 |
| 15 | E-commerce / OMS | 11 | 2 | 0 | 9 |
| 16 | HR & Payroll | 5 | 0 | 0 | 5 |
| 17 | Marketing | 8 | 0 | 0 | 8 |
| 18 | AI Content Engine | 8 | 0 | 1 | 7 |
| 19 | SEO, AEO & AIO | 3 | 0 | 0 | 3 |
| 20 | Projects & Collaboration | 7 | 0 | 0 | 7 |
| 21 | Dashboard & BI | 5 | 3 | 0 | 2 |
| 22 | AI Assistant, Agents & Automation | 5 | 0 | 0 | 5 |
| | **Total** | **113** | **16** | **2** | **95** |

## A4 · MODULE-TO-MODULE WIRING

Every arrow here is a real, required data flow — not an aspiration. This is the fabric that turns 21 modules into one system.




### Reading the important arrows.

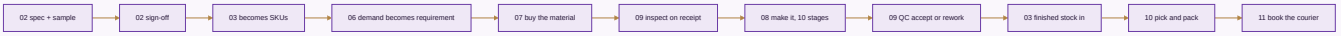
1. **CRM → Sales.** A won lead becomes a sales order carrying the customer, their price list and their credit tier — nothing re-typed.
2. **Sales → Inventory.** The order reserves and then deducts the single stock number, so the last piece sold anywhere disappears everywhere in the same instant.
3. **Sales → Accounting.** The invoice posts Dr Debtor/Bank, Cr Sales plus output GST, with CGST+SGST or IGST decided from the two GSTINs' state codes — never picked from a dropdown.
4. **OMS → Settlement → Accounting.** Payouts reconcile per line; commission, fees, TCS and TDS post to the books; any variance opens a claim with its evidence.
5. **Purchase → Inventory + Accounting.** The GRN adds stock; the three-way match gates the payable; the GST becomes input credit.
6. **Manufacturing → Inventory + HR + Accounting.** Production adds finished stock, computes karigar piece-rate earnings, posts wages, and rolls cost per piece into design profit.
7. **Accounting → Dashboard.** Every dashboard figure is a query on the ledger, never a separate counter. This is the integrity rule the whole system exists to keep.

---

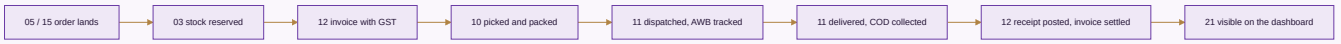
## A5 · THE FIVE END-TO-END FLOWS

Four of these are the journeys the business actually runs on. Each crosses many modules, which is exactly the point — no single module completes any of them alone.

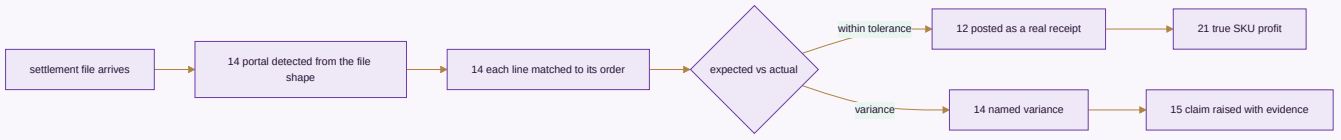
### Flow 1 · Design to dispatch




### Flow 2 · Order to cash

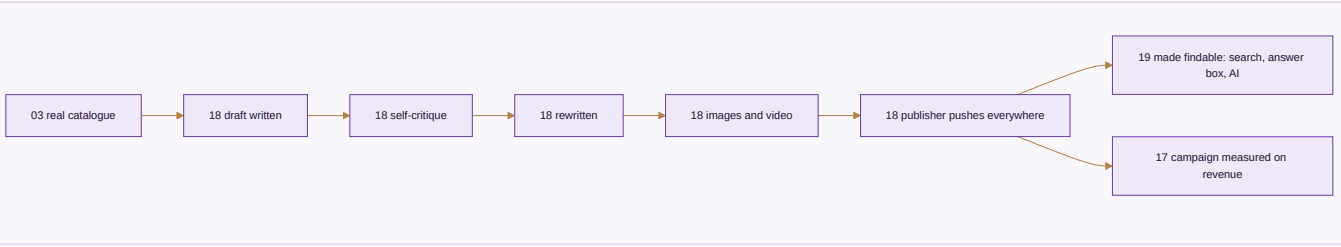



### Flow 3 · Settlement to books




### Flow 4 · Karigar to payroll


### Flow 5 · Content to published




---

## A6 · WHY THIS BUILD ORDER

The order is not a preference. Each module needs something the previous one produces, and building out of order means building against data that does not exist yet.



**Four decisions worth explaining, because they differ from the obvious arrangement.**

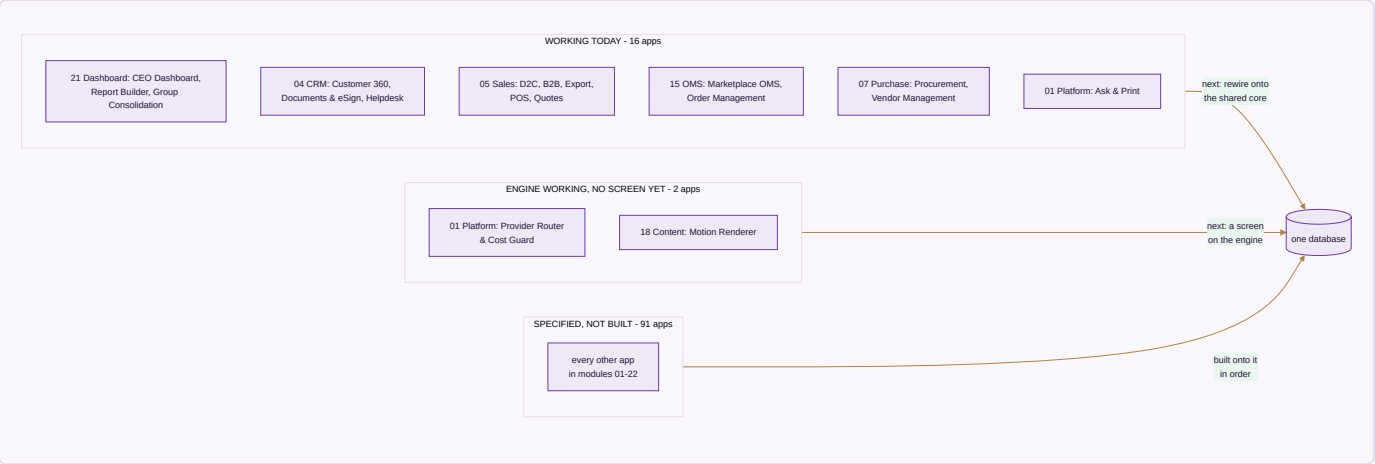
- **Platform is first, not last.** Identity, permissions and the audit trail are what every other module writes through. Retrofitting an audit trail means going back through every module.

- **Design & Sampling comes before Inventory.** A product must be specified and approved before it can exist as a catalogue record. Building the catalogue first means inventing styles with no real specification behind them.
- **Quality is its own module, not a step inside Manufacturing.** Rejecting goods at receipt and rejecting goods on the floor are the same discipline, and the certificates that prove a standard is followed belong beside the inspections that back them.
- **Dashboard is next to last, and the AI layer is last.** Dashboard has nothing true to show until the other modules produce real records; the assistant and the agents above it have nothing to read or act on until the dashboard's figures exist. Both fail the same way if built early. A dashboard built first can only display invented figures.

**What a module does when a later module is not built yet.** It is finished against the data that exists on its day, and shows more as later modules come online. Module 21's dashboard is real from the moment it is built and grows richer as Module 12's ledger fills.

---

## A7 · THE HONEST STATE TODAY




**What "built" means here.** Sixteen single-file apps that open in a browser, carry their own self-tests, and pass a full click-through audit with zero console errors in both editions. That is verified, not claimed.

**What "engine working" means, and why it is a third word rather than a generous reading of the first.** Two apps have their hard part written and passing its own tests on the command line, with no screen on them yet. They are not counted among the sixteen, because the sentence beside that number promises a browser check these have not had. They are not called "specified" either, because the arithmetic exists and runs. Anyone can check both in the time it takes to read this:

```
node brand/suite/router.js --selftest                  # 31 passed, 0 failed
node brand/suite/studio/motion_render.js --selftest    # 14 passed, 0 failed
```

**What is honestly not done.** Those sixteen apps still run on their own storage. The first work of each module is rewiring its built apps onto the shared core so they read and write the same records as everything else. Until that happens they are good tools, not yet one system.

## A7b · WHERE THE SIX NEWEST APPS CAME FROM

Ten open-source projects were read to answer one question: what do they have that this does not? Not to copy — nothing from any of them is in this codebase, and the licences below are the reason that distinction is written down rather than assumed. Six gaps were real enough to specify, and two of the six were built rather than merely described.

| Project | Licence | What was taken |
|---|---|---|
| OmniRoute | MIT | The mechanisms behind **Provider Router & Cost Guard** — cascade, breaker, backoff, budget |
| HyperFrames | Apache-2.0 | Deterministic frame-seeking, now the **Motion Renderer** |
| voicebox | MIT | The shape of **Narration Studio** — chunked long text, many languages, local by default |
| easydiffusion | CreativeML Open RAIL-M | The shape of the **Image Generation Slot** — queue, preview, inpaint, upscale |
| n8n | fair-code | The idea behind **Automation Studio** — a visual when-X-then-Y over an event bus |
| Odoo | LGPL | Read for gap-finding only; it surfaced the missing **Website & Page Builder** |
| OpenMontage | AGPLv3 | Idea only — copying any of it would force this codebase to be published |
| ideogram4 | **Non-Commercial** | Nothing usable. The model may not be used in a commercial product at all; only the capability it demonstrates is described |
| palmier-pro | GPLv3, macOS-only | Nothing — wrong platform |
| higgsfield | — | Nothing — GPU training infrastructure, unrelated to this business |

**Why these six and not others.** Provider Router was first because this document already *claimed* no capability depends on one outside service, and a claim with nothing enforcing it is the kind of gap that only shows up on the evening a courier API stops answering. Motion Renderer was built because the two things it needs — a headless browser and an ffmpeg binary — are already in this repository for other reasons, so it was buildable today rather than someday. The other four are specified honestly: a website builder is a large piece of work, image generation needs hardware this system does not have, and saying so is cheaper than discovering it later.

## A8 · WHAT THE MASTER SPEC HAD THAT THIS DOCUMENT DID NOT

This plan was checked against the Vastrangam ERP Complete Master Prompt v3.0 — all 63 pages — and the check is recorded here rather than summarised, because a gap-check nobody can audit is just a reassurance.

**What already agreed.** The one law and its cascades. Companies and channels as rows rather than constants. Karigar pooled set-completion. Money as integer paise. Effective-dated salary. Audit everything, delete nothing. Group consolidation with inter-company elimination. And provider-agnosticism — §A.3.1 of the master spec says no provider SDK may be called from business logic, and the Provider Router in Module 01 is what enforces it at

runtime. The spec's 16 domains map onto these 22 modules with nothing dropped, and every competitive addition it flagged as missing (kit/combo SKUs, repricing, NDR workflow, listing manager, subscriptions, NPS, knowledge base, events) was already here.

**What was genuinely missing — the whole execution half.**

| The master spec had | This document had |
|---|---|
| The pinned stack: Supabase, Next.js, Interakt, n8n, Capacitor, RLS, Shiprocket, Razorpay | nothing |
| A ~100-table PostgreSQL schema | a 19-table SQLite core |
| A REST surface and five inbound webhooks | nothing |
| 8 phases over 32 weeks, each with a Definition of Done and a gate | module dependency order only |
| BUSY cutover, opening balances, a 60-day parallel run, decommission criteria | nothing |
| p95 performance targets, RTO/RPO, backup policy | nothing |
| A risk register, success metrics, and daily-to-annual runbooks | nothing |
| The UI shell, five role dashboards, PWA and Capacitor | nothing |
| The formulas: daily rate, return costs, closing stock, realisation, CRM tiers, three-way match | almost none |
| Acceptance figures from the owner's own files | one of them |

Three apps were missing outright: **Customisation & Made-to-Measure**, the **WhatsApp Command Console**, and **Recruitment**. All three are now in the module list.

**What closed it.** PART IV of this document (E1–E12) is the execution half, written from that spec. `core/schema.postgres.sql` is the production schema, 113 tables, gated by `core/tests/schema.test.js`. Thirty-four formula rules were added to the rulebook. The counts above are not a summary of the fix — they are what the check found, kept so the next reader can see what was wrong and judge whether it is now right.

**One deliberate difference from the spec, stated rather than slipped in.** The spec asks for money as `numeric(14,2)`. This system stores money as an integer count of paise, in both schemas and in the engine. That satisfies the requirement behind the instruction — never a float, never a rounding decision — more strictly rather than less, and it removes the one place the database and the engine could still round differently. Converting to `numeric(14,2)` for any report is a division of an exact integer, so there is no rounding decision left to get wrong. `core/tests/schema.test.js` fails the build if any money column in either schema is ever declared as a float, a decimal or a numeric.

---

# PART II — THE 22 MODULES

## THE RULEBOOK AT A GLANCE

**285 rules across 22 modules. 86 of them are enforced by a test that runs today; the rest are specified.** Every rule states what happens, and what the system will *not* do instead — because the refusal is the half a business can actually rely on. A rule marked ENFORCED names the file and the test that proves it, and `brand/site/checkrules.js` fails if that test cannot be found, so a rule here cannot claim a proof it does not have.

The specified ones are not filler — they are the build queue, in the order the modules are built. That is the number to watch: it is meant to fall, build by build, and it can be counted rather than claimed.

| Module | Rules | Enforced | Specified |
|---|---|---|---|
| 01 · Platform | 25 | 20 | 5 |
| 02 · Design & Sampling | 7 | 0 | 7 |
| 03 · Inventory & Catalog | 14 | 7 | 7 |
| 04 · CRM | 9 | 0 | 9 |
| 05 · Sales | 18 | 4 | 14 |
| 06 · Planning & Requirements (MRP) | 8 | 0 | 8 |
| 07 · Purchase | 12 | 0 | 12 |
| 08 · Manufacturing | 20 | 9 | 11 |
| 09 · Quality & Compliance | 7 | 0 | 7 |
| 10 · Warehouse | 8 | 0 | 8 |
| 11 · Logistics | 11 | 0 | 11 |
| 12 · Accounting & GST | 24 | 16 | 8 |
| 13 · Treasury & Financial Planning | 8 | 1 | 7 |
| 14 · Settlement | 13 | 0 | 13 |
| 15 · E-commerce / OMS | 19 | 10 | 9 |
| 16 · HR & Payroll | 22 | 8 | 14 |
| 17 · Marketing | 10 | 0 | 10 |
| 18 · AI Content Engine | 11 | 2 | 9 |
| 19 · SEO, AEO & AIO | 6 | 0 | 6 |
| 20 · Projects & Collaboration | 9 | 1 | 8 |
| 21 · Dashboard & BI | 9 | 6 | 3 |
| 22 · AI Assistant, Agents & Automation | 15 | 2 | 13 |
| **Total** | **285** | **86** | **199** |

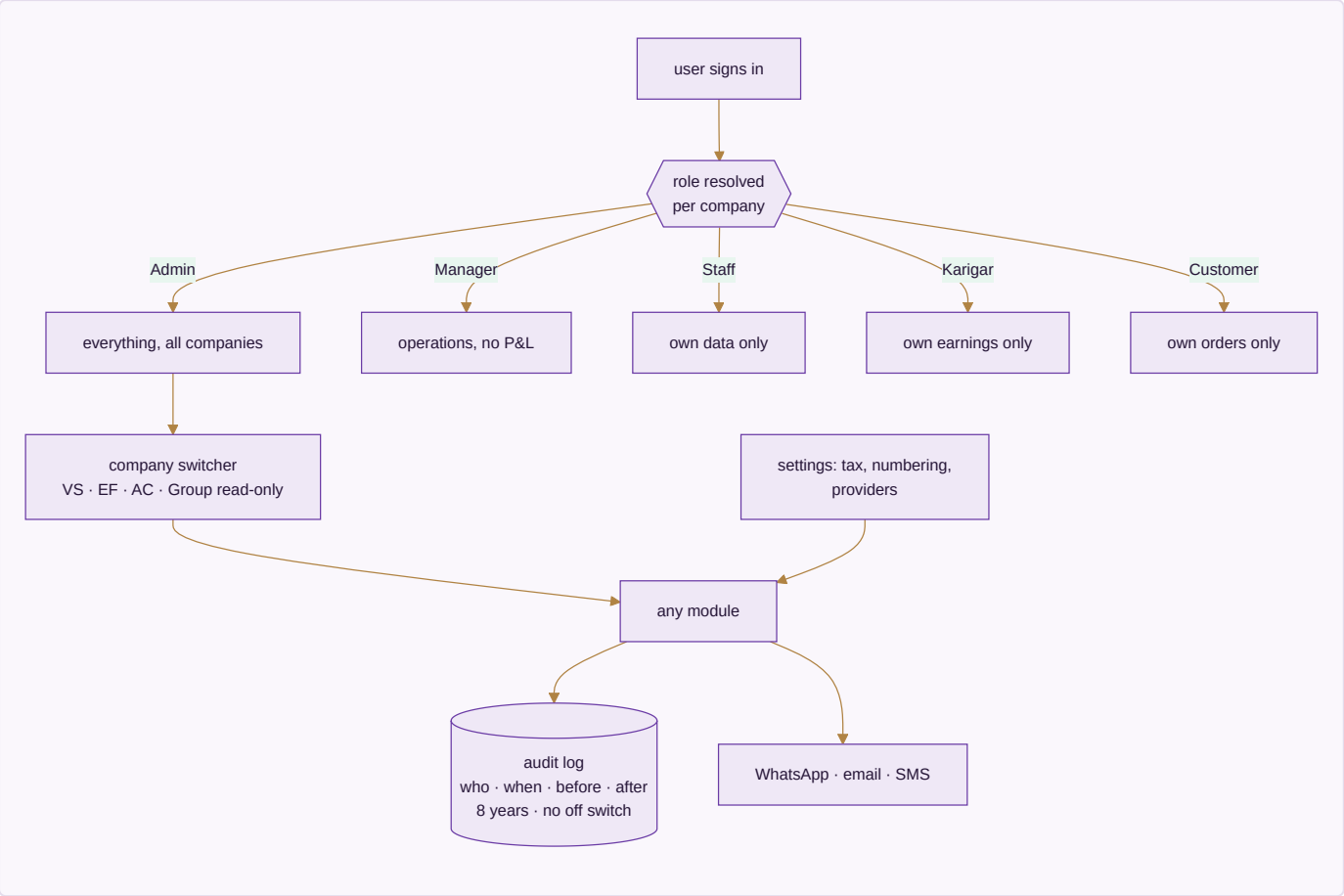
Each module below carries: what it is, a diagram of how it functions, its apps with an honest built or spec mark, the data it owns, the rules that actually govern it, what it reads and writes, and the condition that decides it is finished.

# MODULE 01 · PLATFORM

*The spine every module runs on*

**What it is.** Not a module you open — the layer underneath the other twenty-one. Who can see what, how the business is configured, and an unalterable record of everything that ever happened. It is built first because every module above it writes through it, and an audit trail added later is an audit trail with a hole in it.

## How it works




## The apps (6)

| App | State | What it does |
|---|---|---|
| Identity, Settings & Audit | SPEC | Users, per-company per-role permissions, company switcher, tax and numbering setup, provider config with an integration-health view, and the browser over the audit trail |
| **Provider Router & Cost Guard** | **ENGINE WORKING** | The no-single-provider rule enforced instead of promised: an ordered fallback per capability, a breaker that trips a failing provider out, backoff between retries, and a spend ceiling in paise that refuses rather than warns |
| Ask & Print | **BUILT** | Ask from a phone — a ledger, a bill, today's packing slips — and get a PDF back, or print at the office with nothing plugged into the phone |
| Communications | SPEC | WhatsApp command console, broadcasts, email and SMS, and the scheduled jobs that carry a nudge without anyone remembering to send it |
| Data Privacy & Consent | SPEC | Consent captured where it is given and honoured downstream; retention and erasure tracked as the two different policies they are |
| Payment Data Scope | SPEC | The written statement of which systems ever see a card or bank credential — and which never do |

**What it owns.** `companies` · `users` · `user_companies` · `audit_log` · `integration_errors` · `settings_environment` · `whatsapp_messages` · `whatsapp_broadcasts` · `email_campaigns` · `notifications` · `consents`

### The rules that matter

- Permissions are a matrix — per company, per role — not one global level. A group of sister companies is exactly where a single global level leaks data across a boundary it should not cross.
- The audit trail has no off switch. Eight years, before-and-after values, MCA rule.
- Every capability runs through at least three interchangeable providers. **A provider named as the source of a figure is a bug** — providers move messages and money; the ledger originates numbers.
- Broadcasts go through the official API, warm up at 200/day, and honour a STOP keyword.
- **This system never asks anyone for a marketplace, bank or account password.** A tool that asks for those has become the thing it should protect its users from.

### The rulebook — 25 rules, 20 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R01.1 | **Every business record names the company it belongs to** | any record is written — a sale, a movement, a voucher, an employee | its company is stored on the row itself | inferring the company from who happens to be logged in, which silently mis-files every record made by someone who works across two of them | **ENFORCED** · core/tests/core.test.js · the schema loads and the three companies keep three different codes |
| R01.2 | **One company cannot read another company's records** | a query runs for a user scoped to one company | rows belonging to any other company are not returned at all | filtering in the screen while the data is already loaded — a filter can be removed, a scope cannot | **ENFORCED** · core/tests/core.test.js · one company cannot read another company |
| R01.3 | **The audit trail has no off switch** | anything touching money, stock, price, tax, pay or master data changes | the change and its before-image are written in the same transaction as the change itself | allowing a setting, a role or a migration to disable it — either both land or neither does | **ENFORCED** · core/tests/core.test.js · an audited insert leaves a before/after trail |
| R01.4 | **An update records what it was, not only what it became** | a row is changed | the current row is read first so the before-image is what was really there | trusting the caller's idea of the old value, which makes the trail a record of intentions rather than of facts | **ENFORCED** · core/tests/core.test.js · an update records what it was as well as what it became |
| R01.5 | **A table nobody thought to audit is refused** | code writes to a table that is not on the audited list | the write is refused and the table is named | letting it through quietly, which is how a money column ends up outside the trail without anyone deciding that | **ENFORCED** · core/tests/core.test.js · a table nobody thought to audit is refused, rather than slipping through |
| R01.6 | **Deletion is a reversal, never a removal** | a user deletes anything | the record is voided, marked, and still readable with the reason | removing the row — eight years of trail cannot survive a DELETE | **ENFORCED** · core/tests/core.test.js · voiding is the only removal, and it is reversible |
| R01.7 | **A module that is not in the canonical list cannot join the bus** | code subscribes to a business event | the module number is checked against modules.js and refused if unknown | letting an unregistered listener attach, which is how a cascade gains a step nobody can find later | **ENFORCED** · core/tests/core.test.js · a module not in modules.js cannot subscribe |
| R01.8 | **A cascade is all of it or none of it** | one business event fans out to stock, ledger, | every step commits together, or the whole thing is rolled back | leaving stock moved and the ledger unposted, which is the exact state no report can ever explain | **ENFORCED** · core/tests/core.test.js · a sale moves stock and posts to the ledger, or does neither |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | | customer and documents | | | |
| R01.9 | **A handler that throws takes the transaction with it** | any subscriber to an event fails | the emitting transaction fails too | swallowing the error so the originating action appears to have succeeded | **ENFORCED** · core/tests/core.test.js · a handler that throws takes the whole transaction with it |
| R01.10 | **No capability depends on a single outside service** | any capability is used — books, courier, payments, AI, storage, GST | an ordered list of interchangeable providers is tried, ending on one that needs nothing connected | having one provider whose outage stops the work, however good that provider is | **ENFORCED** · brand/suite/router.js · no spend ceiling can exhaust any cascade (a free option is always in it) |
| R01.11 | **A failing provider is taken out of the list, not hammered** | a provider fails repeatedly | it is tripped open, skipped entirely, and retried once after a cooldown | retrying into a dead service on every call while a working alternative sits further down the list | **ENFORCED** · brand/suite/router.js · three consecutive failures trip the breaker open |
| R01.12 | **A spend ceiling refuses, it does not warn** | a paid call would take spending past the ceiling set for it | that provider is refused and the work completes on a free one | letting it through with a warning nobody reads, and never refusing only the first provider while the same spend reroutes to the next | **ENFORCED** · brand/suite/router.js · a ceiling below the price refuses every paid option, not just the first |
| R01.13 | **The system never asks for a marketplace, bank or account password** | any integration is connected, by any module, including a chatbot or an agent | a scoped, revocable key is requested instead, cancellable from the provider's side without changing the login | accepting, storing, echoing or transmitting an account password — there is no screen, no import and no support flow that takes one | SPECIFIED |
| R01.14 | **Card and bank credentials never reach application code** | a payment needs a card or bank detail | the provider's own secured field takes it directly | passing it through this system, even in transit, even unlogged — what is never received cannot be leaked | SPECIFIED |
| R01.15 | **Consent and retention are two different clocks** | a person's data is held | why it may be used and how long it is kept are tracked separately, and an erasure request is resolved against both | treating a legal retention period as consent to keep using the data for anything else | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R01.16 | **A scoped key is revocable without touching the login** | an outside service is connected | a key limited to what that capability needs is stored, and the connection records which capability it serves | storing a credential that can do more than the capability requires, because the day it leaks is the day that difference matters | SPECIFIED |
| R01.17 | **A webhook is verified, idempotent and never silently dropped** | a payment, courier, storefront or messaging provider calls in | the signature is checked, the external id makes a repeat delivery a no-op, and a failure is logged with its payload for retry | trusting an unsigned call, and never processing the same external id twice — a duplicated payout or a duplicated order is indistinguishable from a real one afterwards | SPECIFIED |
| R01.18 | **A trade is added as data, never as a version of the software** | a business in a trade the system has never seen signs up | its vocabulary, stages, extra fields, documents, rule switches and starting reference data arrive as one configuration file, and every screen reads back in that trade's words | a branch, a fork or a bespoke build per industry — that is a consultancy with software attached, and it is the thing that stops a product from being one | **ENFORCED** · core/tests/packs.test.js · GATE · it loads from a JSON string with no code change |
| R01.19 | **A pack is data and can never be code** | a pack is loaded from any source | every value in it is inspected, at any depth, and a function anywhere inside it refuses the whole pack | letting configuration carry behaviour — the moment a pack can run code, adding a trade is a code change again and the guarantee in R01.18 is worthless | **ENFORCED** · core/tests/packs.test.js · a pack containing a function |
| R01.20 | **A pack may rename a concept, never invent one** | a pack declares its vocabulary | each entry is matched against the fixed list of concepts the engine has, and an unknown one refuses the pack | accepting an unrecognised word as a new concept, which turns "vocabulary" into a place to put anything and leaves the screens with a name for something that does not exist | **ENFORCED** · core/tests/packs.test.js · renaming a concept the engine does not have |
| R01.21 | **A pack extends tables that exist, and nothing else** | a pack adds fields | the table is checked against the real schema and the field type against the types the engine can store | creating a table on a customer's behalf from a configuration file, which puts the shape of the database | **ENFORCED** · core/tests/packs.test.js · adding a field to a table that does not exist |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | | | | outside the reach of the schema test that guards it | |
| R01.22 | **Money in a pack is money everywhere else** | a pack adds a field whose name reads as an amount, a price, a cost, a total, a fee or a rate | it must be declared in paise, and a plain number refuses the pack | letting a trade introduce a floating-point rupee through the side door after the whole schema was built to keep them out | **ENFORCED** · core/tests/packs.test.js · money declared as a plain number |
| R01.23 | **No pack can switch off a guarantee** | a pack sets a rule off | the rule id is checked against the rulebook, and against the list of rules no pack may touch — company scoping, the audit trail, the posting rules, group elimination and roster privacy | a trade opting out of the things that make the books trustworthy; it may call an invoice whatever it likes and may not decide its trail is optional | **ENFORCED** · core/tests/packs.test.js · switching OFF the audit trail |
| R01.24 | **A rule a pack never mentions is on** | a rule is looked up for a trade | the rulebook is the default and the pack is read as an exception list — silence means the rule applies | treating the pack as a permission list, which would mean every rule added after a pack was written silently applies to nobody who is using it | **ENFORCED** · core/tests/packs.test.js · a rule the pack never mentions is ON — a pack is an exception list, not a permission list |
| R01.25 | **An invalid pack is refused whole, never half-loaded** | a pack fails any check | every problem in it is reported at once and none of it is applied | partially loading a trade, which leaves a system whose vocabulary and rules disagree with each other and no way to tell which half is live | **ENFORCED** · core/tests/packs.test.js · a refused pack is refused whole — nothing is half-applied |

**Reads** ← every module · **Writes** → every module

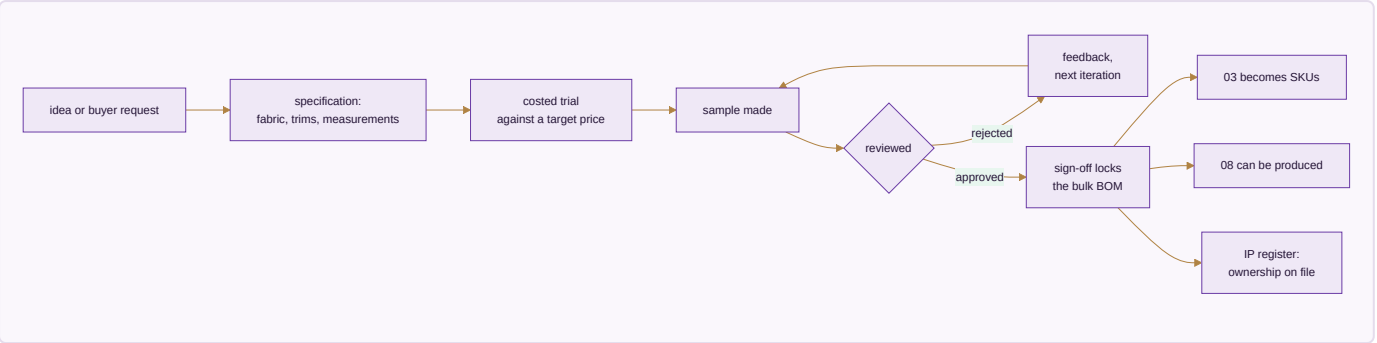
**Done when.** A karigar sees only their own earnings, an admin switches all three companies and every figure changes with the company, and every edit made during the test is already in the audit browser.

# MODULE 02 · DESIGN & SAMPLING

*A style exists on paper before it exists as stock*

**What it is.** Where a product moves from a first idea to something the business can actually make and sell — specification, sample rounds, costed trials, sign-off — before it is ever entered as a catalogue record. Building this before Inventory means the catalogue is never populated with styles that have no real specification behind them.

### How it works




### The apps (2)

| App | State | What it does |
|---|---|---|
| PLM & Development | SPEC | Specification → sample rounds → costed trials → sign-off, every version kept, so last season's costing is still there |
| Design / IP Register | SPEC | Trademark and copyright status, the date a design was first shown, and a flag when a near-identical listing appears elsewhere |

**What it owns.** `designs` (the development record) · `samples` · `sample_iterations` · `design_ip_register` · `tech_packs`

### The rules that matter

- Every version is kept, never overwritten. A design's cost history is the evidence behind this season's price.
- Sign-off is the event that locks the bulk bill of materials. Sample BOMs and bulk BOMs carry different wastage percentages and are not the same record.
- A design with no ownership record on file is a design nobody can defend — which matters most for exactly the designs Module 18 is built to promote at scale.

### The rulebook — 7 rules, 0 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R02.1 | **A style becomes a SKU only after sign-off** | someone tries to create a catalogue record for a design | the design must already have passed sample sign-off | letting a SKU exist for something with no agreed specification, which puts an unmakeable item on sale | SPECIFIED |
| R02.2 | **Every version of a specification is kept** | a sample round changes a measurement, a fabric or a trim | a new version is written and the old one stays readable | editing the specification in place — a karigar paid against last month's spec must still be able to show what it said | SPECIFIED |
| R02.3 | **A costed trial carries the date its rates came from** | a sample is costed | the rate and the date it was in force are both stored on the trial | recosting an old trial with today's rates and presenting the result as what it cost then | SPECIFIED |
| R02.4 | **A design with no ownership record is flagged, not blocked** | a design reaches sign-off with no trademark or copyright status on file | it proceeds and is listed as unprotected | silently treating it as protected, which is only discovered when a near-identical listing appears and there is nothing to act on | SPECIFIED |
| R02.5 | **The first-shown date is recorded when it happens** | a design is first shown publicly — an exhibition, a listing, a lookbook | that date is stamped and never editable afterwards | backdating it later, which is precisely the field a dispute turns on | SPECIFIED |
| R02.6 | **A rejected sample keeps its reason** | a sample round is rejected | the reason is recorded against the version | closing it with a status alone, which loses the only information that stops the same mistake in the next round | SPECIFIED |
| R02.7 | **A specification cannot be deleted while stock exists against it** | someone removes a design that has ever been made | it is archived and stays linked to every piece produced from it | orphaning finished stock from the specification it was made to | SPECIFIED |

**Reads** ← CRM · **Writes** → Inventory & Catalog · Manufacturing

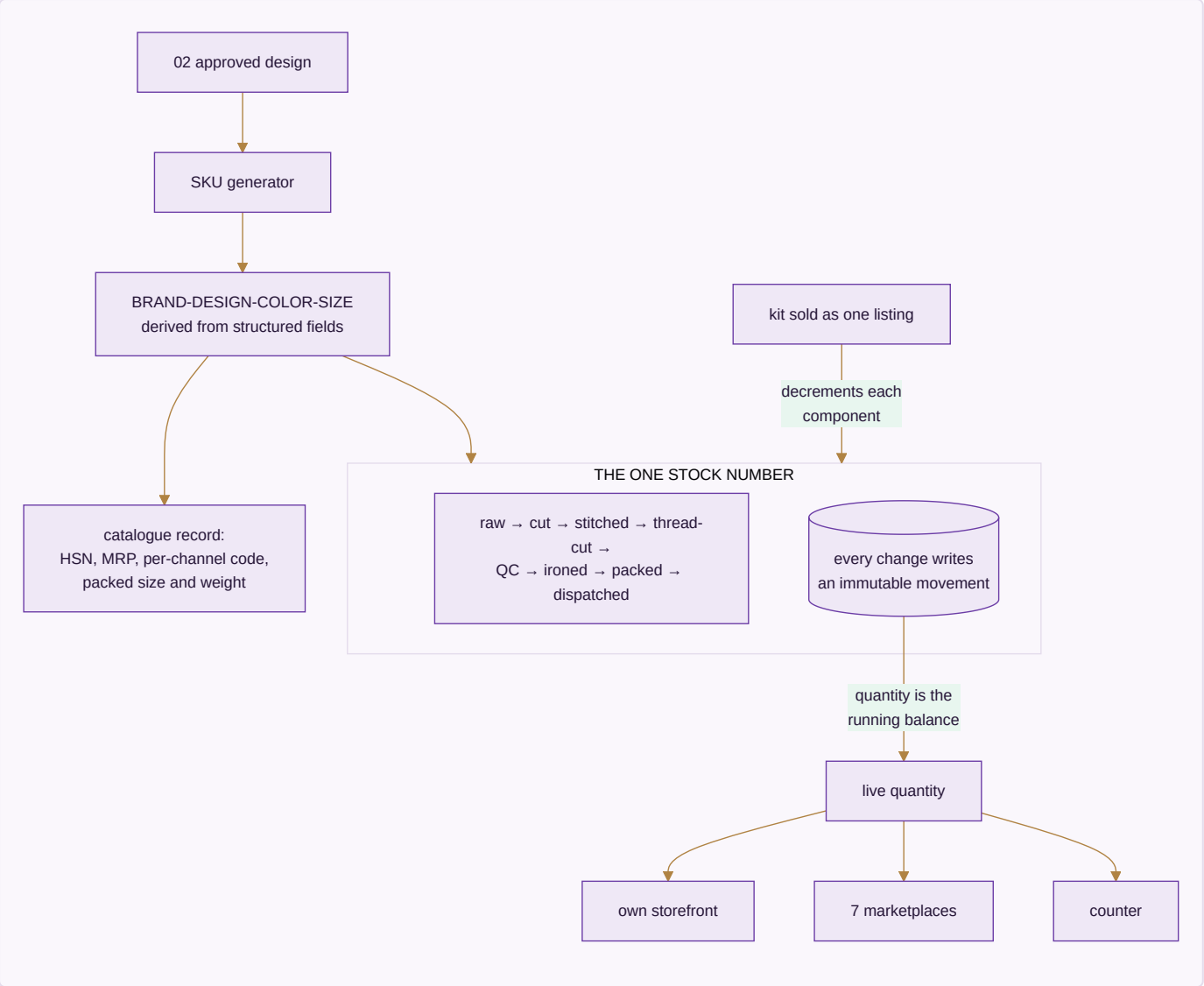
**Done when.** A design goes idea → sample → rejection → second sample → approval, and only the approved version can generate SKUs.

# MODULE 03 · INVENTORY & CATALOG

*One number everyone trusts*

**What it is.** The most important number in the system: one quantity per SKU, per location, per stage — read and written by every other module. And one product record that every channel lists from.

## How it works




## The apps (4)

| App | State | What it does |
|---|---|---|
| Stock | SPEC | Live quantity by SKU, location and stage, with reorder alerts, batches and dead-stock ageing |
| Catalog / PIM | SPEC | One product record with each channel's own code mapped to yours, and the packed size and weight that decide courier rate and settle weight disputes |
| Kit & Combo SKU | SPEC | A sellable SKU made of component SKUs; selling it decrements every component |
| Master-Data Hygiene | SPEC | Duplicate detection and merge across customers, vendors and designs |

**What it owns.** `designs` · `colors` · `sizes` · `items` · `item_aliases` · `kit_items` · `stock` · `stock_movements` · `batches` · `opening_stock` · `hsn_codes` · `gst_rates` · `locations`

### The rules that matter

- **Movements are the ledger; the quantity is its running balance.** Every transition writes an immutable row, so "how did we get to 4?" is always answerable.
- Stock is event-driven and **not per channel**. The last piece sold on one marketplace leaves every other in the same instant — not three hours later as a cancellation, because cancellations are what a seller rating is lost to.
- The SKU string is *derived* from structured fields. Search, grouping and analytics use the fields, never substring-matching on the string.
- Negative stock is a data fault, not a state the business can be in. An issue beyond what exists is refused and recounted.
- A wrong return is dead stock and is **never added back**.
- Stock valuation sets the balance-sheet figure — the two must be equal, or one of them is wrong.

### The rulebook — 14 rules, 7 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R03.1 | **Stock is one number per SKU, per location, per stage** | any module asks how much there is | it reads the one quantity, with the channel recorded on the movement rather than on the stock | keeping a separate stock figure per channel — the last piece sold on one marketplace has to vanish from the other ten at the same instant, which per-channel inventory cannot do | **ENFORCED** · core/tests/core.test.js · stock is one number per SKU, with the channel recorded on the movement |
| R03.2 | **Negative stock is a fault, not a state** | an issue would take a quantity below zero | the issue is refused | recording a negative balance and leaving someone to explain it at month-end | **ENFORCED** · core/tests/core.test.js · issuing more than exists is refused — negative stock is a fault, not a state |
| R03.3 | **Selling a kit decrements every component** | a kit or combo SKU is sold | each component SKU is decremented at order time | decrementing only the kit, which leaves the components sellable twice | **ENFORCED** · core/tests/core.test.js · selling a kit decrements every component |
| R03.4 | **A kit with no components is refused** | an item is marked a kit but lists nothing | the record is refused and named | accepting it and silently decrementing nothing on every sale | **ENFORCED** · core/tests/core.test.js · a kit that lists no components is refused, not silently sold as nothing |
| R03.5 | **Stock value ties to the item cost, always** | stock is valued | the value is computed from the quantity and the item cost | storing a valuation that can drift from the quantity it is supposed to describe | **ENFORCED** · core/tests/core.test.js · stock value ties to the item cost |
| R03.6 | **Every movement has a source, a destination, or both** | a stock movement is recorded | at least one end is named | accepting a movement from nowhere to nowhere, which is how quantity appears without a cause | **ENFORCED** · core/tests/core.test.js · a movement with neither a source nor a destination is refused |
| R03.7 | **A quantity is a whole number above zero** | a movement is written | a non-integer or non-positive quantity is refused | accepting a negative movement as a shorthand for a reversal — a reversal is its own movement with its own reason | **ENFORCED** · core/tests/core.test.js · a quantity must be a whole number above zero |
| R03.8 | **Goods in someone else's warehouse are still yours** | stock sits in a channel's own warehouse under consignment or sale-or-return | that warehouse is a location like any other and the stock is counted, valued and aged there | letting it drop off the books until it sells, which understates both stock and exposure | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R03.9 | **Fabric in metres and pieces in numbers share one item master** | an item is defined | its unit of measure is a property of the item | building a second item master for a second unit, which splits the one stock number this module exists to protect | SPECIFIED |
| R03.10 | **A listing needs the packed size and weight before it can go out** | a product is pushed to a channel | packed dimensions and weight must be present | listing without them, because that is the field every courier weight dispute is settled on | SPECIFIED |
| R03.11 | **The channel's own code for a product is mapped, not assumed** | a product exists on a marketplace | that channel's identifier is stored against ours | matching on name or on a code we invented, which mis-posts every settlement line for that product | SPECIFIED |
| R03.12 | **A duplicate master record is merged, never left as two** | the same customer, vendor or design is detected twice | they are merged and both old identifiers keep resolving | leaving two live records, which splits every total that record appears in | SPECIFIED |
| R03.13 | **A price is per channel and dated** | a channel price is set | it is stored against that channel with the date it takes effect | holding one price and reading it as if it applied everywhere and always | SPECIFIED |
| R03.14 | **Dead stock is named as dead stock** | an item has not moved for the period set for it | it appears on the dead-stock register with its age and carrying value | leaving it inside the general stock figure where it reads as healthy inventory | SPECIFIED |

**Reads** ← Design & Sampling, every module · **Writes** → every module

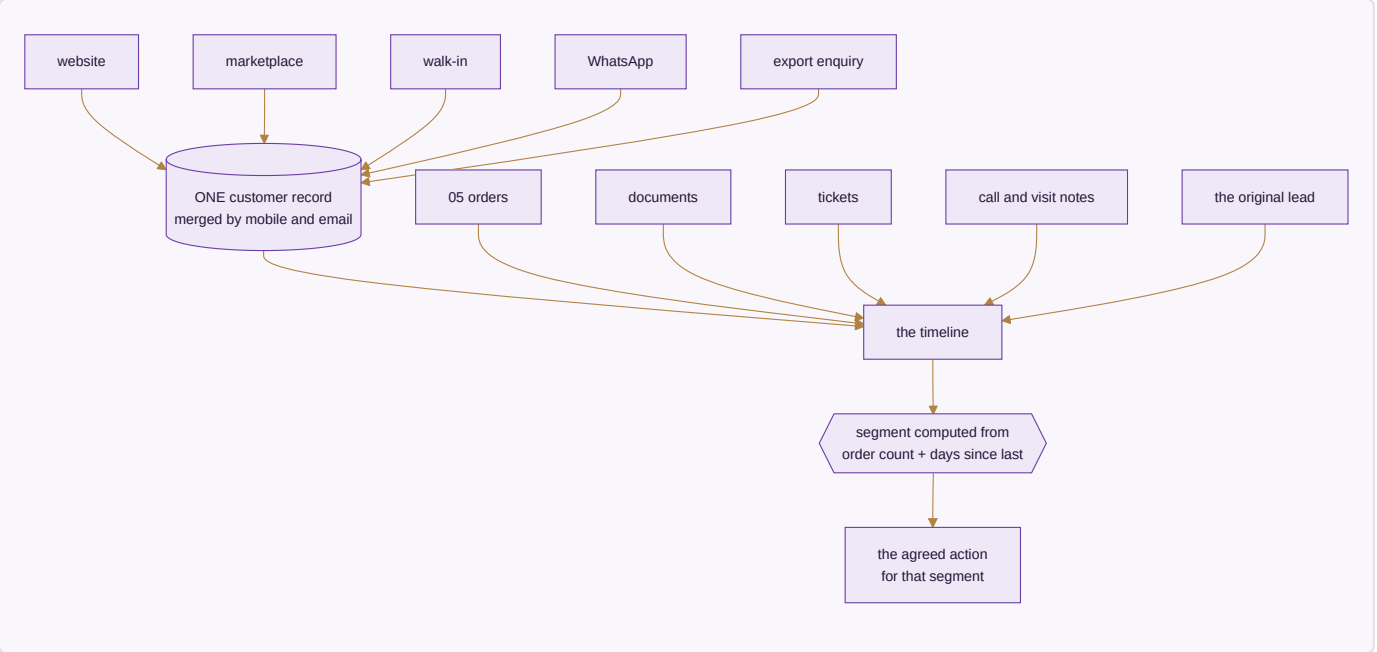
**Done when.** Stock is one number across every channel, a kit sale decrements all components, and stock valuation equals the balance sheet.

# MODULE 04 · CRM

*Know every customer completely — and answer them fast*

**What it is.** One record per customer carrying every lead, order, return, document and conversation, whichever channel it arrived on. Whoever picks up the next question can already see everything that came before it.

### How it works




### The apps (4)

| App | State | What it does |
|---|---|---|
| CRM & Customer 360 | **BUILT** | Lead to won, then the full lifetime — orders, returns, value, and what to offer next, on one screen |
| Documents & eSign | **BUILT** | Every agreement and certificate filed against the record it belongs to; send for signature and the signed copy files itself back |
| Helpdesk & Live Chat | **BUILT** | Questions become tickets tied to the order they are about, with the whole history already open |
| Forms & Feedback (NPS) | SPEC | Post-delivery feedback attached to the *design*, so a complaint-prone design surfaces as a pattern |

**What it owns.** `customers` · `customer_addresses` · `customer_interactions` · `customer_lifecycle_events` · `loyalty_ledger` · `documents` · `tickets` · `nps_responses`

### The rules that matter

- Nobody tags anybody. A customer's segment is computed from two facts only — how many orders and how long since the last — so a buyer moves segment by themselves the moment they buy or go quiet. Hand-tagged segments are always six months out of date.
- D2C, B2B, export, walk-in and WhatsApp merge to one customer by mobile and email. Marketplace customers stay separate because the channel does not share their details, but tie by pattern.
- The pipeline advances Lead → Qualified → Quoted → Negotiation and stops there; won or lost is a separate deliberate act.

- Feedback attaches to the design, not only the buyer — that is what turns a scatter of complaints into a fault Manufacturing can fix.

### The rulebook — 9 rules, 0 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R04.1 | **One customer, one record, whichever channel they arrived by** | the same person orders on a marketplace and later at the counter | both land on one record with the channel noted on each order | creating a second customer per channel, which makes lifetime value meaningless | SPECIFIED |
| R04.2 | **A document is filed against the record it belongs to** | any agreement, receipt, certificate or scan is stored | it is attached to the order, party, case or employee it concerns | filing it in a folder that has to be remembered rather than found | SPECIFIED |
| R04.3 | **A signed copy files itself back** | a document sent for signature is signed | the signed version returns to the same record automatically | leaving the signed copy in an inbox while the record still shows it as pending | SPECIFIED |
| R04.4 | **A ticket carries the order it is about** | a question arrives by chat, email or phone | it is tied to the order or account it concerns, with the history already on screen | opening a ticket with no link, which makes the first reply a request to explain again | SPECIFIED |
| R04.5 | **Feedback attaches to the item, not only the buyer** | a rating or complaint arrives after delivery | it is attached to the design or item it is actually about | holding it only against the customer, which hides a complaint-prone item as a scatter of unrelated gripes | SPECIFIED |
| R04.6 | **A customer's consent travels with their data** | a customer record is used for marketing or profiling | the consent captured at the point it was given is checked first | assuming that having the data implies permission to use it for anything | SPECIFIED |
| R04.7 | **A merged customer keeps both histories** | two customer records are merged | every order, ticket and document from both survives on the surviving record | discarding the shorter history to make the merge simple | SPECIFIED |
| R04.8 | **Credit state is read at the moment of the order** | a B2B order is placed | the customer's outstanding and limit are evaluated then | using a figure cached from the last sync, which is how a party goes past its limit between refreshes | SPECIFIED |
| R04.9 | **A closed ticket keeps what resolved it** | a ticket is closed | the resolution is recorded on it | closing with a status alone, which loses the answer the next identical question needs | SPECIFIED |

**Reads** ← every module · **Writes** → Sales · E-commerce/OMS · Marketing

**Done when.** One customer's whole cross-channel history is on one screen, and crossing an order threshold moves their segment without anyone touching it.

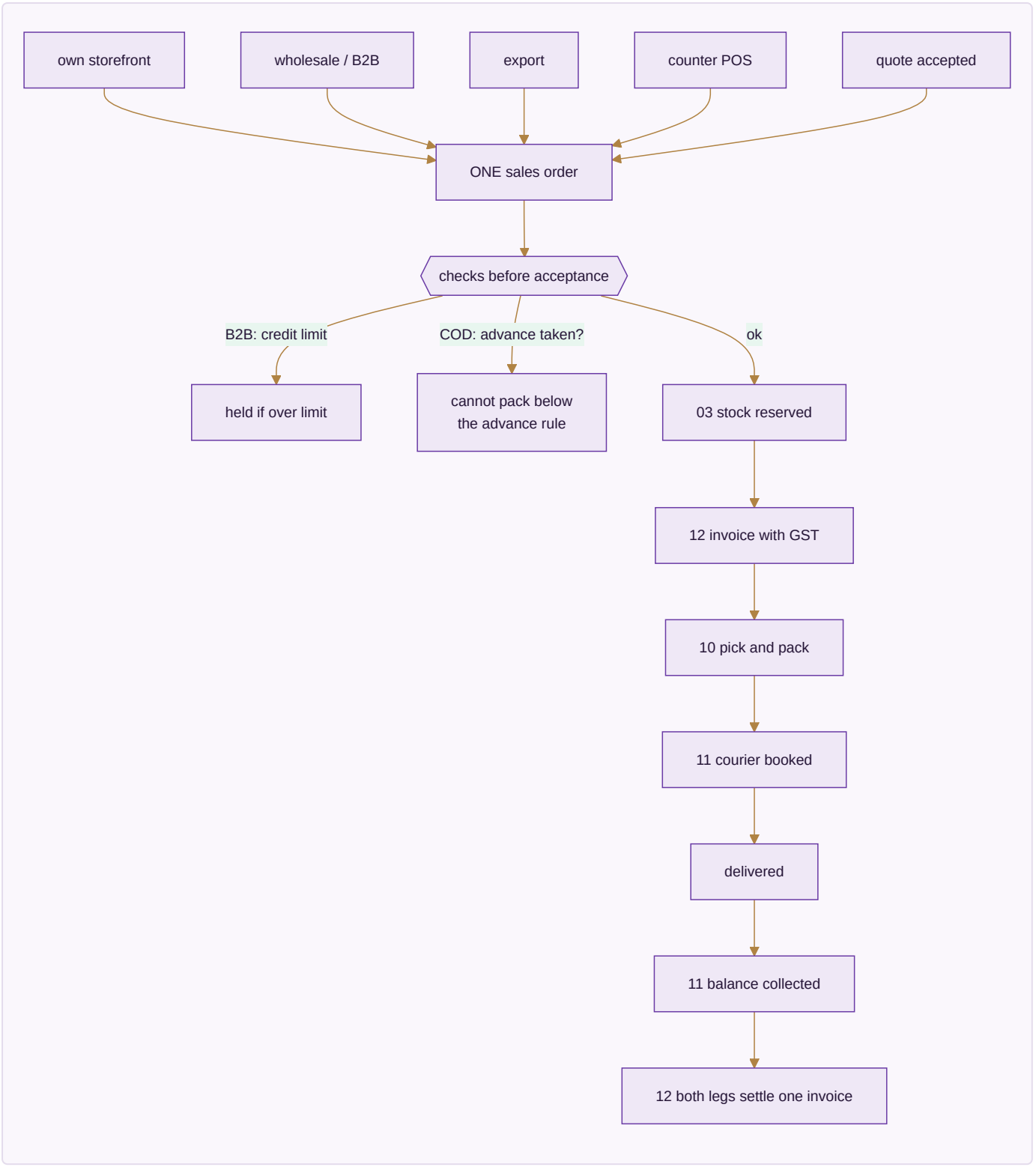
---

# MODULE 05 · SALES

*Every way you sell, one order book — to the doorstep*

**What it is.** Counter, wholesale, export and your own website all writing to the same order and drawing on the same stock number. A sale is not finished when it is billed — it is finished when it is delivered and the cash-on-delivery money is in.

### How it works


own storefront
wholesale / B2B
export
counter POS
quote accepted
ONE sales order
checks before acceptance
B2B: credit limit
COD: advance taken?
ok
held if over limit
cannot pack below the advance rule
03 stock reserved
12 invoice with GST
10 pick and pack
11 courier booked
delivered
11 balance collected
12 both legs settle one invoice

## The apps (7)

| App | State | What it does |
|---|---|---|
| D2C Sales | **BUILT** | Own-storefront orders cart to dispatch, with loyalty and partial COD |
| B2B & Credit | **BUILT** | Wholesale orders with credit limits, tier pricing and outstanding ageing |
| Export | **BUILT** | Commercial invoice, packing list, LUT bond and IGST-refund tracking |
| POS | **BUILT** | Counter billing drawing on the same stock as the website |
| Quotes & Proforma | **BUILT** | Send a quote, convert it to a confirmed order in one step |
| Couriers & AWB | SPEC | Compare couriers on the order, print the label, follow the AWB to the door |
| Subscriptions | SPEC | A schedule that raises its own invoice and follows up when a payment fails |

**What it owns.** `sales_orders` · `sales_order_items` · `invoices` · `invoice_items` · `b2b_orders` · `b2b_credit_ledger` · `export_orders` · `customization_orders` · `subscriptions`

### The rules that matter

- **A COD order cannot be packed until it carries its advance.** A refused COD parcel costs the courier fee both ways and comes back handled; a customer who has paid something almost always accepts. This one rule is the cheapest defence the business has.
- **Partial COD reconciles both legs to one invoice** — the advance taken online and the balance the courier remits.
- Credit is checked *before* acceptance, not discovered later as a bad debt. Reminders at −3 days, +1 day, and a +7-day soft block.
- The delivery date is **derived** — cut-off plus transit from the warehouse that actually holds stock — never typed. An order no warehouse can serve gets no date at all, because a promise nobody can keep is worse than no promise.
- Export lines are zero-rated under LUT; FX difference between billing rate and realisation rate posts to FX gain/loss.

### The rulebook — 18 rules, 4 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R05.1 | **Every sale carries its company and its channel** | an order is created on any channel | both are written on the order | leaving either to be inferred later from the document number or the warehouse | **ENFORCED** · core/tests/core.test.js · every one of the hundred cells posted its own figure, channel by channel |
| R05.2 | **A sale posts stock and ledger together** | a sale is confirmed | stock is deducted, the invoice is raised, and the ledger is posted in one transaction | invoicing without moving stock, or moving stock without posting | **ENFORCED** · core/tests/core.test.js · a sale moves stock and posts to the ledger, or does neither |
| R05.3 | **If the ledger refuses, the stock never moved** | the posting half of a sale fails | the stock movement is rolled back with it | leaving the goods gone and the books untouched | **ENFORCED** · core/tests/core.test.js · and if the ledger refuses, the stock never moved |
| R05.4 | **A quote becomes an order without being retyped** | a quotation is accepted | the order is created from it, carrying the same lines and prices | re-entering the lines, which is where the price on the quote and the price on the invoice start to differ | SPECIFIED |
| R05.5 | **A price below the floor needs an approval, not a note** | a line is priced under the floor set for it | the order waits in the approvals queue with the rule that stopped it named | letting it through with a comment box, which is a discount policy nobody can enforce | SPECIFIED |
| R05.6 | **An export invoice knows it is an export** | an order ships outside the country | the LUT or IGST treatment, currency and shipping terms are set on the order itself | treating it as a domestic invoice and correcting the tax afterwards | SPECIFIED |
| R05.7 | **A counter sale is the same order record** | someone buys at the counter | the same order table records it, with the counter as the channel | running the till on a separate book that has to be merged later | SPECIFIED |
| R05.8 | **A credit sale reserves the credit at the moment it is taken** | a B2B order is accepted on credit | the exposure is committed against the party immediately | counting it only when the invoice is raised, which lets several orders each fit inside the same limit | SPECIFIED |
| R05.9 | **A dispatch cannot exceed** | a shipment is prepared | quantities are checked against the order line | shipping over, which becomes an invoice | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | **what was ordered** | | | the customer never agreed to | |
| R05.10 | **A cancelled order releases what it held** | an order is cancelled | reserved stock and committed credit are both released | leaving stock reserved against a dead order, which shows the business as out of goods it actually has | SPECIFIED |
| R05.11 | **An AWB belongs to the shipment, not the courier integration** | a tracking number is recorded, typed in or fetched | it is stored on the shipment | making the number reachable only through whichever courier API produced it, which loses it the day that courier is dropped | SPECIFIED |
| R05.12 | **A subscription renewal is a new order** | a subscription renews | a fresh order is created with its own stock, invoice and posting | extending the original order, which makes the revenue of two periods indistinguishable | SPECIFIED |
| R05.13 | **A sale to a sister company is marked as one** | the counterparty is another company in the group | the counterparty company is recorded on the entry | posting it as an ordinary outside sale, which inflates the group turnover by trade it never did | **ENFORCED** · core/tests/core.test.js · an entry cannot be its own counterparty |
| R05.14 | **A quote or proforma number carries its type and financial year** | a quotation or proforma is raised | it is numbered Q-{FY}-#### or PI-{FY}-####, sequential within that company and year | sharing one sequence between quotations and proformas, which makes a proforma indistinguishable from a quote in the register | SPECIFIED |
| R05.15 | **A quote line with no description, no quantity or a negative rate is not a line** | a quotation is totalled | only lines with a description, a quantity above zero and a rate of zero or more are counted | letting a half-filled row contribute a number to the total | SPECIFIED |
| R05.16 | **An export line carries no GST** | a quotation or invoice is marked export under LUT | the GST percentage is zero and the document says why | applying the domestic rate and correcting it after the buyer queries the total | SPECIFIED |
| R05.17 | **A made-to-measure order has two money** | a customisation order is accepted | the advance and the balance are recorded as separate amounts | showing one payment at the end, which hides money already | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | **legs, and both are visible** | | with their own dates, and the balance stays owed until dispatch | taken and work already owed | |
| R05.18 | **A customisation quote keeps every round of the negotiation** | a price is revised during a bespoke enquiry | each quoted figure is kept in order with what changed | overwriting the earlier figure, which is the one the customer remembers agreeing to | SPECIFIED |

**Reads** ← Inventory & Catalog · CRM · Warehouse · Logistics · **Writes** → Inventory & Catalog · Accounting & GST · Warehouse · Logistics

**Done when.** A storefront order appears within a minute with stock reserved and an invoice raised, and a partial-COD order reconciles both legs by itself.

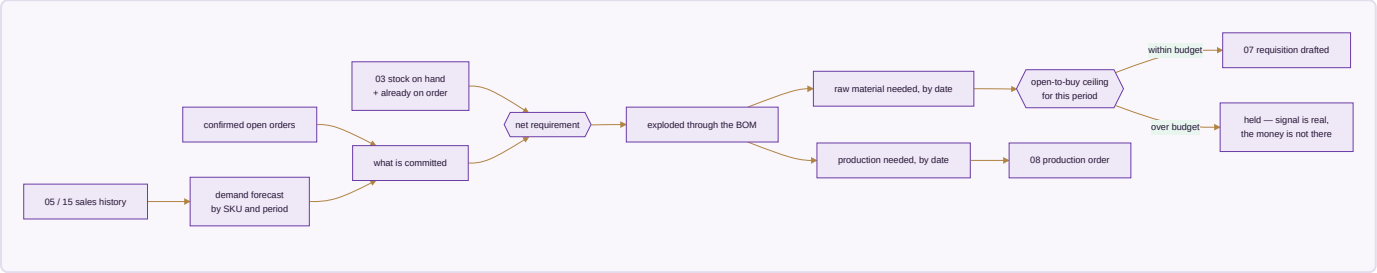
---

# MODULE 06 · PLANNING & REQUIREMENTS (MRP)

*Turn what is selling into what to buy and make*

**What it is.** The module that sits between demand and supply. Confirmed orders and sales history become a requirement before Purchase buys anything or Manufacturing starts anything — otherwise both are guessing.

### How it works




### The apps (3)

| App | State | What it does |
|---|---|---|
| Demand Forecast & Signal | SPEC | What sold, by SKU and period, turned into a short-term forecast — the number every downstream requisition is measured against |
| Requirement Explosion (MRP run) | SPEC | Demand exploded through the bill of materials into what to buy and what to make, and by when |
| Open-to-Buy / Budget Ceiling | SPEC | A spending ceiling per period that a demand signal alone cannot exceed |

**What it owns.** `demand_forecast` · `mrp_runs` · `mrp_requirements` · `open_to_buy_budgets`

### The rules that matter

- Net requirement is demand minus stock on hand minus what is already on order. Buying against gross demand is how a business ends up holding two seasons of the same fabric.
- Every requisition traces back to the run that produced it, and that run back to the demand behind it. A purchase order whose only justification is "stock looked low" is exactly what this module exists to eliminate.
- **The budget ceiling is a hard guardrail, not a warning.** A genuine demand signal can still commit more money than the business decided to risk this season, and without a ceiling it will.

### The rulebook — 8 rules, 0 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R06.1 | **A forecast is labelled a forecast wherever it appears** | a projected figure is shown beside actuals | it is visually and structurally distinct | letting a forecast total sit in the same column as a real one, which is how a plan becomes a reported result | SPECIFIED |
| R06.2 | **A requirement run reads live stock, not a snapshot** | the MRP run explodes requirements | it reads the current quantity at the moment it runs | planning against a nightly copy, which orders material the business already has | SPECIFIED |
| R06.3 | **A requirement names what caused it** | the run produces a shortfall | the order, forecast or reorder level that generated it is recorded on the line | producing a bare quantity nobody can trace back to a demand | SPECIFIED |
| R06.4 | **Stock already on order counts against the shortfall** | the run computes what to buy | open purchase orders are netted off first | ignoring them and ordering the same material twice | SPECIFIED |
| R06.5 | **A budget ceiling refuses, it does not warn** | a proposed purchase would exceed the open-to-buy ceiling | it is held for approval with the ceiling named | raising it with a warning, which makes the ceiling advisory and therefore not a ceiling | SPECIFIED |
| R06.6 | **A lead time is per vendor and per item** | a run works out when to order | it uses the lead time recorded for that vendor and that item | applying one global lead time, which under-orders the slow lines and over-orders the fast ones | SPECIFIED |
| R06.7 | **A run is kept, not overwritten** | the MRP run executes again | the previous run stays readable with its inputs | replacing it, which makes it impossible to see why last week’s decision was taken | SPECIFIED |
| R06.8 | **A seasonal signal cannot silently become a permanent one** | a festival or season inflates demand | the period it applies to is stored with the signal | folding a spike into the baseline, which keeps ordering for a festival all year | SPECIFIED |

**Reads** ← Sales · E-commerce/OMS · Inventory & Catalog · **Writes** → Purchase · Manufacturing

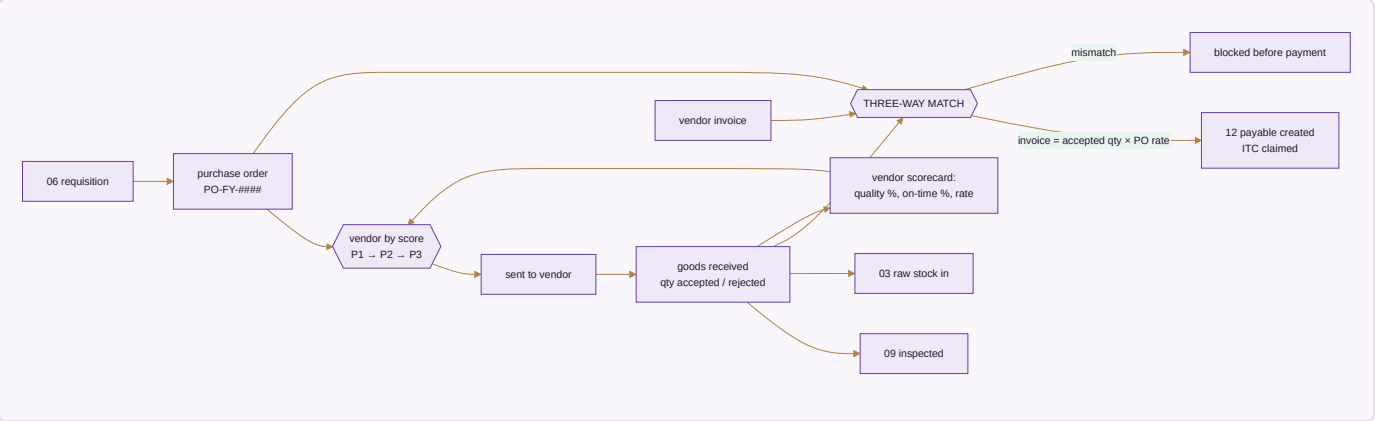
**Done when.** An MRP run turns a month of real demand into requisitions and production orders that each trace back to the orders that caused them, and the ceiling blocks the run that exceeds it.

# MODULE 07 · PURCHASE

*Nothing over-billed gets paid*

**What it is.** The buy side end to end — mills, dyers, job workers, packing suppliers — and the control that stops you paying for goods you rejected.

### How it works




### The apps (3)

| App | State | What it does |
|---|---|---|
| Procurement | **BUILT** | Requisition → PO → GRN, with a strict three-way match before any bill is paid |
| Vendor Management | **BUILT** | Vendor 360, payables, ageing, a real risk score, and sourcing that follows performance |
| Insurance Register | SPEC | Cover over stock in transit, stock in the warehouse and product liability, matched against what is actually moving |

**What it owns.** `purchase_requisitions` · `purchase_orders` · `purchase_order_items` · `grn` · `grn_items` · `vendor_invoices` · `three_way_match` · `vendors` · `vendor_materials` · `third_party_services` · `insurance_policies`

### The rules that matter

- **The three-way match is the whole point.** You ordered 100 metres, 100 arrived, quality accepted 96 — the bill clears for 96 and not a metre more. The system flags GRN ≠ PO, invoice ≠ accepted × rate, and any invoice arriving before its GRN.
- The vendor scorecard is computed per transaction from real outcomes — accept rate, on-time delivery, rate against market — and it drives the priority order for the next requisition. Sourcing follows performance rather than habit.
- Real stock value moves through purchase and the warehouse continuously; cover that nobody tracks is cover nobody can claim on.

### The rulebook — 12 rules, 0 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R07.1 | **Nothing is paid without a three-way match** | a vendor invoice is approved | the purchase order, the goods received note and the invoice must agree | paying on the invoice alone, which pays for goods that never arrived | SPECIFIED |
| R07.2 | **A short or damaged receipt is recorded as received short** | the GRN quantity is below the PO quantity | the difference is recorded with its reason and the payable follows the received quantity | receiving the full quantity to make the match pass | SPECIFIED |
| R07.3 | **Input tax credit is claimed against a real document** | ITC is taken on a purchase | the vendor invoice and its tax detail are on file | claiming credit from a payment record alone, which is the claim that fails reconciliation | SPECIFIED |
| R07.4 | **Landed cost reaches the item, not just the P&L** | freight, duty or insurance is attached to a purchase | it is apportioned into the cost of the items received | expensing it separately, which understates the cost of every piece made from that material | SPECIFIED |
| R07.5 | **A vendor price is dated** | a rate is agreed with a supplier | it is stored with the date it takes effect | overwriting the old rate, which makes last month's purchase look mispriced | SPECIFIED |
| R07.6 | **A purchase order over its approval level waits** | a PO exceeds the value a role may approve | it goes to the approvals queue naming the rule and the level | splitting it into smaller orders to fit under the limit — the split is detected and the parts are assessed together | SPECIFIED |
| R07.7 | **A vendor with no active record cannot be paid** | a payment is raised | the vendor must exist, be active, and have its bank detail verified | paying to detail typed onto the payment itself, which is the single most common route for payment fraud | SPECIFIED |
| R07.8 | **A change to vendor bank detail is treated as high risk** | a vendor's bank account is changed | the change is approved by a second person and the old detail is kept | accepting a change from an email instruction alone | SPECIFIED |
| R07.9 | **A job-work despatch stays on the books** | material is sent to a contractor | it moves to a job-work location and remains this company's stock | writing it out on despatch, which loses material the business still owns | SPECIFIED |
| R07.10 | **An insurance policy is linked to what it covers** | a policy is recorded | the stock, premises or shipment it covers is named on it | holding policies as documents with no link, which is discovered only at the moment of a claim | SPECIFIED |
| R07.11 | **The three-way match is arithmetic, not a judgement** | a vendor invoice is checked | the payable equals the received quantity × the purchase-order rate, and the purchase order, the goods receipt and the invoice must | passing an invoice whose value exceeds received quantity × agreed rate, and never letting an override | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | | | all agree on quantity and value | happen without recording who made it and why | |
| R07.12 | **A material is sourced down a ranked list, not from whoever answers** | a material has to be bought | the vendors ranked for that material are approached in their priority order | defaulting to the last vendor used, which is how a price rise becomes permanent without anyone deciding | SPECIFIED |

**Reads** ← Inventory & Catalog · Planning/MRP · Manufacturing · **Writes** → Inventory & Catalog · Accounting & GST · Quality & Compliance

**Done when.** A vendor invoice for more than was accepted is blocked before payment, and the rejection reason is on the record.

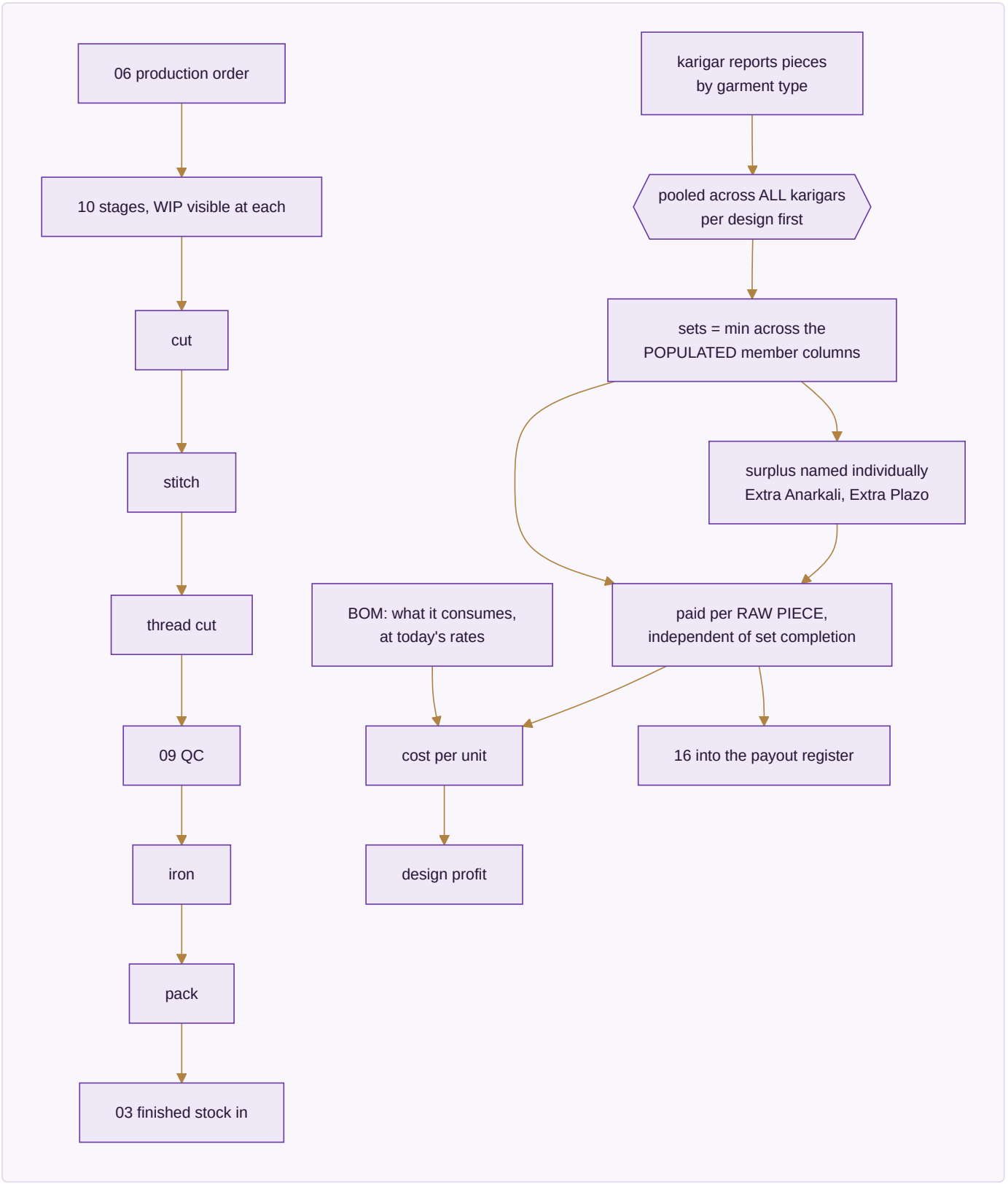
---

# MODULE 08 · MANUFACTURING

*Know what a unit really costs to make*

**What it is.** From material in the door to the finished unit — every operation, every worker's earning, and what each design actually cost before it was priced. Design sign-off happens upstream in Module 02; inspection and the compliance record live downstream in Module 09. This module is purely the making.

**How it works**


06 production order
10 stages, WIP visible at each
cut
stitch
thread cut
09 QC
iron
pack
03 finished stock in
karigar reports pieces by garment type
pooled across ALL karigars per design first
sets = min across the POPULATED member columns
surplus named individually Extra Anarkali, Extra Plazo
BOM: what it consumes, at today's rates
paid per RAW PIECE, independent of set completion
cost per unit
16 into the payout register
design profit

## The apps (4)

| App | State | What it does |
|---|---|---|
| Production Orders | SPEC | Your own stages from first operation to finished goods, with work-in-progress visible at each |
| Piece-rate & Contractors | SPEC | Output-based pay: pooled set completion, per-unit rates, rework and advances resolved into one payout |
| BOM & Consumption | SPEC | What each design consumes — fabric, zari, lining, trims, packing — costed at today's rates |
| Maintenance | SPEC | Machines and tools: what is due for service, what it cost, what stopped while it was down |

**What it owns.** `production_orders` · `production_stages` · `bom` · `bom_items` · `karigar_assignments` · `karigar_reports` · `performance_flags`

### The karigar costing rules — the heart of this module

These are settled and already proven in a working engine. They are unusual enough to be worth stating exactly, because getting any one of them wrong changes what a person gets paid.

1. **23 garment columns collapse to 13 set types.** A set is a named combination, not an arbitrary grouping.
2. **Pool across all karigars per design first, then apply the rule.** Pooling after applying the rule produces a different, wrong answer.
3. **Sets = the minimum across the *populated* member columns of that set type.** A column nobody worked on is not a zero that drags the set count to nothing — it is simply not part of the calculation.
4. **Extras are named individually** — Extra Anarkali, Extra Plazo — never a generic leftover bucket, and there is **no combined "Set + Extra" total column**, because that column would double-count.
5. **Cost is per raw piece, independent of set completion.** A surplus piece that completes no set is still work somebody did, and is still paid.
6. **A missing rate posts ₹0 and raises a flag — never a guessed rate.** A guessed rate is a silent error in someone's wages.
7. Alteration earning is alteration hours × ₹100; an alteration caused by the worker's own mistake is ₹0.
8. A performance flag fires when the same person, design and task takes more than 1.2× the previous hours, and asks why by WhatsApp.

### The acceptance gate — figures that must reproduce to the rupee

| Source | Designs | Karigar units | Sets | Pieces | Total | Flagged |
|---|---|---|---|---|---|---|
| The owner's hand-made report, Apr 2025 – Jun 2027 | 143 | 29 | **25,307** | **59,110** | **₹26,90,062** | 5 designs with no rate |
| The engine, run today on the workbooks as they now stand | 128 | 20 | **16,662** | **36,229** | **₹17,45,911** | 0 designs with no rate |

**Why the two rows differ, stated rather than reconciled away.** The FY2026-27 workbook has since been restructured into one payment sheet per karigar team, each sheet named for its team, and it no longer carries a

design grid at all, so that year's rows cannot be read from it. The five previously unrated designs have since been given rates, which is why nothing is flagged now.

The verification does not weaken the gate to make it pass. It places **every** design in the reference report into exactly one bucket with a named cause — matched exactly, changed at source, rate added since, incomplete-set rule, or present only in the FY2026-27 grid — prints the buckets, and fails on any design whose difference has no explanation. There are currently none. A mismatch with no cause is a bug, not a rounding difference; an input the engine cannot read is a stated limitation, not a passing test.

**What closing this needs.** A reader for the per-team sheet layout, so the FY2026-27 year is costed from the file the business actually keeps today rather than from one it no longer maintains.

### The rulebook — 20 rules, 9 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R08.1 | **Sets are pooled across every karigar before the minimum is taken** | completed sets are counted for a design | every karigar's pieces for that design are pooled first, and the set count is the minimum across the populated member columns of the pool | counting sets per karigar row and adding them up, which loses every set completed by two people between them | **ENFORCED** · brand/suite/studio/verify_studio.js · pooling happens before the minimum, not per karigar row |
| R08.2 | **A surplus piece is paid for, and is not a set** | a karigar makes more of one garment than the set needs | the extra is named individually and paid at its own piece rate | adding it to the set count, and never leaving it unpaid because it did not complete a set — the person made it either way | **ENFORCED** · brand/suite/studio/verify_studio.js · a surplus piece is named, is still paid for, and is never added to the sets |
| R08.3 | **A design counts on the garments it actually has** | a design is made of fewer garment types than the usual set | it is counted on the members it does have | returning zero because an optional member is absent, which silently unpays a whole design | **ENFORCED** · brand/suite/studio/verify_studio.js · an Anarkali-only design counts on what it has, not zero |
| R08.4 | **A missing rate posts zero and is flagged, never guessed** | a design has no entry in the stitching rate master | it costs zero and the design is named in the summary | inferring a rate from a similar design — a guessed rate is a wrong payment to a real person | **ENFORCED** · brand/suite/studio/verify_studio.js · a missing rate posts zero and is flagged, never guessed |
| R08.5 | **A two-row heading is read as two rows** | the production grid uses a merged heading over garment columns | both header rows are read so repeated garment names stay distinct columns | reading only the first row, which collapses three Dupatta columns into one and undercounts the work | **ENFORCED** · brand/suite/studio/verify_studio.js · the two-row heading is read, so three Dupatta columns stay three garments |
| R08.6 | **A karigar written as a pair stays one unit** | two names share one row as a working pair | they are treated as a single paying unit | splitting them into two karigars, which halves each person's recorded output and breaks the payout | **ENFORCED** · brand/suite/studio/verify_studio.js · a karigar written as a pair stays one unit |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R08.7 | **Several years of grids pool into one set of figures** | more than one production workbook is supplied | their grids pool into a single costing | reporting each file separately, which double-counts nothing but hides the sets completed across a year boundary | **ENFORCED** · brand/suite/studio/verify_studio.js · several years of grids pool into one set of figures |
| R08.8 | **Cost per piece is independent of set completion** | the cost of a design is worked out | each raw piece is costed at its own rate | costing by completed sets, which values an unfinished set at nothing while the labour has already been spent | **ENFORCED** · brand/suite/studio/verify_studio.js · the grand total is the sum of the designs, and of the karigars |
| R08.9 | **A production report moves stock and pay together** | a karigar production report is accepted | finished stock comes in, the payout is raised in HR, wages post to the ledger, and the design cost updates — in one transaction | taking the stock in and settling the pay in a separate pass, which is how the two disagree | SPECIFIED |
| R08.10 | **Material issued to production leaves raw stock at the moment it is issued** | a production order consumes material | raw stock is reduced and work in progress increases | consuming at completion, which shows material as available while it is already cut | SPECIFIED |
| R08.11 | **A bill of materials is versioned with the design** | a production order is created | it captures the BOM version in force at that moment | reading the current BOM when costing an old order, which recosts history | SPECIFIED |
| R08.12 | **Wastage is recorded, not absorbed** | consumption exceeds the BOM | the excess is recorded as wastage against the order with its reason | quietly increasing the BOM to match what was used, which destroys the only signal that something is going wrong | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R08.13 | **A stage cannot be skipped without being recorded as skipped** | work moves past a defined stage without that stage being marked | the skip is recorded on the order | letting the stage silently complete, which makes every stage-time figure fiction | SPECIFIED |
| R08.14 | **An advance to a karigar is a balance, not a deduction from nowhere** | an advance is paid | it is held against that karigar and recovered from later payouts, with the running balance visible | deducting an amount at payout time that cannot be traced to a specific advance | SPECIFIED |
| R08.15 | **A rework carries the cost of the rework** | a piece is returned to a stage to be redone | the additional labour is costed to the design that caused it | costing it as new production, which makes a failing design look as profitable as a good one | SPECIFIED |
| R08.16 | **Material consumed is the average per piece times the pieces made** | consumption is costed against a production run | consumption equals the average consumption per piece × pieces produced, and the difference against the bill of materials is recorded as wastage | back-fitting the average to whatever was issued, which makes wastage mathematically impossible to see | SPECIFIED |
| R08.17 | **A set type comes from the rate master, and an inferred one says so** | a design is classified into a set type | the rate master's Set column decides it; when the design is absent, the type is inferred from which garment columns actually carry pieces and the design is flagged as inferred | presenting an inferred classification as though it came from the master | **ENFORCED** · brand/suite/studio/verify_studio.js · the two-row heading is read, so three Dupatta columns stay three garments |
| R08.18 | **An alteration caused by the karigar's own** | a piece is reworked because of an error by the | the alteration hours are recorded and paid at zero | paying for the rework at the standard alteration rate, and never | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | **mistake is unpaid** | person who made it | | leaving the hours unrecorded — the time still happened and the design still bore the cost | |
| R08.19 | **Alteration time is paid at the alteration rate, not the piece rate** | admin-assigned alteration hours are settled | they are paid at the hourly alteration rate in force and added to that karigar's payout | folding alteration hours into the piece count, which corrupts both the production figure and the earnings figure at once | SPECIFIED |
| R08.20 | **A contract worker paid by the hour has no attendance row** | a contract role is settled | payment is hours worked × the agreed hourly rate, recorded against the person without an attendance record | forcing a contract worker through the salaried attendance model, which produces a monthly figure nobody agreed to | SPECIFIED |

**Reads** ← Purchase · Planning/MRP · Design & Sampling · **Writes** → Inventory & Catalog · HR & Payroll · Accounting & GST · Quality & Compliance

**Done when.** Three production orders run to completion — self-made, full job work, partial job work — and the acceptance-gate totals reproduce exactly.

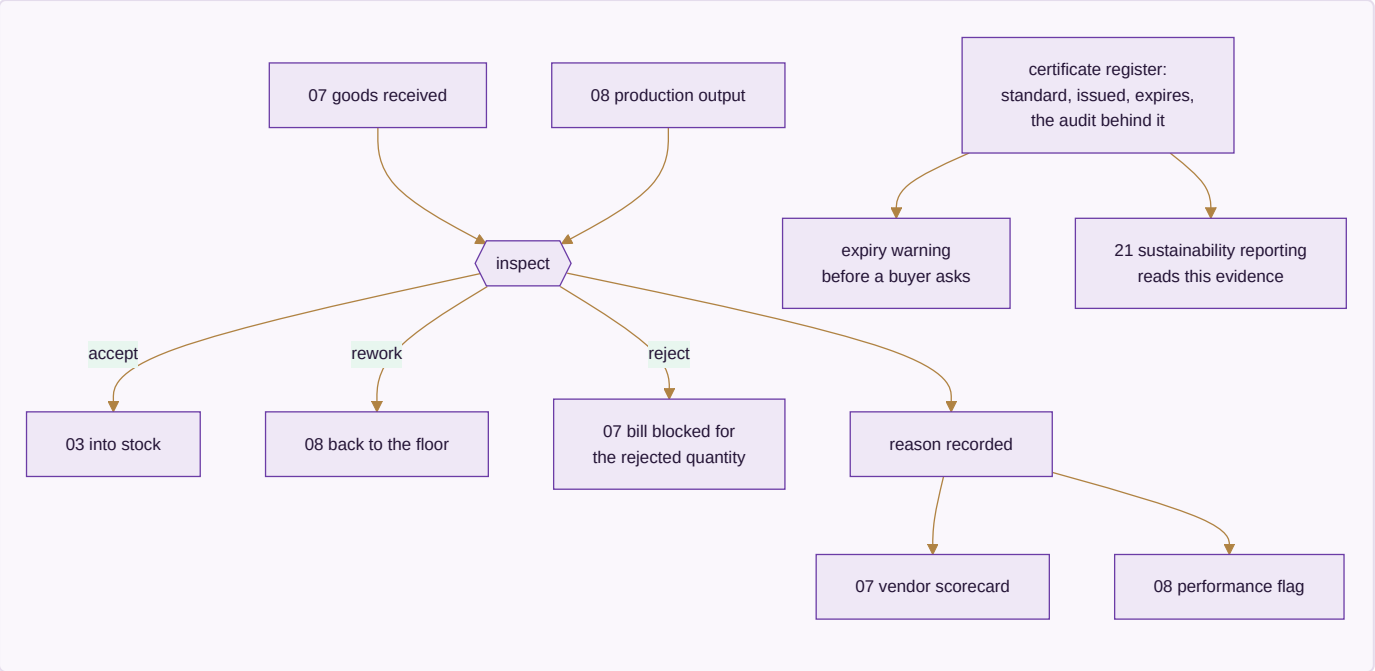
---

# MODULE 09 · QUALITY & COMPLIANCE

*Certify what was received and what was made*

**What it is.** Inspection used to be one step buried inside Manufacturing. It stands on its own here because rejecting goods at receipt and rejecting goods on the floor are the same discipline — and because the certificates that prove a standard is followed belong beside the inspections that back them up, not scattered across email.

**How it works**




## The apps (2)

| App | State | What it does |
|---|---|---|
| Quality Control | SPEC | Accept, reject or rework — on goods received and goods made — with reasons that feed the vendor scorecard and the performance flags alike |
| Certificate & Compliance Register | SPEC | Every standard held, with issue date, expiry and the audit that backs it |

**What it owns.** `qc_records` · `qc_defect_categories` · `compliance_certificates` · `compliance_audits`

## The rules that matter

- A rejection is never just a number — it carries a reason, and that reason is what makes the vendor scorecard and the performance flag meaningful rather than decorative.
- The quantity accepted at QC, not the quantity delivered, is what the three-way match pays against. This is the join between this module and the money.
- A certificate about to lapse is visible *before* a buyer asks for it and finds it expired.
- What this register holds is precisely what a sustainability report is later built from — which is why the reporting app in Module 21 reads it rather than collecting evidence separately.

## The rulebook — 7 rules, 0 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R09.1 | **A failed check blocks the next stage** | an inspection fails | the batch cannot progress until it is passed, reworked or written off | letting it move with the failure noted, which sends a known defect to a customer | SPECIFIED |
| R09.2 | **A check names the person who did it** | any inspection is recorded | the inspector, the time and the sample size are stored | accepting an anonymous pass, which cannot be investigated when the complaints arrive | SPECIFIED |
| R09.3 | **An expiring certificate warns before it expires** | a certificate approaches its expiry | it is raised while there is still time to renew | discovering the lapse at the moment a buyer asks for it | SPECIFIED |
| R09.4 | **A rejected batch cannot be sold as first quality** | a batch is rejected | it is marked and can only be sold through a channel that accepts seconds | letting it re-enter the ordinary sellable pool | SPECIFIED |
| R09.5 | **A defect is attached to the design and the stage** | a defect is recorded | both the design and the stage that produced it are named | recording it against the batch alone, which loses the pattern that would have prevented the next one | SPECIFIED |
| R09.6 | **A compliance document is evidence, not a checkbox** | a compliance requirement is marked met | the document proving it is attached | accepting a tick with nothing behind it, which is what fails an audit | SPECIFIED |
| R09.7 | **A sustainability figure comes from the same evidence** | an ESG figure is reported | it is computed from the certificate and audit records already on file | assembling it separately once a year from numbers nobody can trace | SPECIFIED |

**Reads** ← Purchase · Manufacturing · **Writes** → Purchase · Manufacturing · Inventory & Catalog

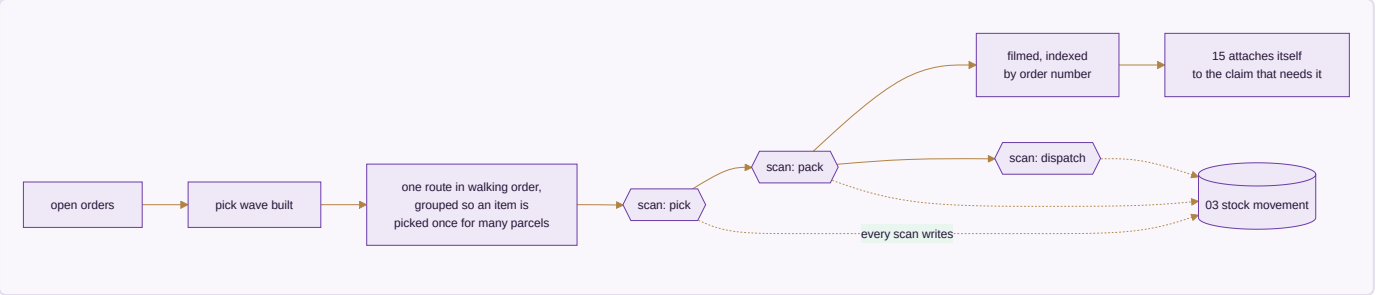
**Done when.** A rejected receipt blocks payment for exactly the rejected quantity, and an expiring certificate raises a warning before its expiry date.

# MODULE 10 · WAREHOUSE

*Pick right the first time — and prove what you sent*

**What it is.** Bin-level instructions and barcode scanning so the right item leaves the building and stock stays honest — and a recording of each parcel as it is packed, so an argument about what was in it is settled by footage instead of memory.

## How it works




## The apps (3)

| App | State | What it does |
|---|---|---|
| Picking & Bins | SPEC | Pick lists that name the bin, in walking order, so nobody crosses the floor twice |
| Barcode Operations | SPEC | Scan to pick, pack, dispatch and count from a phone — the same scan whatever channel the order came from |
| Packing Video | SPEC | Every parcel filmed as it is packed and indexed by order number |

**What it owns.** `bins` · `pick_lists` · `pick_list_lines` · `barcode_scans` · `packing_videos`

## The rules that matter

- A picker is never sent to an empty rack — the wave is built from the stock record, not from hope.
- **Every scan writes a stock movement.** That is what keeps the running balance real rather than a number someone updates at the end of the day.
- The day is grouped by product, so one design is picked once for eleven parcels instead of eleven times.
- The footage attaches to the claim by order number automatically. A wrong-item claim answered with the clip is a claim won on evidence rather than argument.

## The rulebook — 8 rules, 0 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R10.1 | **A pick is confirmed against the bin it came from** | an item is picked | the bin is recorded on the movement | decrementing a warehouse total with no bin, which makes the next cycle count unexplainable | SPECIFIED |
| R10.2 | **A short pick stops the pack, it does not silently reduce the order** | the picker cannot find the full quantity | the shortage is raised against the order and the pack waits | packing what was found and invoicing for it as though that was the order | SPECIFIED |
| R10.3 | **A scan is the same event as a keyed entry** | a code is captured by scanner, phone camera or typing | the same movement is written | having a scanning path that writes different records from the manual path | SPECIFIED |
| R10.4 | **A cycle count adjustment names a reason** | a count differs from the system | the adjustment records the reason and the person | writing the system down to the counted figure with no explanation, which hides theft and damage equally well | SPECIFIED |
| R10.5 | **The packing video is linked to the shipment** | a parcel is recorded on video at packing | the recording is attached to that shipment | keeping the footage in a folder by date, which makes it unusable in the dispute it exists for | SPECIFIED |
| R10.6 | **A bin holds a location, not a guess** | stock is put away | the destination bin is captured at put-away | assigning a default bin so the step can be skipped | SPECIFIED |
| R10.7 | **A dispatch cut-off is per channel** | a channel has a handover deadline | the queue is ordered and warned against that channel's own cut-off | applying one cut-off to all of them, which misses the earliest and idles for the latest | SPECIFIED |
| R10.8 | **A returned parcel is inspected before it is anything else** | a return arrives at the warehouse | it is booked into a return-inspection location first | restocking on arrival, which puts an unchecked item back on sale | SPECIFIED |

**Reads** ← Sales · E-commerce/OMS · Inventory & Catalog · **Writes** → Inventory & Catalog · Sales · E-commerce/OMS

**Done when.** A parcel is picked from the right bin, filmed, and its clip is already attached when the claim arrives.

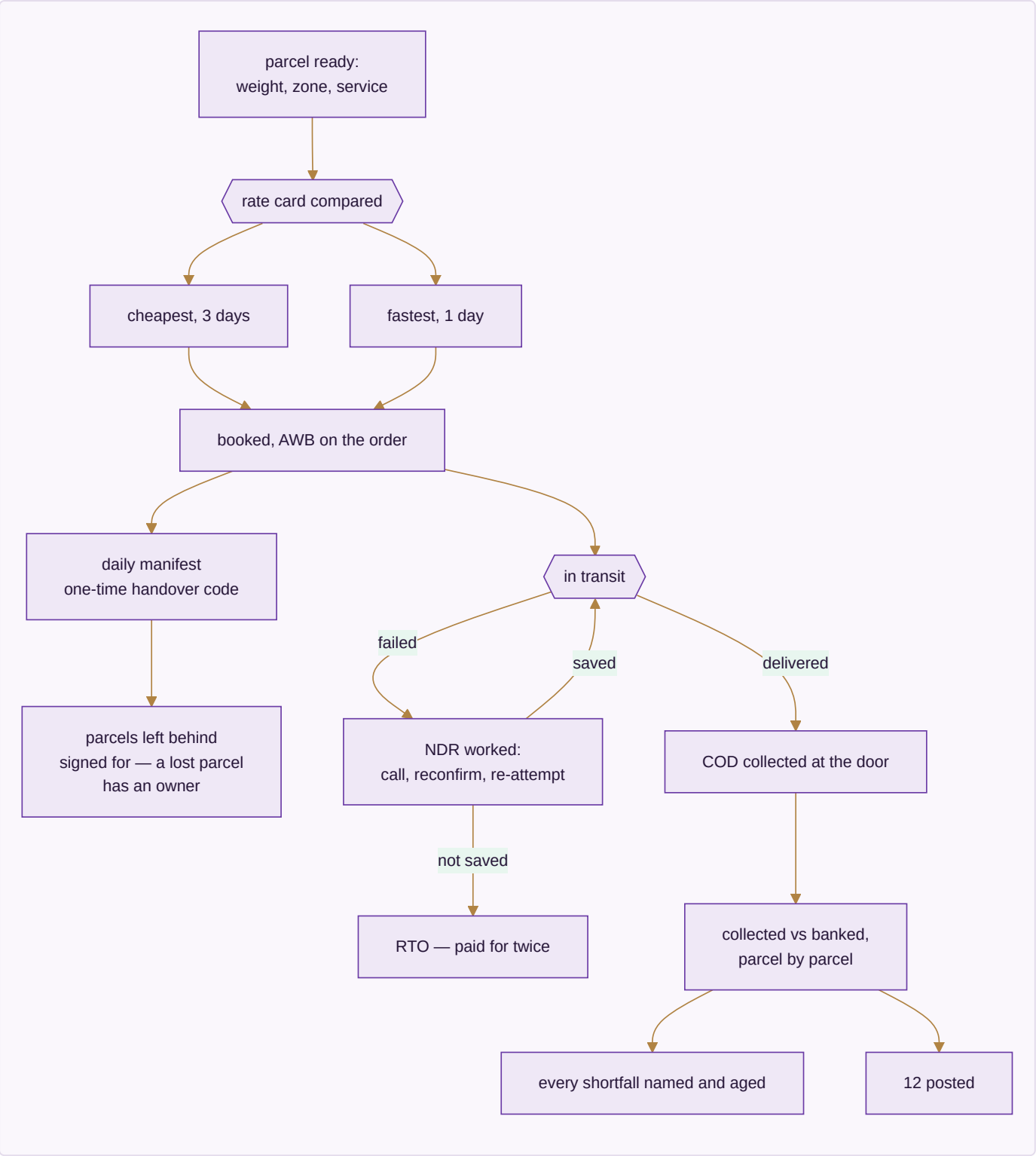
---

# MODULE 11 · LOGISTICS

*The courier network — rates, failed deliveries and the COD money*

**What it is.** Booking one parcel happens on the order, in Sales. This module is the network behind it: what each courier charges before you pick one, what happens to a delivery that fails, and whether the cash collected at the door actually reached your bank.

## How it works




## The apps (5)

| App | State | What it does |
|---|---|---|
| Rates & Zones | SPEC | Every courier's rate card by zone, weight slab and service — cheapest and fastest both known before booking |
| NDR & RTO Rescue | SPEC | A failed delivery worked while it can still be saved, before it becomes a return you pay for twice |
| COD Remittance | SPEC | What was collected at the door against what reached the bank, parcel by parcel |
| Handover & Manifest | SPEC | What went out against what the courier actually took, with a one-time code and a signed record of what was left |
| Fleet | SPEC | Own vehicles, if any — trips, cost, service due. Optional; most run couriers only |

**What it owns.** `courier_rates` · `shipments` · `ndr_cases` · `cod_remittance` · `manifests` · `vehicles`

### The rules that matter

- Both the cheapest and the fastest option are known before booking. Choosing without seeing both is choosing blind.
- **An NDR is worked while it can still be saved.** An RTO costs freight in both directions and returns handled goods — the rescue attempt is almost always cheaper than the return.
- COD is reconciled parcel by parcel, and every shortfall is named and aged. A lump-sum remittance that roughly matches is not reconciliation.
- The manifest carries a signed record of what was *not* taken, so a parcel lost between the packing table and the van has an owner.

### The rulebook — 11 rules, 0 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R11.1 | **The courier rate is checked against the packed weight** | a courier bills for a shipment | the billed weight is compared with the packed weight recorded at packing | accepting the courier's weight without comparison, which is the most consistently overcharged line in the business | SPECIFIED |
| R11.2 | **A weight dispute is raised with the evidence attached** | billed and packed weight differ beyond tolerance | a dispute is raised carrying the packing record | absorbing the difference because each one is small | SPECIFIED |
| R11.3 | **An undelivered parcel is chased before it becomes a return** | a delivery attempt fails | the NDR is actioned within the window the courier allows | letting it lapse into a return, which costs the freight twice and the sale once | SPECIFIED |
| R11.4 | **COD collected is a receivable until it is remitted** | a COD parcel is delivered | the amount is a receivable from the courier | treating delivery as payment, which reports cash the business does not have | SPECIFIED |
| R11.5 | **A remittance is matched parcel by parcel** | a courier remits COD | each parcel in the remittance is matched individually | accepting the total, which is how short remittances go unnoticed for months | SPECIFIED |
| R11.6 | **A manifest is a record, not a printout** | parcels are handed over | the handover is recorded against each shipment with the time and the person | keeping only a signed sheet, which cannot be queried when a parcel is disputed | SPECIFIED |
| R11.7 | **An RTO parcel is stock again only after inspection** | a return to origin is received | it goes through inspection before it can be sold | restocking it automatically on scan | SPECIFIED |
| R11.8 | **Freight cost reaches the order it belongs to** | a shipment is costed | the freight is attributed to the order | holding freight only as a monthly expense, which makes per-order and per-channel profit fiction | SPECIFIED |
| R11.9 | **A courier can be changed without losing history** | a courier is switched off | every past shipment, AWB and dispute stays readable | making history depend on an integration that is still connected | SPECIFIED |
| R11.10 | **A zone and rate card are dated** | courier rates change | the new card is stored with its effective date | overwriting the card, which makes every past shipment look mischarged | SPECIFIED |
| R11.11 | **A partial-COD order has two collections and both are tracked** | an order is placed with an advance online and the balance on delivery | the advance is a receipt now and the balance is a receivable from the courier until it is remitted | treating the advance as the whole payment, which makes every such order look settled while most of the money is still outstanding | SPECIFIED |

**Reads** ← Sales · E-commerce/OMS · Warehouse · **Writes** → Accounting & GST · Sales · E-commerce/OMS

**Done when.** COD collected reconciles to COD banked parcel by parcel, and an NDR is worked before it turns into an RTO.

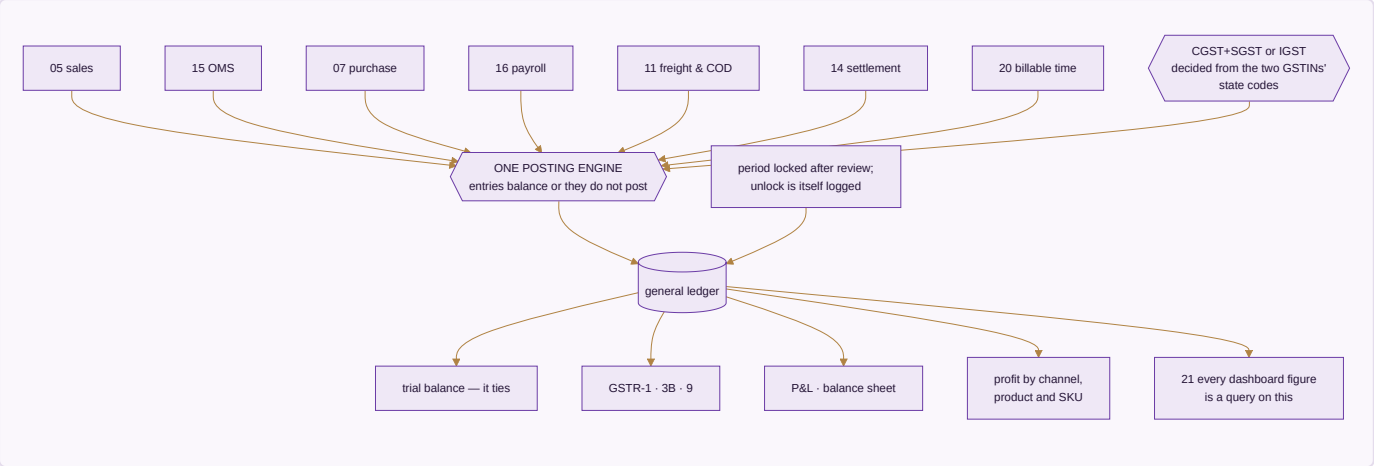
---

# MODULE 12 · ACCOUNTING & GST

*Books that always balance*

**What it is.** A full double-entry ledger built for Indian compliance — not a tax report bolted onto a spreadsheet. This system keeps the books on its own; no other accounting package is required, ever. Existing packages stay available as connectors for anyone already running one, but no figure in the business is ever *sourced* from them.

## How it works




## The apps (9)

| App | State | What it does |
|---|---|---|
| Accounting | SPEC | Chart of accounts, nine voucher types, and the one posting engine every voucher writes through |
| Invoicing | SPEC | GST tax invoices computed from the lines to the paise, with round-off and e-invoice IRN |
| Expenses | SPEC | Spend by category with approvals, and bill OCR to save typing |
| GST & Tax | SPEC | CGST, SGST, IGST, TDS, TCS, input credit and the GSTR returns, filed per registration |
| ITC Reconciliation | SPEC | Purchases matched against the portal's own GSTR-2A/2B before a return is filed |
| Receivables, Payables & PDC | SPEC | Payments allocated against named invoices, and post-dated cheques posting on realisation |
| Fixed Assets & Depreciation | SPEC | The asset register with both straight-line and written-down-value tracked side by side |
| Year-End Close & Period Lock | SPEC | Carry-forward at year end, and a locked period no backdated edit can touch |
| Finance Reports | SPEC | P&L, balance sheet, and profit by channel, product and SKU |

**What it owns.** `chart_of_accounts` · `voucher_series` · `journal_entries` · `journal_lines` · `gst_returns` · `gst_input_credit` · `tds_entries` · `tcs_entries` · `bank_accounts` · `bank_transactions` · `fixed_assets` · `depreciation_entries` · `post_dated_cheques` · `bill_allocations` · `period_locks`

### The rules that matter

- **One posting engine.** Every voucher writes the ledger through it, and an entry that does not balance does not post — it is refused, not saved half-done.
- **The tax type is derived, never chosen.** Intra-state versus inter-state comes from comparing the two GSTINs' state codes. A hand-picked tax type is a hand-made error.
- Rates default from the HSN and are **versioned by effective date**, so an invoice from last year keeps last year's rate after a rate change.
- Input credit is reconciled against the government's own data before GSTR-3B. Without it, a filing is a guess about how much credit you may legitimately claim.
- **Round-off posts to its own ledger account** — never absorbed into the sale amount, because absorbing it corrupts the GST calculation underneath.
- A closed period locks. No backdated edit without an admin unlock, and the unlock is itself logged.

**Build order inside this module** — fixed by the nature of the thing: posting engine and audit first, wrapping every write from day one; then sales and purchase invoices, verified against the trial balance; then the other seven voucher types; then year-end close and period lock; then GST returns and 2A/2B reconciliation.

### The rulebook — 24 rules, 16 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R12.1 | **Money is an integer count of paise** | any amount is held, added or compared | it is an integer number of paise, becoming a decimal string only where a person reads it | holding money in a floating-point number, where ₹0.10 + ₹0.20 is not ₹0.30 and a trial balance stops balancing | **ENFORCED** · core/tests/core.test.js · the classic float error cannot happen here |
| R12.2 | **An amount finer than a paisa is refused, not rounded** | a computation produces a fraction of a paisa | it is refused and the caller must round deliberately | rounding silently, which is how two sides of the same figure drift apart and nobody can say which is right | **ENFORCED** · core/tests/core.test.js · an amount finer than a paisa is refused rather than silently rounded |
| R12.3 | **A split sums back to the original, exactly** | an amount is divided — across lines, across companies, across periods | the parts add back to the whole, with the round-off returned as its own figure | losing or inventing a paisa in the split, and never hiding the remainder inside the largest part | **ENFORCED** · core/tests/core.test.js · a split always sums back to the original — no paisa lost or invented |
| R12.4 | **An unbalanced entry is refused, with the gap named** | a voucher is posted whose debits and credits differ | it is refused and the difference is stated | posting it to a suspense account to make it balance, which converts an error into a permanent record | **ENFORCED** · core/tests/core.test.js · an unbalanced entry is refused, with the gap named |
| R12.5 | **A line cannot be a debit and a credit at once** | a posting line carries both | it is refused | netting the two into whichever is larger | **ENFORCED** · core/tests/core.test.js · a line cannot be a debit and a credit at once |
| R12.6 | **The trial balance is computed, never stored** | the trial balance is asked for | it is summed from the posting lines at that moment | reading a maintained total, which is a number that can be wrong without anything looking wrong | **ENFORCED** · core/tests/core.test.js · the trial balance is computed from the lines, never stored |
| R12.7 | **A locked period refuses a backdated entry** | a voucher is dated inside a closed period | it is refused and the lock that stopped it is named | posting it into the current period instead, which silently moves last year's result into this one | **ENFORCED** · core/tests/core.test.js · a locked period refuses a backdated entry |
| R12.8 | **Unlocking a period is itself recorded** | a closed period is reopened | who reopened it, when and why is written to the trail | allowing a quiet reopen, which is the one action that could | **ENFORCED** · core/tests/core.test.js · unlocking a period is itself recorded |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | | | | undo every other guarantee here | |
| R12.9 | **A tax rate resolves on the date of the document** | tax is computed for any invoice | the rate in force on that document's date is used | applying today's rate to an old invoice, which makes correct history look like an error | **ENFORCED** · core/tests/core.test.js · a tax rate resolves on a date, so old invoices stay correct |
| R12.10 | **Two rates covering one date is ambiguous, not a coin toss** | two effective-dated rows overlap for the same date | the resolution is refused and the overlap is named | picking the newer one, which makes the answer depend on insertion order | **ENFORCED** · core/tests/core.test.js · two rows covering one month is ambiguous, not a coin toss |
| R12.11 | **A voided entry is reversed, never erased** | a posted voucher is wrong | a reversing entry is posted and both stay visible | editing or deleting the original, which is the difference between a correction and a cover-up | **ENFORCED** · core/tests/core.test.js · voiding is the only removal, and it is reversible |
| R12.12 | **Every figure clicks down to the record that produced it** | any total appears on any screen | it is a live query that can be opened down to its vouchers and their documents | showing a figure that cannot be traced — an untraceable number is a defect, not a rounding difference | **ENFORCED** · core/tests/core.test.js · the trial balance is computed from the lines, never stored |
| R12.13 | **An invoice number is sequential per company and per series** | an invoice is raised | it takes the next number in that company's series | reusing, skipping or back-filling a number, which is the first thing a tax audit tests | SPECIFIED |
| R12.14 | **A GST return is built from vouchers, not from a summary** | GSTR-1 or 3B is prepared | it is computed from the underlying invoices | accepting a typed summary figure, which cannot be reconciled when the portal disagrees | SPECIFIED |
| R12.15 | **ITC is claimed only where the supplier has filed** | input credit is taken | it is matched against the supplier's filed data and the unmatched part is held | claiming everything and reversing later, which turns a reconciliation into a liability | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R12.16 | **A place of supply decides the tax, not the billing address** | GST is computed | the place of supply determines CGST/SGST or IGST | defaulting to the billing address, which mis-splits the tax on every drop-ship | SPECIFIED |
| R12.17 | **A credit note references the invoice it reverses** | a credit note is raised | the original invoice is named on it | issuing a free-standing credit note, which cannot be matched in either set of books | SPECIFIED |
| R12.18 | **Depreciation is posted, not just calculated** | a period closes | depreciation is posted as an entry like any other | showing it as a computed figure on a report while the ledger disagrees | SPECIFIED |
| R12.19 | **A company with no tax registration is still a company** | a group company has no registration of its own | it keeps its own books and joins the group figures | dragging it into a return it does not belong in, and never leaving it out of the group result | SPECIFIED |
| R12.20 | **Year-end close locks, and the lock is the record** | a financial year is closed | the period is locked and the closing balances are carried forward as an entry | leaving the year open indefinitely so late entries can drift in unnoticed | **ENFORCED** · core/tests/core.test.js · a locked period refuses a backdated entry |
| R12.21 | **Every voucher type posts through one engine** | a sale, purchase, credit note, debit note, payment, receipt, journal, contra or counter sale is recorded | all nine post through the same ledger routine | giving a voucher type its own posting logic — this is where home-built accounting breaks and the modules stop agreeing about the same figure | **ENFORCED** · core/tests/core.test.js · a balanced entry posts |
| R12.22 | **Net GST is input against output, per period, per company** | the GST position for a period is computed | it is output tax less eligible input credit for that company and that period | netting across companies, which offsets one registration's liability with another's credit and is not a return anyone may file | SPECIFIED |
| R12.23 | **Money never becomes a float, in any layer** | an amount is stored, moved between the engine and the | it stays an integer count of paise end to | a real, double, float or an unlabelled decimal column | **ENFORCED** · core/tests/schema.test.js · no money column is a float, in either schema |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | | database, or exported | end, converted for display only | anywhere a money value lives | |
| R12.24 | **A money column says what unit it is in** | a column holds an amount | its name ends in paise | a column called total, amount or cost with no unit — the same name read as rupees by one developer and paise by the next is a factor of a hundred in the books | **ENFORCED** · core/tests/schema.test.js · no column is named amount/price/cost without saying what unit it is in |

**Reads** ← every module · **Writes** → Finance Reports · Treasury

**Done when.** One month of books closes cleanly, the trial balance ties, and GSTR-1 and GSTR-3B generate and verify.

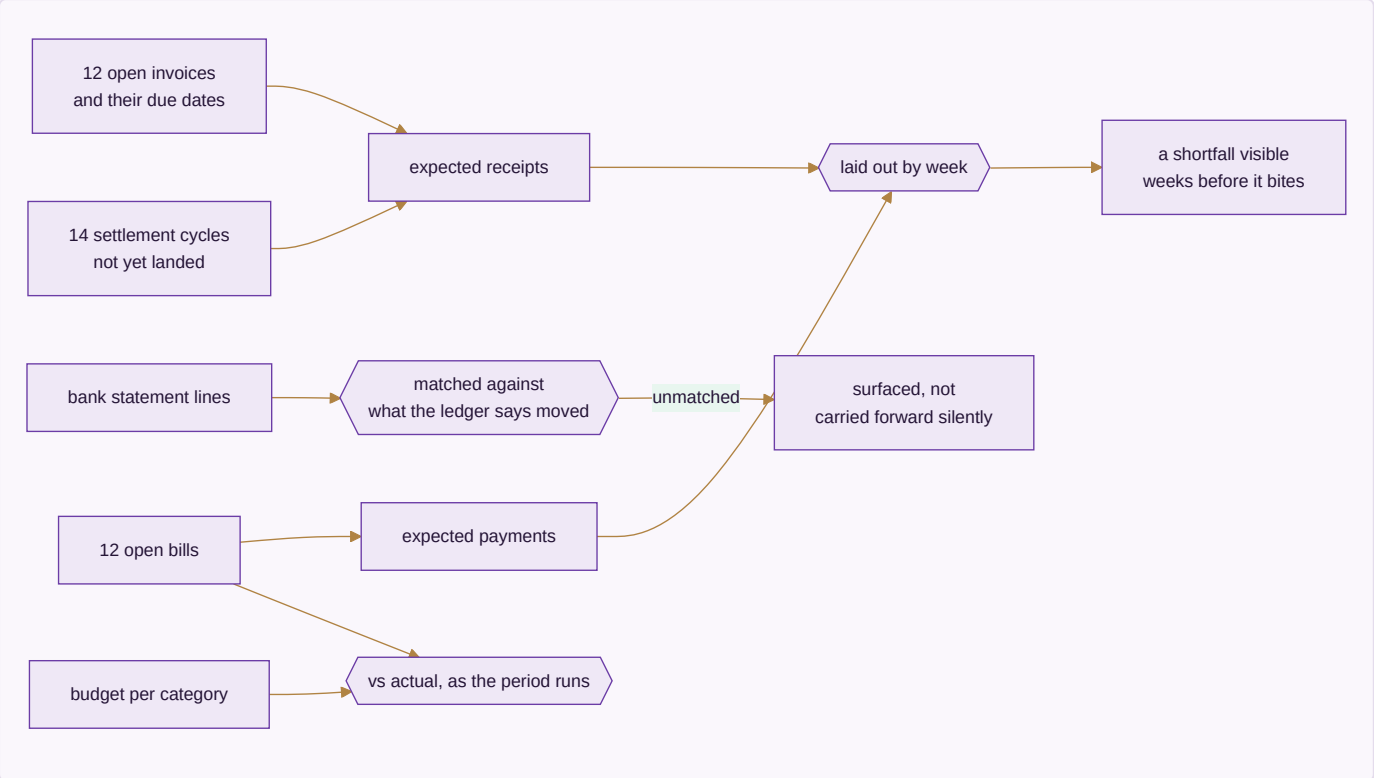
---

# MODULE 13 · TREASURY & FINANCIAL PLANNING

*Know what cash is coming, not just what already arrived*

**What it is.** Accounting records what happened. This module is about what happens next — how much cash is genuinely expected and when, and whether spending against a budget is on track while there is still time to act.

## How it works



### The apps (3)

| App | State | What it does |
|---|---|---|
| Cash Flow Forecast | SPEC | Expected receipts and payments by week, drawn from open invoices and bills rather than typed in |
| Banking & Reconciliation | SPEC | Statement lines matched against the ledger's own record, with anything unmatched surfaced |
| Budget vs Actual | SPEC | A budget per category and period, tracked against real spend while the period is still running |

**What it owns.** `cash_forecasts` · `bank_reconciliation_sessions` · `budgets` · `budget_lines`

### The rules that matter

- The forecast is **derived from open invoices and bills**, not typed in. A forecast someone maintains by hand is out of date the day after it is written.
- Marketplace payouts that have not landed yet are part of expected cash — that is what makes the forecast honest for a business selling on seven channels.
- An unmatched bank line is surfaced, never quietly carried forward. Carried-forward differences are how a reconciliation becomes fiction.
- Budget variance is shown **while the period runs**. A variance discovered after close is history; the only thing left to do with it is explain it.

### The rulebook — 8 rules, 1 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R13.1 | **A forecast never posts to the ledger** | a cash-flow projection is produced | it is held as a projection, separate from posted entries | writing an expected receipt into the books, which reports money that has not arrived | SPECIFIED |
| R13.2 | **A bank line is matched to a voucher, not to a total** | a bank statement is reconciled | each line is matched to the entry that caused it | reconciling on the closing balance alone, which hides two errors that happen to cancel | SPECIFIED |
| R13.3 | **An unmatched bank line stays visible until it is explained** | a statement line cannot be matched | it stays on the unreconciled list with its age | writing it off to a sundry account to clear the screen | SPECIFIED |
| R13.4 | **A PDC is a commitment before it is cash** | a post-dated cheque is received | it is tracked as a commitment until it clears | recognising it as cash on receipt | SPECIFIED |
| R13.5 | **Budget versus actual compares like with like** | a variance is shown | both sides use the same period, company and account basis | comparing a full-year budget against a part-year actual without saying so | SPECIFIED |
| R13.6 | **A cash forecast names its assumptions** | a projection is produced | the collection and payment assumptions behind it are stored with it | presenting a projection whose basis cannot be recovered a month later | SPECIFIED |
| R13.7 | **Inter-company funding is recorded on both sides** | one group company funds another | both companies post it, naming each other as counterparty | recording it in one set of books only, which leaves the group permanently out of balance | **ENFORCED** · core/tests/core.test.js · an entry cannot be its own counterparty |
| R13.8 | **A currency amount keeps the rate it was converted at** | a foreign-currency transaction is recorded | the original amount, the currency and the rate used are all stored | storing only the converted figure, which cannot be revalued or explained afterwards | SPECIFIED |

**Reads** ← Accounting & GST · Sales · Purchase · **Writes** → Accounting & GST

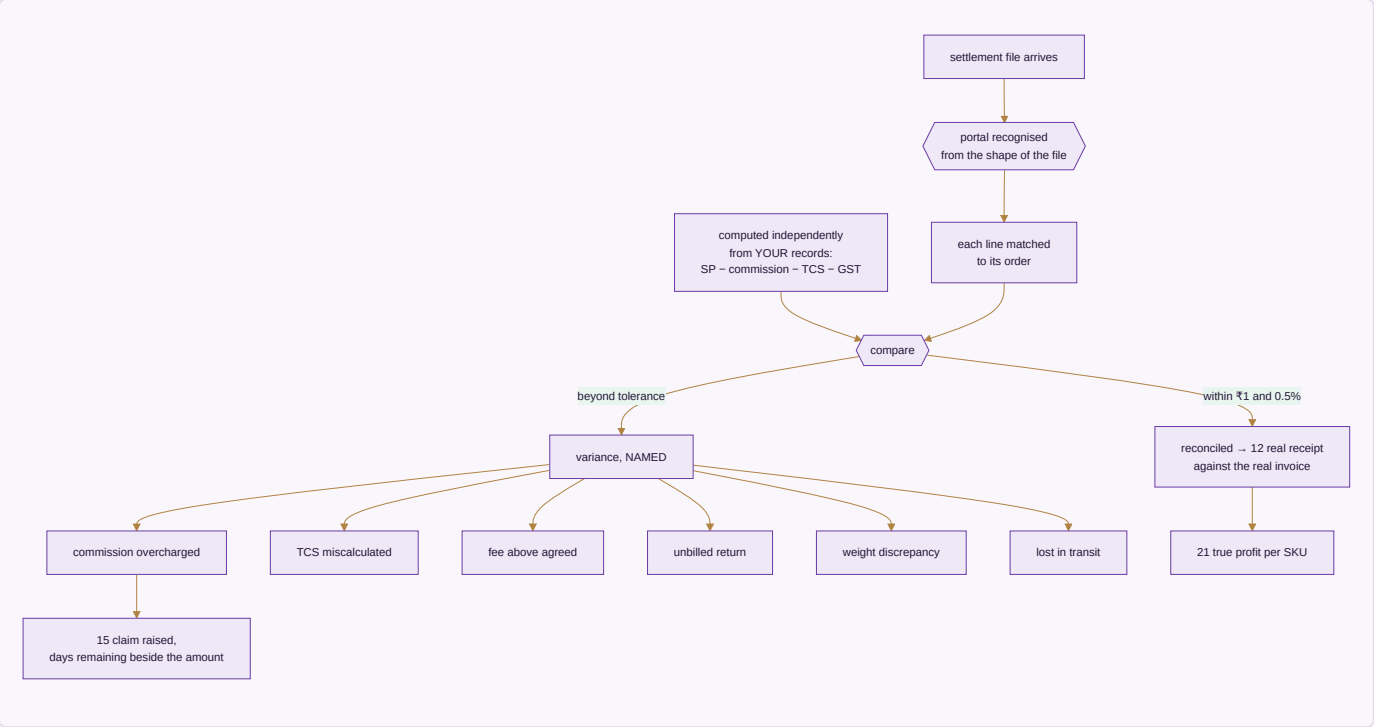
**Done when.** A twelve-week cash forecast is produced with no manual entry, and a bank statement reconciles with every unmatched line named.

# MODULE 14 · SETTLEMENT

*Get paid what you are owed — cycle by cycle*

**What it is.** A marketplace does not pay you what the customer paid. It pays selling price minus a commission, minus a collection fee, minus a shipping charge it decided, minus a return it may or may not have handled, minus tax it withheld — and hands you a file with no total you can trust. This module reads that file, works out what each line *should* have been, and puts the two side by side.

## How it works




## The apps (3)

| App | State | What it does |
|---|---|---|
| Payout Cycles | SPEC | What each cycle should pay, what actually landed, and on which day — so a late payout is visible the day it is late |
| Fee & Commission Audit | SPEC | The published rate against the rate actually charged, by category, SKU and tier |
| TCS & TDS Register | SPEC | Every rupee withheld, matched against the portal's own figures |

**What it owns.** `settlement_cycles` · `fee_audit_lines` · `tcs_tds_register`, reading `marketplace_settlements` from Module 15

## The rules that matter

- **The expected figure is computed from your own records, not read back from the file.** The whole point is comparing the file against an answer the file cannot influence.

- **Commission is read from the file, then challenged against the published rate.** A system that assumes the commission is whatever the file says can never detect an overcharge — it has already agreed to it.
- A line is flagged only past **₹1 or 0.5%**, so the variance list holds real problems rather than a thousand one-paise arguments.
- **Every variance has a named kind.** You dispute a weight discrepancy differently from a lost parcel; the name is what makes the claim actionable.
- Days remaining sit beside the rupees at stake. A valid claim that lapses because nobody saw the clock is money given away.

**The gate.** A real settlement file must reconcile at **98% or better** automatically, and per-SKU profit must land within **₹10** of your own records. Below that the module is not trustworthy enough to run money on — and a settlement tool you cannot trust is worse than none, because it lends confidence to a wrong number.

### The rulebook — 13 rules, 0 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R14.1 | **A payout is matched line by line to orders** | a marketplace settlement file arrives | every line is matched to the order it belongs to | accepting the net credited amount, which is how a short payment becomes invisible | SPECIFIED |
| R14.2 | **Every deduction is identified before the payout is accepted** | commission, shipping, penalty, TCS or TDS is deducted | each is posted to its own account | posting the deductions as one lump, which makes an overcharge impossible to find | SPECIFIED |
| R14.3 | **A variance beyond tolerance raises a claim** | the settled amount differs from the expected amount | a claim is raised carrying the order, the expectation and the difference | absorbing it because it is small — the small ones are the recurring ones | SPECIFIED |
| R14.4 | **A claim has a deadline and the deadline is tracked** | a claim is raised | the channel's filing window is stored and warned on | letting a valid claim expire unfiled | SPECIFIED |
| R14.5 | **An expected settlement exists from the moment of the sale** | an order is confirmed on a marketplace | a settlement expectation is created then | waiting for the payout to discover what should have arrived | SPECIFIED |
| R14.6 | **TCS and TDS are receivables, not costs** | a marketplace deducts tax at source | it is posted as a receivable against the tax authority | expensing it, which understates profit and loses the credit | SPECIFIED |
| R14.7 | **A settlement is reconciled to the bank, not just to the file** | a payout is recorded | it is matched to the actual bank credit | treating the settlement report as proof that the money arrived | SPECIFIED |
| R14.8 | **A re-sent settlement file does not double-post** | the same settlement file is imported twice | already-matched lines are recognised and skipped | posting them again, which doubles both revenue and deductions | SPECIFIED |
| R14.9 | **A fee schedule is dated and compared against** | a commission is deducted | it is checked against the agreed rate in force on that date | accepting whatever rate the file states, which is the single largest silent leak in marketplace trade | SPECIFIED |
| R14.10 | **A settled order is profitable or unprofitable at the SKU** | a payout is fully matched | the true net per SKU is computed after every deduction | judging profitability on the listed price, which ignores the third of it that never arrives | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R14.11 | **A claim that is paid closes against the original variance** | a channel credits a claim | it is matched back to the variance it settles | posting the credit as unrelated income, which leaves the variance open forever | SPECIFIED |
| R14.12 | **A settlement figure never overwrites a sale figure** | the settlement disagrees with the order | both are kept and the difference is the variance | adjusting the original sale to match the payout, which erases the evidence of the shortfall | SPECIFIED |
| R14.13 | **The realisation on a marketplace sale is the price minus every deduction** | what a channel sale actually earned is computed | it is the selling price less shipping, commission, fixed fee, GST on those fees, TCS and TDS — each taken from the settlement file | judging a sale on its listed price, which ignores the part of it that never arrives, and never applying an assumed commission percentage when the file states the real one | SPECIFIED |

**Reads** ← E-commerce/OMS · Accounting & GST · **Writes** → Accounting & GST

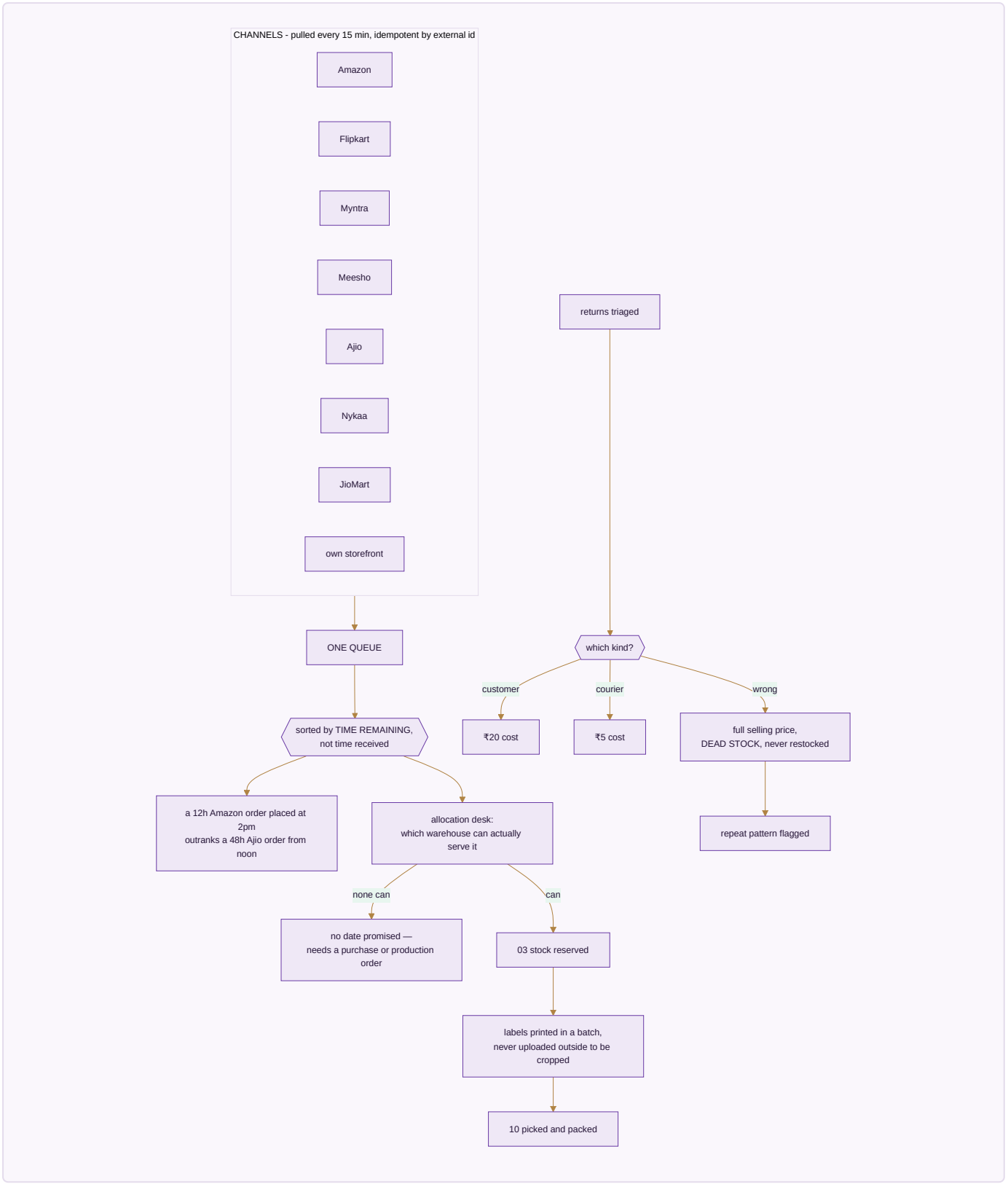
**Done when.** A real settlement file reconciles at 98% or better and every variance it raises is one you agree is genuinely real.

# MODULE 15 · E-COMMERCE / OMS

*Every marketplace and your own website, one queue*

**What it is.** Stop logging into seven seller panels and your own store admin. Every order lands in one pipeline and one stock number goes back out to all of them — then the money side closes in the same module: what each channel paid, what it kept, what came back, and what it still owes.

**How it works**


CHANNELS - pulled every 15 min, idempotent by external id
Amazon
Flipkart
Myntra
Meesho
Ajio
Nykaa
JioMart
own storefront
ONE QUEUE
sorted by TIME REMAINING, not time received
a 12h Amazon order placed at 2pm outranks a 48h Ajio order from noon
allocation desk: which warehouse can actually serve it
none can
no date promised — needs a purchase or production order
can
03 stock reserved
labels printed in a batch, never uploaded outside to be cropped
10 picked and packed
returns triaged
which kind?
customer
courier
wrong
₹20 cost
₹5 cost
full selling price, DEAD STOCK, never restocked
repeat pattern flagged

## The apps (11)

| App | State | What it does |
|---|---|---|
| Marketplace OMS | **BUILT** | Every channel in one order queue, with the right cut-off counting down on each order |
| Order Management | **BUILT** | One pipeline new to delivered, with the allocation desk and the promise-date engine |
| Manual Data Check | SPEC | Upload the sheets you already download and read ten cross-checks back, every figure clickable to the transactions behind it |
| Reconciliation | SPEC | Match every payout to the order line that earned it, and expose the gap |
| Claims & Disputes | SPEC | Shortfalls, weight disputes and lost parcels filed with evidence, answered before the clock runs out |
| Returns / RMA | SPEC | Customer, courier and wrong returns kept apart — only one of the three is really your fault |
| Channels & Storefronts | SPEC | Connect a channel once and it stays in step: catalogue out, price out, stock out, orders in |
| Labels & Documents | SPEC | The channel's PDF turned into something a packing table can work from, in one batch |
| Listing & Catalog Manager | SPEC | Bulk create and edit listings across channels, catching listed-but-out-of-stock mismatches |
| Size / Fit Recommendation AI | SPEC | A fit suggestion from the item's measurements and the return history by size |
| AR / Virtual Try-On | SPEC | Seeing drape and colour before buying, where a flat photo leaves too much to guess |

**What it owns.** `marketplace_orders_raw` · `marketplace_settlements` · `marketplace_settlement_lines` · `returns` · `claims` · `channels` · `channel_listings`

### The rules that matter

- **The queue sorts by time remaining, not time received.** Channels have different dispatch windows; sorting by arrival breaches the tight ones while the loose ones sit safe.
- Orders are pulled on a schedule and are **idempotent by external ID**, so the same order never doubles no matter how often the pull runs.
- **An order no warehouse can serve gets no promised date.** It needs a purchase or a production order, and nothing is gained by giving the customer a date in the meantime.
- **A wrong return is dead stock and is never restocked.** Booking it back into sellable stock is how a business sells the same damaged piece twice and loses the customer permanently.
- **Labels are never uploaded to an outside website to be cropped.** That is a customer's name and address leaving your control for a convenience.
- Commission is read from the settlement file, never assumed — the challenge to it happens in Module 14.

### The rulebook — 19 rules, 10 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R15.1 | **Companies and channels are read from the data, never from a list in the code** | orders or sheets from any number of companies and channels are processed | the companies and channels present are discovered and each gets its own columns | writing a fixed set of companies or channels into the software, which caps the business at whatever it happened to have on the day the code was written | **ENFORCED** · brand/suite/studio/verify_studio.js · the companies are found from the sheets, not from a hardcoded list |
| R15.2 | **A tenth or eleventh channel needs no code change** | a new marketplace or company is added | it is a row, and every figure, column and consolidation follows | requiring a release to sell somewhere new | **ENFORCED** · core/tests/core.test.js · an eleventh company and an eleventh channel need no code change |
| R15.3 | **A channel belongs to a company** | two companies both sell on the same marketplace | each has its own channel record, and both may use the same short code | sharing one channel across companies, which merges two companies' sales into one figure | **ENFORCED** · core/tests/core.test.js · a channel belongs to a company — two companies may both call one AMZN |
| R15.4 | **A price is never invented for an item that has none** | an item has no price on file | it is reported as having no price and named in the summary | substituting an average or a similar item's price, which quietly fabricates revenue | **ENFORCED** · brand/suite/studio/verify_studio.js · the price status matches, and no price was ever invented |
| R15.5 | **Net is sale minus return, and inventory is net plus wrong return** | quantities are rolled up | net sale is sale minus return, and the inventory figure adds back the wrong returns | treating a wrong return as ordinary saleable stock, because it is not the item that was sent out | **ENFORCED** · brand/suite/studio/verify_studio.js · sale minus return is the net, and net plus wrong return is the inventory |
| R15.6 | **A blank cell is blank, not a value** | a column contains only whitespace | it is read as empty | treating a lone space as a marked entry, which converts | **ENFORCED** · brand/suite/studio/verify_studio.js · a lone space in the Wrong Return column is not a wrong return |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | | | | formatting into business fact | |
| R15.7 | **An item that only ever came back is still reported** | an item has returns but no sales in the period | it appears with its returns | dropping it because it has no sale line, which hides the worst-performing items entirely | **ENFORCED** · brand/suite/studio/verify_studio.js · an item that only ever came back is still reported |
| R15.8 | **A totals row is the sum of the rows above it** | a report shows a total | it equals the rows it sits under | computing the total by a different route from the detail, which is how a report disagrees with itself | **ENFORCED** · brand/suite/studio/verify_studio.js · the totals row is the sum of the rows above it |
| R15.9 | **A marketplace order pull creates a real order** | orders are fetched from a channel | a sales order is created, stock is reserved, and the pick list follows | holding channel orders in a staging area that has to be re-entered to become real | SPECIFIED |
| R15.10 | **A cancelled channel order releases its reservation** | the channel cancels an order | the reservation is released and the cancellation recorded | leaving stock reserved against an order the channel has already dropped | SPECIFIED |
| R15.11 | **A wrong return is dead stock, not stock** | a return is inspected and found to be a different or damaged item | it is written to dead stock with its cost recognised as a loss | restocking it as first quality, which sells a customer the same problem twice | SPECIFIED |
| R15.12 | **A listing rejected by a channel says why** | a push to a channel fails | the rejection and its reason are reported back against the listing | reporting a push as successful when part of it failed, which leaves the business believing it is | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | | | | present where it is not | |
| R15.13 | **A manual data check is a recorded step, not a habit** | figures are checked by hand before a cycle closes | the check, the person and the outcome are recorded | relying on someone remembering to look | SPECIFIED |
| R15.14 | **A channel-specific SKU code never becomes the master code** | a channel uses its own identifier | it is stored as a mapping against our SKU | adopting the channel's code as the item code, which breaks the moment a second channel does the same | SPECIFIED |
| R15.15 | **A size recommendation is advice, never a silent substitution** | a fit suggestion is offered | it is shown as a recommendation the customer chooses | changing the size on an order on the customer's behalf | SPECIFIED |
| R15.16 | **An order held past its cut-off is escalated, not queued** | an order approaches the channel's dispatch deadline | it is raised to the person who can act, naming the deadline | letting it age quietly into a penalty | SPECIFIED |
| R15.17 | **Closing stock is opening plus in minus out** | a stock position is computed for a period | closing = opening + receipts − issues, from the movements themselves | carrying a maintained closing figure that can drift from the movements that produced it | **ENFORCED** · core/tests/core.test.js · a receipt then an issue leaves the right number |
| R15.18 | **Courier return, customer return and wrong return cost three different things** | a return is processed | a courier return costs repacking only, a customer return costs alteration plus iron plus packing at the rate set for that design, and a wrong return is written off at the full selling price | applying one blended return cost to all three, which hides the expensive kind inside the cheap kind | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R15.19 | **A wrong return is never added back to stock** | a return is found to be a different item from the one sent | it becomes dead stock and the selling price is recognised as a loss | restocking it, at any value, however sellable it looks | **ENFORCED** · brand/suite/studio/verify_studio.js · sale minus return is the net, and net plus wrong return is the inventory |

**Reads** ← Inventory & Catalog · CRM · Sales · Accounting · Logistics · Settlement · **Writes** → Inventory & Catalog · Accounting · Warehouse · Logistics · Settlement

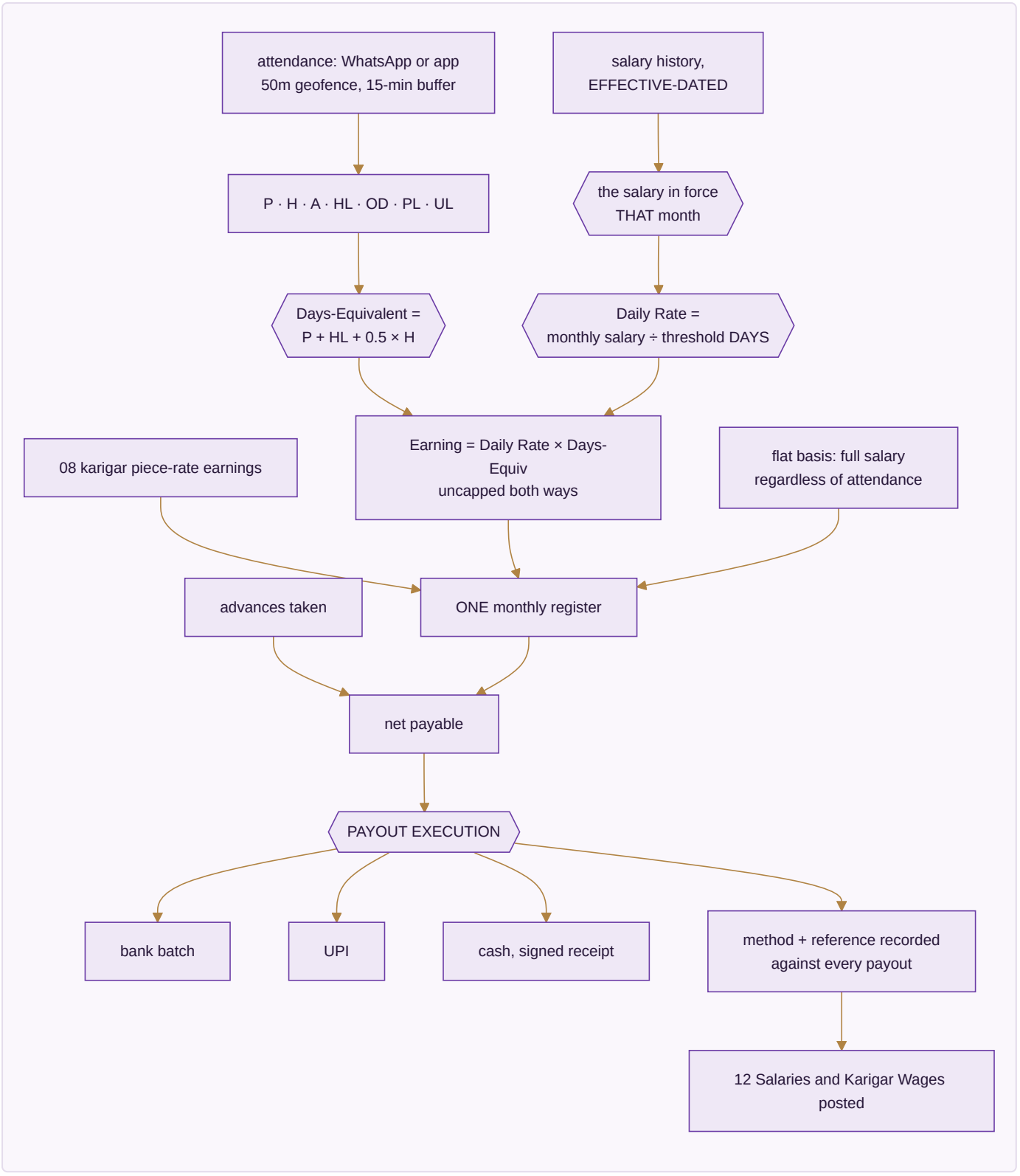
**Done when.** A full week runs with every channel live, settlements reconciled, and no panel oversells.

# MODULE 16 · HR & PAYROLL

*Pay people right, on time*

**What it is.** Salaries and output-based earnings in one register, with attendance driving both — whether people are on a monthly wage, an hourly rate, or paid by what they finish. And, unlike the version of this that stops at a calculated figure, it carries through to the money actually leaving the business.

**How it works**


attendance: WhatsApp or app
50m geofence, 15-min buffer
salary history,
EFFECTIVE-DATED
P · H · A · HL · OD · PL · UL
the salary in force
THAT month
Days-Equivalent =
P + HL + 0.5 × H
Daily Rate =
monthly salary ÷ threshold DAYS
08 karigar piece-rate earnings
Earning = Daily Rate × Days-
Equiv
uncapped both ways
flat basis: full salary
regardless of attendance
advances taken
ONE monthly register
net payable
PAYOUT EXECUTION
bank batch
UPI
cash, signed receipt
method + reference recorded
against every payout
12 Salaries and Karigar Wages
posted

## The apps (4)

| App | State | What it does |
|---|---|---|
| Staff & Contractors | SPEC | Attendance, effective-dated salary and output-based earnings in a single register |
| Time-off & Advances | SPEC | Leave, festival advances, and exactly how they change this month's payout |
| Appraisal & Hiring | SPEC | Performance reviews and a hiring pipeline that ends in an employee record |
| Payout Execution | SPEC | Where the calculation becomes money leaving the business, with method and reference on every payout |

**What it owns.** `staff_salary_history` · `attendance` · `eod_reports` · `leave_requests` · `advance_requests` · `payroll_runs` · `payroll_slips` · `karigar_earnings_summary` · `piece_rates` · `task_threshold_rates` · `payout_batches`

### The pay rules, exactly

1. Attendance codes are **P · H · A · HL · OD · PL · UL**.
2. **Days-Equivalent = P + HL + 0.5 × H.** A holiday pays a full day.
3. **Daily Rate = the resolved monthly salary ÷ the resolved threshold DAYS** — both effective-dated.
4. **Earning = Daily Rate × Days-Equivalent, uncapped in both directions.** Working beyond the threshold earns beyond the salary; working under it earns under.
5. A flat-basis worker draws full salary regardless of attendance. A piece-rate worker is hours × flat rate with no attendance row at all.
6. Three month-states are distinguished and never conflated: **Not employed**, **No data**, and a real month. Treating the first two as zero is how someone gets paid nothing for a month they worked.

*Worked example.* Salary ₹15,000 with a 26-day threshold gives a daily rate of ₹576.92. A month of 24 P, 1 HL and 2 H is 24 + 1 + 1.0 = 26.0 days-equivalent, so ₹15,000.00 exactly. The same person on 28.0 days-equivalent earns ₹16,153.85 — it scales past the threshold because the rule is uncapped.

**The rule that closes the loop.** A salary edit is made **once, with an effective-from date**. The previous row closes automatically, past months keep their old rate, and a future-dated raise activates itself when that month's payroll runs. No historical payroll is ever rewritten.

### The rulebook — 22 rules, 8 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R16.1 | **A raise closes the old row, it does not overwrite it** | a salary or rate changes | the row in force is closed on the day before, and a new row opens | editing the existing figure, which rewrites what the person was actually paid last year | **ENFORCED** · core/tests/core.test.js · a raise closes the open row instead of overwriting it |
| R16.2 | **History resolves to what was actually in force** | a past month is recomputed | the rate in force in that month is used | recomputing an old payslip at today's rate | **ENFORCED** · core/tests/core.test.js · history still resolves to what was actually in force |
| R16.3 | **A future-dated raise activates by itself** | a raise is entered with a future date | it takes effect when that month arrives, with nobody remembering to apply it | requiring a manual step, which is how an agreed raise is missed | **ENFORCED** · core/tests/core.test.js · a future-dated raise activates by itself when that month arrives |
| R16.4 | **A month with nothing in force raises, and never returns zero** | no rate covers the month being computed | the computation is refused and the gap is named | returning zero, which pays a real person nothing and looks like a valid answer | **ENFORCED** · core/tests/core.test.js · a nothing-in-force month raises, and never returns zero |
| R16.5 | **Backdating over an open row is refused** | a change is entered with a date inside a period already settled | it is refused | silently rewriting history that has already been paid and posted | **ENFORCED** · core/tests/core.test.js · backdating over an open row is refused — that would rewrite history |
| R16.6 | **A person can leave and come back** | someone rejoins after a break | the spell log holds both periods and the gap between them | creating a second employee record, which splits their history and their service | **ENFORCED** · core/tests/core.test.js · a spell log lets a person leave and come back |
| R16.7 | **Month spans handle February and the year end** | a period is computed across month or year boundaries | the real calendar is used | assuming thirty-day months, which is wrong twelve times a year and badly wrong in February | **ENFORCED** · core/tests/core.test.js · month spans handle February and the year end |
| R16.8 | **Staff and piece-rate workers sit in one register** | payroll is prepared | monthly staff and piece-rate karigars are computed in the same run and paid from the same register | running two payrolls that have to be added together by hand | SPECIFIED |
| R16.9 | **An advance is recovered** | a deduction is made at payout | it names the advance it is | deducting an amount that cannot be traced to a | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| | **against a named advance** | | recovering and reduces that balance | specific advance | |
| R16.10 | **Attendance drives pay, and both are visible together** | a payout is computed | the attendance it was computed from is shown beside it | presenting a pay figure whose basis the person being paid cannot see | SPECIFIED |
| R16.11 | **Identity documents are read, never stored in a file that leaves** | Aadhaar, PAN, bank or UPI detail is used for a computation | it is used and not serialised into any exported or committed artifact | writing personal identifiers into a report, a backup file or a repository | SPECIFIED |
| R16.12 | **A payout that fails to post does not mark as paid** | the bank transfer or the ledger posting fails | the payout stays unpaid and the failure is raised | marking it paid on submission, which loses a real person's wages in the gap | SPECIFIED |
| R16.13 | **The daily rate is the monthly salary divided by twenty-seven** | a day of attendance is priced | the daily rate is that month's salary ÷ 27, using the salary in force in that month | using calendar days, working days, or a rate carried over from a month with a different salary | SPECIFIED |
| R16.14 | **Attendance codes have fixed multipliers and a blank is absent** | earned pay is computed from attendance | present, holiday, on-duty and paid leave count 1, a half day counts 0.5, absent and unpaid leave count 0, and an empty cell counts as absent | treating a blank as present, or as unknown to be filled in later — a blank that pays is a blank that will be left blank | SPECIFIED |
| R16.15 | **Threshold hours do not move when salary moves** | a raise takes effect | the monthly hour threshold for that role stays as it was | scaling the threshold with the salary, which silently changes what the person is expected to work in exchange for a raise | SPECIFIED |
| R16.16 | **Productivity cost is that month's salary over the threshold, times hours worked** | the cost of a person's time is charged to work | it is (salary in force that month ÷ threshold hours) × the hours actually active | using a single annual figure, which misprices every month on either side of a raise | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R16.17 | **A holiday is paid and produces no hours** | a holiday is marked | it pays a full day and contributes zero productive hours | counting holiday hours as production, which flatters every efficiency figure that reads them | SPECIFIED |
| R16.18 | **A half day is half the hours, from the same start** | a half day is marked | it starts at the normal in-time and its hours are half the full shift for that person's pattern | assuming a fixed midday finish for everyone, when the male and female shift lengths differ | SPECIFIED |
| R16.19 | **The festival flag drives leave and nothing else** | a religion is recorded against a person | it is used only to match a festival-leave request | using it as a filter, a grouping or a report dimension anywhere else in the system | SPECIFIED |
| R16.20 | **A geofence failure flags, it does not refuse** | attendance is marked outside the radius set for the unit, or outside the grace window | it is recorded with the flag and raised to the manager | refusing the mark — a system that locks someone out of being paid for standing at the wrong gate has failed at its actual job | SPECIFIED |
| R16.22 | **A shared document carries the pay rules, never the pay roster** | a plan, a specification or any document that leaves this building is generated | it carries the formulas, thresholds and effective-dating that decide pay, and refers to the roster rather than reproducing it | printing an individual's name beside their salary, or their religion at all, into a document that is committed to a repository and travels with every copy — the software needs those fields, a reader of the plan does not | SPECIFIED |
| R16.21 | **An override is allowed and is always recorded** | an administrator corrects attendance, a geofence flag or a payroll figure | the change, the person and the reason go to the audit trail | an override that leaves no trace, which is indistinguishable from the system having been wrong | **ENFORCED** · core/tests/core.test.js · an update records what it was as well as what it became |

**Reads** ← Manufacturing · **Writes** → Accounting & GST

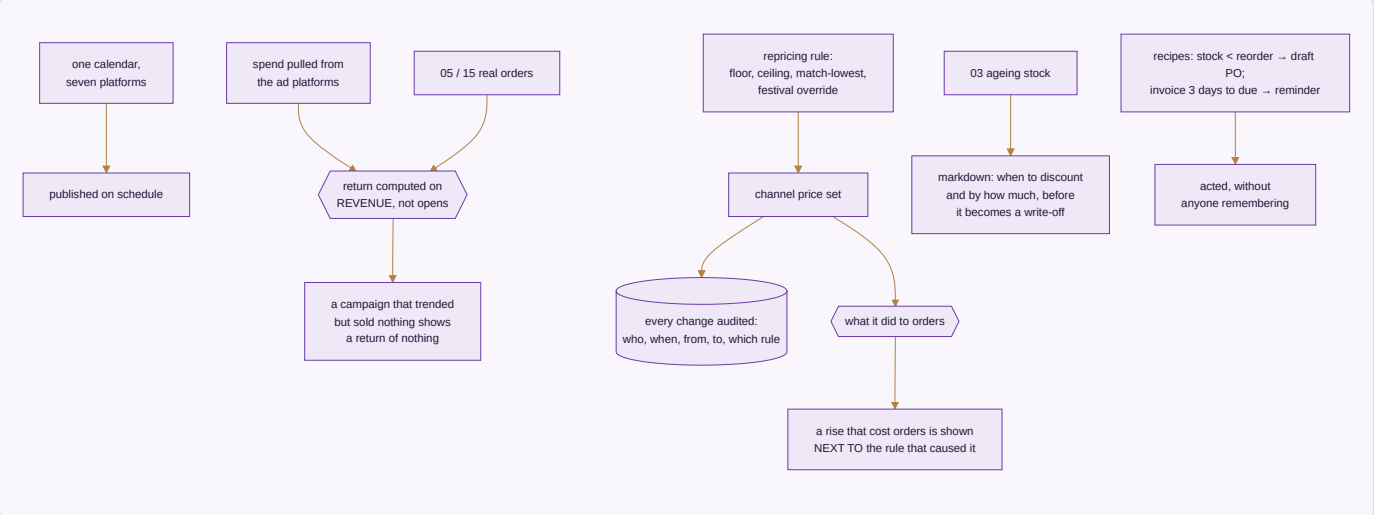
**Done when.** A full month's payroll runs end to end with zero manual touch, reconciles to the owner's own figures, and every rupee paid has a method and a reference against it.

# MODULE 17 · MARKETING

*Sell more without discounting*

**What it is.** The demand-generation side of the same order book. Its discipline is that every lever it pulls is judged by what it did to orders and margin — using the same stock number and the same ledger as everything else.

## How it works




## The apps (8)

| App | State | What it does |
|---|---|---|
| Social Calendar | SPEC | Plan and publish across every platform from one calendar |
| Campaigns | SPEC | Campaigns measured on real revenue, not opens |
| Repricing Engine | SPEC | Rules per channel and SKU, every change audited, and what each one actually did |
| Automation | SPEC | If this happens, do that — across any module, without writing code |
| Blog & Pages | SPEC | Articles and landing pages published to your own site with meta and internal links set |
| Website & Page Builder | SPEC | The storefront itself, built by dragging sections into place, each block reading live from the catalogue rather than from figures someone pasted in |
| Events | SPEC | Trade shows worked as a channel, leads landing straight in CRM |
| Markdown / Clearance Optimization | SPEC | The repricing engine aimed at ageing stock before it becomes a write-off |

**What it owns.** `content_calendar` · `campaigns` · `influencers` · `asset_library` · `repricing_rules` · `automation_recipes` · `events` · `markdown_plans`

## The rules that matter

- **Return is computed on revenue.** Opens, clicks and reach describe attention, not money.

- **A price that rose and took orders down with it is shown as exactly that, beside the rule that raised it.** Pricing is the one lever whose damage is invisible unless you look for it — you simply sell fewer, without being told why.
- Every price change is audited: who, when, from what to what, by which rule. A price is money; an unexplained change to it is the same failure as an unexplained ledger entry.
- **Repricing sets price and only price.** It cannot touch quantity, so no pricing automation can ever oversell by moving a number it had no business touching.
- Discounting ageing stock is a decision with a right time. Too early gives away margin that was available; too late turns a sale into a write-off.

### The rulebook — 10 rules, 0 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R17.1 | **A campaign is measured on revenue, not on opens** | campaign performance is reported | it is attributed to actual orders | reporting opens and clicks as the result, which measures the message rather than the business | SPECIFIED |
| R17.2 | **A repricing rule shows what it did** | a rule changes a price | the change, the rule that made it and the effect on orders are recorded together | changing prices with no record, which makes a bad rule impossible to identify or reverse | SPECIFIED |
| R17.3 | **A price floor is a floor** | a repricing rule would go below the floor set for a SKU | it stops at the floor | undercutting to match a competitor below cost | SPECIFIED |
| R17.4 | **A markdown starts before the stock is dead, not after** | stock reaches the age set for it | the markdown schedule begins | waiting until it is unsellable, which converts a lower-margin sale into a write-off | SPECIFIED |
| R17.5 | **A campaign cannot message someone who has not consented** | a marketing send is prepared | the recipient list is filtered by consent at send time | sending to a list captured before the consent was checked | SPECIFIED |
| R17.6 | **A published page reads live catalogue data** | a page shows a price or a stock state | it reads the same record the order screen reads | pasting a figure into the page, which goes stale the first time the price changes | SPECIFIED |
| R17.7 | **An exhibition is a channel** | leads and sales come from a trade show | they land in CRM and the order book against that channel | collecting them on paper to be entered later, which is where they are lost | SPECIFIED |
| R17.8 | **A marketing automation cannot move money** | a campaign rule fires | it may message, tag, schedule or reprice within its limits | issuing a refund, a credit note or a payment — that is not what this engine is allowed to do | SPECIFIED |
| R17.9 | **A scheduled post that fails is reported as failed** | a scheduled publication does not go out | it is raised with the reason | showing it as published in the calendar while nothing was posted | SPECIFIED |
| R17.10 | **Return on ad spend is measured against real orders** | campaign performance is computed | it is revenue from attributed orders ÷ spend actually incurred | using a platform's own reported conversions as the revenue figure, which counts orders this system has no record of | SPECIFIED |

**Reads** ← Inventory & Catalog · CRM · **Writes** → Sales · E-commerce/OMS

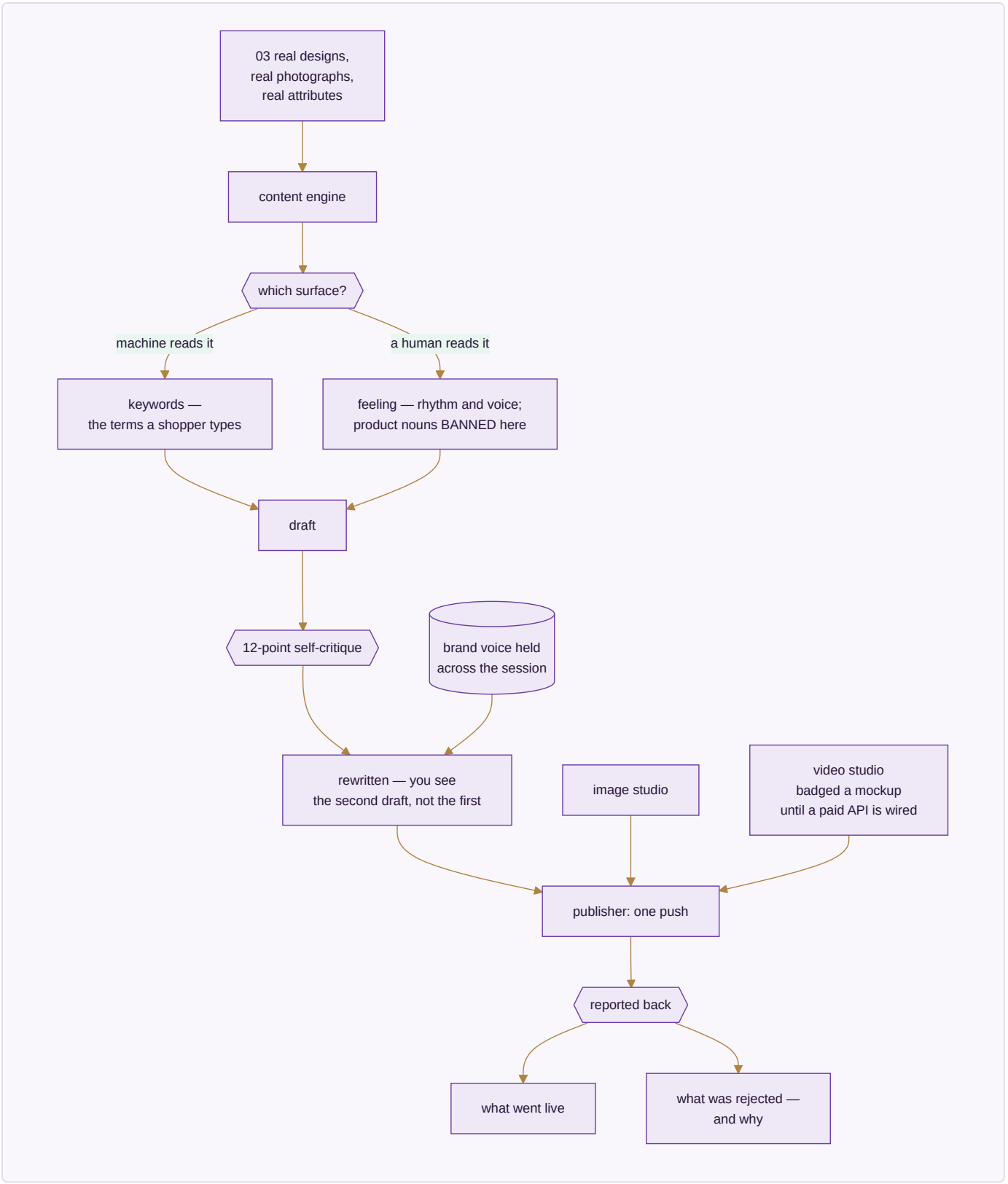
**Done when.** A month of content publishes on schedule, and a repricing rule can be judged by what it actually did to orders.

# MODULE 18 · AI CONTENT ENGINE

*Write it, shoot it, cut it — from the catalogue you already have*

**What it is.** The module the platform is named for, deliberately at eighteen rather than first: it produces content *about* the catalogue, so the catalogue has to exist and be correct before there is anything true to write about. Being wired to the same database as the shop floor is what stops it inventing a product that is not real.

### How it works


03 real designs,
real photographs,
real attributes
content engine
which surface?
machine reads it
a human reads it
keywords —
the terms a shopper types
feeling — rhythm and voice;
product nouns BANNED here
draft
12-point self-critique
brand voice held
across the session
rewritten — you see
the second draft, not the first
image studio
video studio
badged a mockup
until a paid API is wired
publisher: one push
reported back
what went live
what was rejected —
and why

## The apps (8)

| App | State | What it does |
|---|---|---|
| Content Engine | SPEC | Fourteen stages in your own voice, each written from your own catalogue so the words match the thing |
| Image Studio | SPEC | Layers, free transform, background removal, channel presets and alt text — a phone photo becomes a channel-compliant product image |
| Video Studio | SPEC | Text and image to video, reels and ad cuts sized per channel |
| Design Studio | SPEC | A full design surface exporting at whatever size the channel or printer asks for |
| **Motion Renderer** | **ENGINE WORKING** | HTML and CSS rendered frame by frame into a real MP4 on this machine, deterministically — the same scene twice gives the same file to the byte |
| Narration Studio | SPEC | The written script spoken over the reel; the browser's own voice by default, a cloned or branded voice as an interchangeable provider behind it |
| Image Generation Slot | SPEC | Generated imagery as a provider-pluggable capability — queue, preview, inpainting, upscaling. Needs a GPU, which is why it is a slot and not an engine |
| Publisher | SPEC | One push everywhere, reporting what went live and what was rejected, with the reason |

**What it owns.** `ai_runs` · `ai_listings` · `ai_design_analytics` · `asset_projects`

### The rules that matter

- **Structured data gets keywords; anything a human reads gets feelings.** A back-end field is written for a search algorithm; a caption is written for a person. Writing both the same way is what makes machine-written content read as machine-written.
- **Product nouns are banned from creative surfaces.** The right phrase in a search field is exactly the wrong phrase in a caption, and that bleed is the clearest tell of lazy automation.
- **The engine criticises its own draft before showing it.** A model's first attempt is rarely its best; building the criticism into the pipeline is what clears the bar of publishable rather than demonstrable.
- **Generation stays badged a mockup until a real paid API is wired.** Showing a simulated render as a finished one is exactly the dishonesty this whole platform is built to avoid, so the label is not optional.
- **A render is seeked, never recorded.** The Motion Renderer fakes the clock and seeks the animation to the exact instant of each frame before capturing it, rather than playing the scene and recording the screen. A recording is at the mercy of whatever else the machine was doing — one slow frame during the render is a stutter baked into the customer's reel forever, and the same scene rendered twice gives two different files, which means it can never be checked. Seeking makes the output reproducible to the byte, and that is what makes a reel something the business can verify rather than something someone has to watch all the way through and hope about.
- **Image generation says out loud that it needs a graphics card.** Image models cannot run on an ordinary office machine. The queue, the review screen and the provider slot are the honest deliverable; the generating is done by whatever engine it is pointed at. A screen that looks finished and produces nothing is the failure this rule exists to prevent.
- **A cloned voice needs the consent of the person it was cloned from**, recorded and filed against them in Data Privacy & Consent like any other permission — not assumed because the recording was easy to obtain.

- A publish that silently fails on two of six channels leaves you believing you are present where you are not. The report closes that gap.

### The rulebook — 11 rules, 2 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R18.1 | **Content is written from the catalogue, not about the category** | a listing or description is generated | it is generated from that product's own attributes | writing plausible copy about the kind of thing it is, which is how a listing describes features the product does not have | SPECIFIED |
| R18.2 | **Structured fields get keywords; anything a human reads gets feeling** | text is produced for a back-end field versus a caption | each is written for its own reader | writing both the same way, which is the clearest signal of machine-written content | SPECIFIED |
| R18.3 | **Product nouns are banned from creative surfaces** | a caption or a hook is written | the product noun is excluded | letting search vocabulary bleed into copy meant to be felt | SPECIFIED |
| R18.4 | **The engine criticises its own draft before anyone sees it** | a draft is produced | it is put through the self-critique pass and the second draft is what is shown | showing the first attempt, which is rarely the best one | SPECIFIED |
| R18.5 | **A render is seeked, never recorded** | a video is produced from a page | the clock is driven by hand and each frame is captured at its exact instant | playing the animation and recording the screen, which bakes whatever else the machine was doing into the customer's reel | **ENFORCED** · brand/suite/studio/motion_render.js · a second render produces frame-for-frame identical images |
| R18.6 | **The same scene renders to the same file** | a render is repeated | the output is identical to the byte | producing a different file each time, which makes the output impossible to check or approve | **ENFORCED** · brand/suite/studio/motion_render.js · and a byte-identical MP4 |
| R18.7 | **A generated asset is labelled as generated** | an image or video is produced by a model | it carries that fact in the asset record | letting a generated image become indistinguishable from a photograph of the actual product | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R18.8 | **Generation stays badged a mockup until a real provider is wired** | a capability is demonstrated without a live provider behind it | it is labelled a mockup wherever it appears | showing a simulated render as a finished one | SPECIFIED |
| R18.9 | **Image generation states that it needs a graphics card** | the image generation slot is opened with no provider attached | it says so plainly and produces nothing | presenting a finished-looking screen that cannot generate anything | SPECIFIED |
| R18.10 | **A cloned voice needs the consent of the person it came from** | a voice is cloned for narration | that person's recorded consent is on file against them | cloning from a recording merely because it was available | SPECIFIED |
| R18.11 | **A publish reports what actually went live** | content is pushed to several destinations | each result comes back individually, with reasons for rejections | reporting one overall success, which leaves the business absent where it believes it is present | SPECIFIED |

**Reads** ← Inventory & Catalog · **Writes** → Marketing · E-commerce/OMS

**Done when.** A listing generates for one design across six platforms in under twenty seconds and publishes, with every rejection explained.

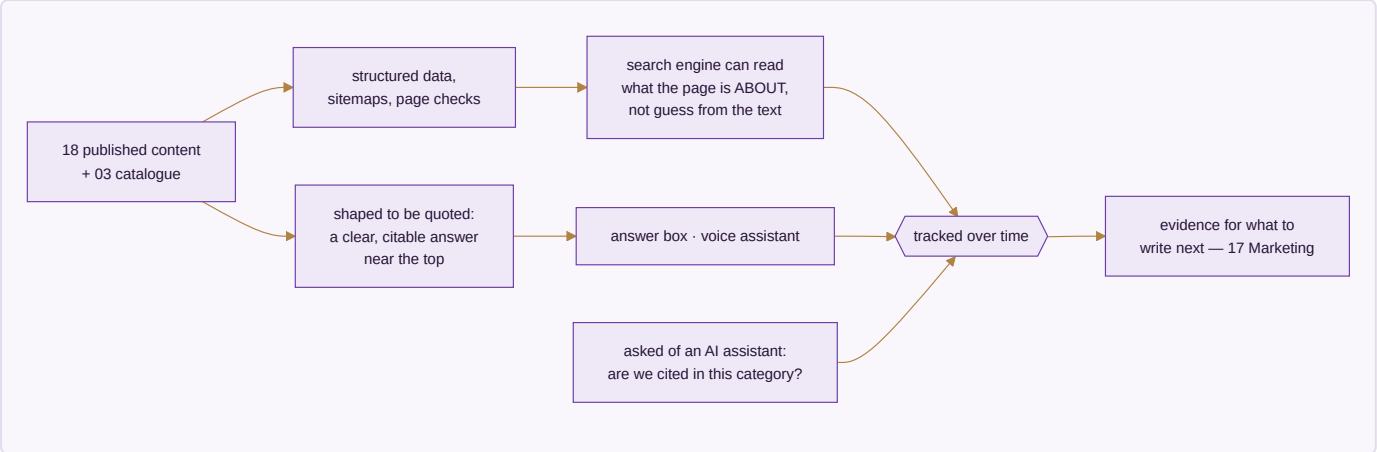
---

# MODULE 19 · SEO, AEO & AIO

*Be found by a search box, an answer box, and an AI*

**What it is.** Content exists by the time this module is reached; here it is made findable — by a traditional search engine, by the answer box above the results, and by the AI assistants now answering shopping questions directly instead of sending someone to a results page.

**How it works**




## The apps (3)

| App | State | What it does |
|---|---|---|
| Technical SEO & Schema | SPEC | Structured data, sitemaps and page-level checks so a search engine can read what a page is about |
| Answer-Engine Optimization | SPEC | Content shaped to be quoted directly rather than written only to be scrolled |
| AI-Engine Visibility Tracking | SPEC | Whether this business is actually cited when someone asks an AI a shopping question in this category |

**What it owns.** `seo_audits` · `schema_templates` · `answer_blocks` · `visibility_checks`

## The rules that matter

- Structured data is how a page stops being guessed at. Prose alone leaves the engine inferring.
- An answer engine quotes; it does not summarise a page written to be scrolled. Content has to be shaped for the quote to exist.
- **AI visibility is rank tracking aimed at a newer kind of results page.** The discipline is the same — measure over time, act on the trend — and ignoring it means being absent from where a growing share of shopping questions now get answered.

## The rulebook — 6 rules, 0 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R19.1 | **Structured data describes what is actually on the page** | schema markup is generated | it is generated from the same record the page renders | marking up a price or availability that differs from the page, which is penalised and deserved | SPECIFIED |
| R19.2 | **A ranking figure names where it was measured** | a position or citation is reported | the engine, the query and the date are stored with it | reporting a bare position, which cannot be compared with anything | SPECIFIED |
| R19.3 | **An answer-shaped page still says the same thing as the product record** | content is shaped to be quoted by an answer box | the claims match the catalogue | writing a more quotable claim than the product supports | SPECIFIED |
| R19.4 | **A technical fix is verified on the live page** | a technical SEO issue is marked resolved | the live page is re-fetched and re-checked | closing it because the change was deployed | SPECIFIED |
| R19.5 | **AI-engine visibility is tracked over time, not sampled once** | citation in an AI answer is measured | it is measured repeatedly and stored as a series | quoting a single lucky result as the position | SPECIFIED |
| R19.6 | **A sitemap lists only pages that exist and are meant to be found** | a sitemap is generated | it contains live, indexable pages | listing archived or blocked pages, which wastes the crawl on nothing | SPECIFIED |

**Reads** ← Inventory & Catalog · AI Content Engine · **Writes** → Marketing

**Done when.** Every product and content page carries valid structured data, and citation in this category is tracked over time rather than assumed.

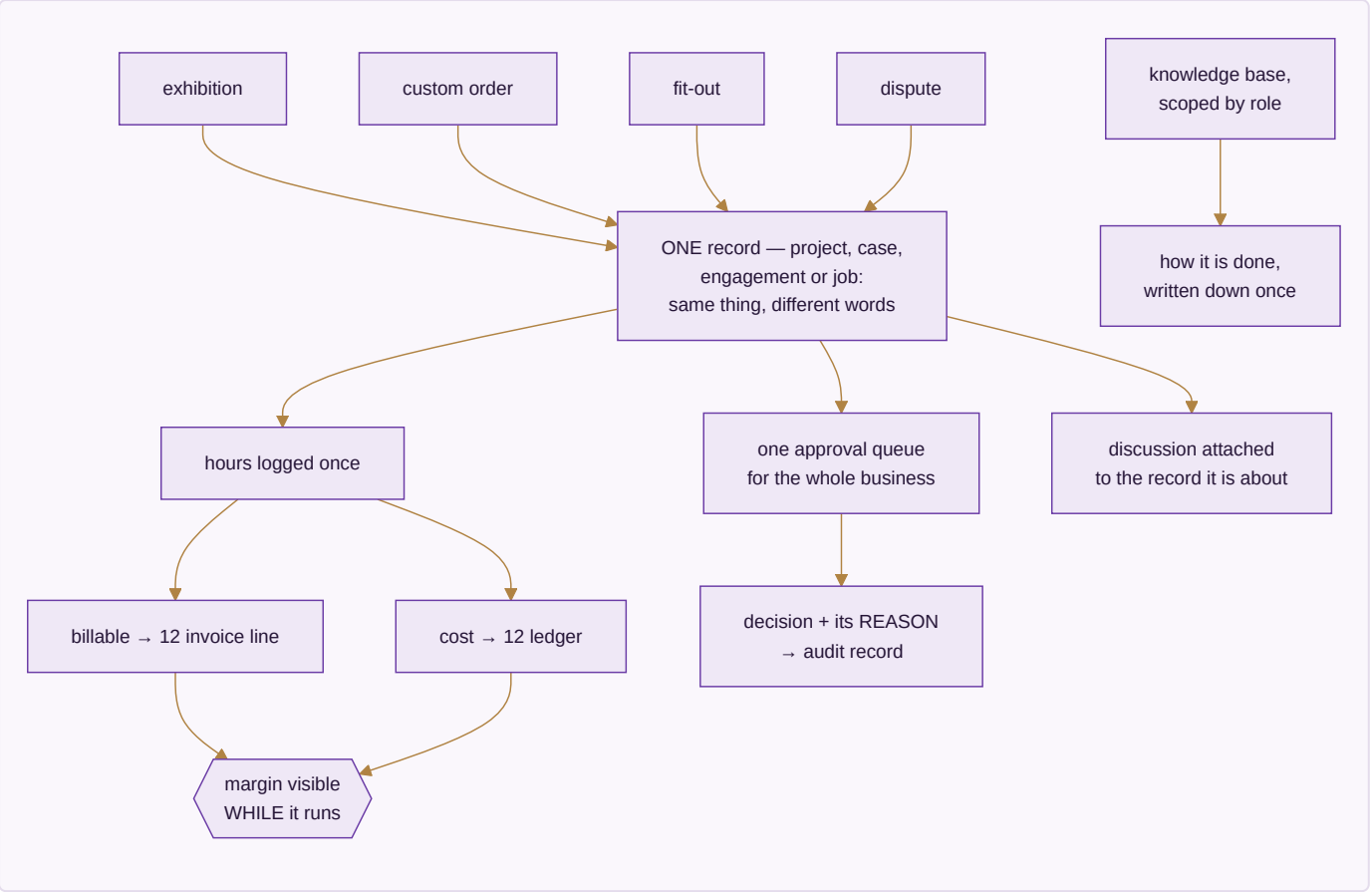
---

# MODULE 20 · PROJECTS & COLLABORATION

*The work that is not an order — and the talking around it*

**What it is.** Not everything a business does is an order. An exhibition, a custom bulk enquiry that runs six weeks before it becomes an invoice, a godown fit-out, a dispute with a supplier. That work has a deadline, a cost and documents, and it belongs on the same records as everything else.

**How it works**




## The apps (7)

| App | State | What it does |
|---|---|---|
| Projects & Cases | SPEC | Stages you define, owners, deadlines, documents, billable time and real cost, on one record the ledger can see |
| Automation Studio | SPEC | "When this happens, do that" built by dragging it out and watching it run, over the event stream every module already writes to — with every run kept, step by step, because an automation nobody can inspect afterwards is a rule the business cannot trust with its money |
| Timesheets & Planning | SPEC | Hours against a project or a machine, billable and non-billable kept apart |
| Approvals | SPEC | One queue for everything waiting on a yes, with the rule that sent it there beside it |
| Forum | SPEC | Questions and answers that outlive a chat |
| Discuss | SPEC | Conversation attached to the record it is about |
| Knowledge Base | SPEC | A role-scoped wiki of standard operating procedures |

**What it owns.** `projects` · `timesheets` · `approvals` · `forum_posts` · `discussions` · `knowledge_base`

## The rules that matter

- **Hours are entered once** and become both an invoice line and a cost. Re-keying billable hours from a timesheet into an invoice is exactly where hours get lost and margin quietly leaks.
- **Margin is visible while the work runs.** A margin discovered at the end is a margin you could not steer.
- **Every approval decision carries its reason into the audit record.** A year later, "why did we approve this" has an answer sitting next to the approval, not in someone's memory.
- One approval queue, not one per module — because approvals scattered across modules are approvals that stall unseen.

### The rulebook — 9 rules, 1 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R20.1 | **Billable time becomes an invoice line without retyping** | approved time exists against a project | the rate card turns it into an invoice line and a real cost | re-entering hours into an invoice, which is where the two figures start to differ | SPECIFIED |
| R20.2 | **Billable and non-billable are separated at entry** | time is recorded | it is marked billable or not as it is entered | deciding at invoice time, which quietly turns unbillable work into a charge | SPECIFIED |
| R20.3 | **An approval shows the rule that demanded it** | anything lands in the approvals queue | the rule that sent it there is displayed beside it | presenting a request with no stated reason, which makes approval a formality | SPECIFIED |
| R20.4 | **An approval decision goes to the audit trail** | a request is approved or refused | the decision, the person and the time are recorded | recording only the outcome on the record, which loses who accepted the risk | **ENFORCED** · core/tests/core.test.js · an update records what it was as well as what it became |
| R20.5 | **An automation run is kept step by step** | a rule fires | what triggered it, each step, and what each step returned are stored | keeping only the outcome — an automation nobody can inspect afterwards is a rule the business cannot trust with its money | SPECIFIED |
| R20.6 | **An automation acts within a named scope** | a rule is built | the records it may read and write are declared on it | letting a rule reach anywhere in the system because it happens to run as an administrator | SPECIFIED |
| R20.7 | **A project cost includes the time and the material** | project profitability is computed | labour, material and expenses booked to it are all included | reporting on revenue and time alone, which shows a loss-making project as profitable | SPECIFIED |
| R20.8 | **A decision is recorded where the decision was made** | a discussion resolves something | it is attached to the record it concerns | leaving the reasoning in a chat thread that will not be found in a year | SPECIFIED |
| R20.9 | **A procedure is scoped to the role it applies to** | a standard procedure is published | it is scoped to the role that performs it | publishing one undifferentiated manual that nobody reads | SPECIFIED |

**Reads** ← CRM · Sales · HR & Payroll · Inventory & Catalog · **Writes** → Accounting & GST · HR & Payroll · CRM

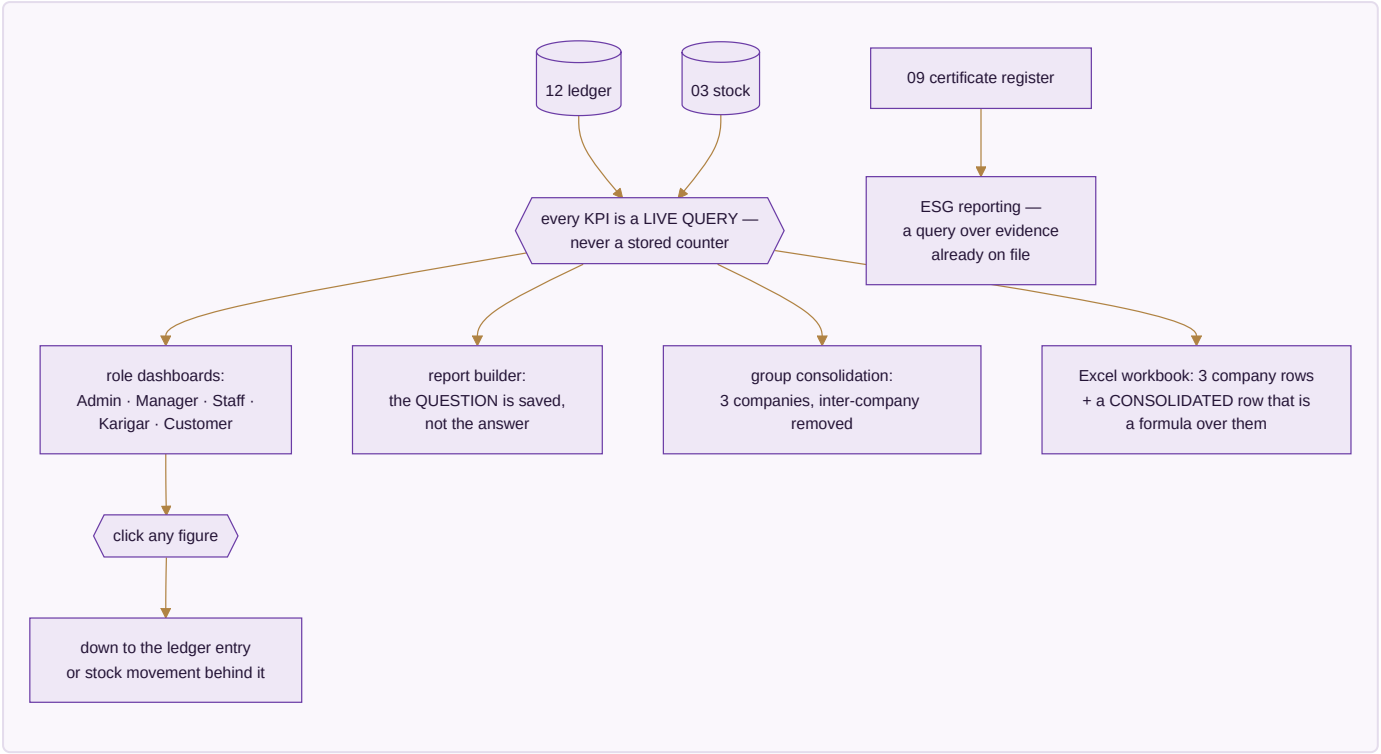
**Done when.** Billable time on a case becomes an invoice and a cost without being re-keyed once.

# MODULE 21 · DASHBOARD & BI

*See the whole business without asking anyone*

**What it is.** Every number in the business on one screen, as work happens. It is built last for a reason: it has nothing true to show until the other modules are producing real records. Two dials govern every screen — which period, and which company.

## How it works




## The apps (5)

| App | State | What it does |
|---|---|---|
| CEO Dashboard | **BUILT** | Cash, sales, stock, profit and alerts on one screen, refreshed as work happens |
| Report Builder | **BUILT** | Drag the fields you want; the question is saved, so next month it answers for next month |
| Group Consolidation | **BUILT** | Three companies as one set of figures, inter-company entries removed |
| Excel Dashboard Builder | SPEC | The full workbook, each company a row and the consolidated row a live formula over them |
| ESG / Sustainability Reporting | SPEC | Water, chemical compliance, waste and packaging, reported from the certificate and audit records already held |

**What it owns.** Nothing. It reads. It stores only saved report definitions and dashboard layouts.

## The rules that matter

- **Every KPI is a live query over the ledger and stock.** There are no stored counters, because a stored counter is a number that can drift from the truth it claims to summarise.
- **The consolidated row is a formula over the three company rows**, never a separately calculated fourth number — so it foots by construction and an auditor can trust it.
- Group profit removes inter-company sales and purchases. Money the group moved from one pocket to another is not turnover.
- **Any figure clicks down to the record behind it.** A dashboard you cannot interrogate is a dashboard you eventually stop believing.
- The financial year is detected from the data. Nothing is hardcoded.

### The rulebook — 9 rules, 6 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R21.1 | **The group figure is the sum minus inter-company trade** | several companies are consolidated | entries naming a counterparty inside the group are eliminated, and gross, eliminated and group are all shown | presenting the plain sum as the group result, which inflates turnover by trade the group never did with the outside world | **ENFORCED** · core/tests/core.test.js · the group is the sum MINUS what the companies sold each other |
| R21.2 | **An entry cannot be its own counterparty** | an entry names a counterparty company | it is refused if that is the same company | allowing a company to trade with itself, which eliminates a figure that was never doubled | **ENFORCED** · core/tests/core.test.js · an entry cannot be its own counterparty |
| R21.3 | **The number of companies is data, not a constant** | the group grows | a company is a row and every consolidation follows | building around a fixed number of companies or channels | **ENFORCED** · core/tests/core.test.js · ten companies and ten channels each is a hundred channels, not a limit |
| R21.4 | **Every dashboard figure is a live query** | a KPI is displayed | it is computed from the ledger and the stock table at that moment | reading a maintained summary table, which can be wrong without looking wrong | **ENFORCED** · core/tests/core.test.js · the trial balance is computed from the lines, never stored |
| R21.5 | **A consolidated row is a formula over the company rows** | a workbook or report shows a consolidated figure | it is computed from the company rows beside it | typing a separate consolidated total, which is a second copy that will disagree | **ENFORCED** · brand/suite/studio/verify_studio.js · the totals row is the sum of the rows above it |
| R21.6 | **A figure a user may not see is not returned** | a report runs for a scoped user | out-of-scope rows are excluded from the query | computing the full figure and hiding part of it in the display | **ENFORCED** · core/tests/core.test.js · one company cannot read another company |
| R21.7 | **An exported report says when it was taken** | a report is exported | the as-at time and the filters are printed on it | producing an undated export, which is quoted months later as though it were current | SPECIFIED |
| R21.8 | **A saved report keeps its definition, not its results** | a report is saved and re-run | the definition re-runs against current data | storing a snapshot and presenting it as live | SPECIFIED |

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R21.9 | **A figure with no drill-down is a defect** | any total is shown | it opens to the records beneath it | shipping a number that cannot be explained by clicking it | SPECIFIED |

**Reads** ← every module · **Writes** → nothing

**Done when.** Any figure on any screen clicks down to the ledger entry or stock movement behind it, in both editions.

---

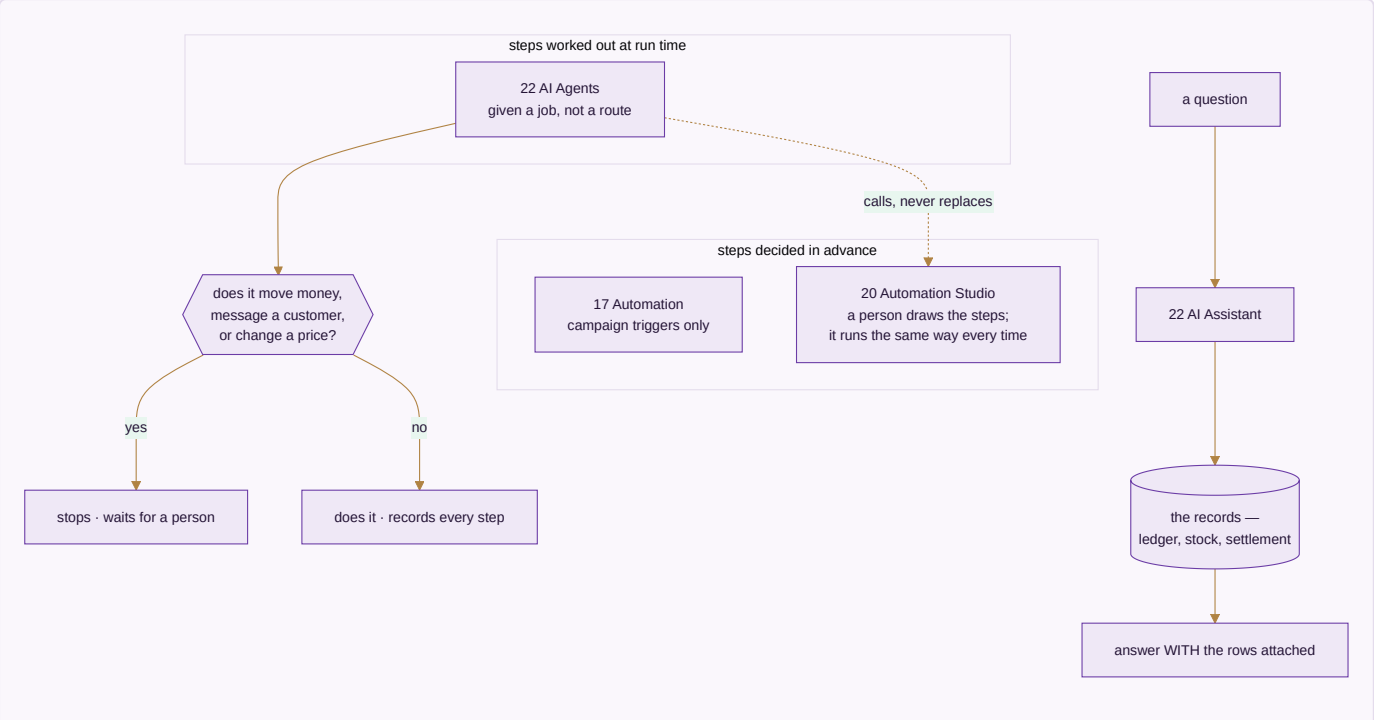
# MODULE 22 · AI ASSISTANT, AGENTS & AUTOMATION

*Ask the business a question — and let the routine work run itself*

**What it is.** Three different things that get confused with each other constantly, so the difference is stated before anything else. An **assistant** answers a question you asked, from the records, with the records attached. A **chatbot** holds that same conversation with your customer instead of with you. An **agent** is given a *job* rather than a question, and works out the steps itself.

It is last for the same reason Dashboard & BI is late: something that answers questions about the whole business can only exist once the whole business is in one place.

**How it is different from what already exists** — this matters, because two modules already automate things and neither is being replaced:



### The apps (5)

| App | State | What it does |
|---|---|---|
| AI Assistant | SPEC | Ask in plain language, get the answer with the rows it came from, each clicking through to the record |
| AI Chatbot | SPEC | The same engine facing the customer on the storefront and WhatsApp, handing over to a person with the whole conversation attached |
| AI Agents | SPEC | Given a job rather than a question; works out the steps and stops where a person has to decide |
| Agent Guardrails & Run Log | SPEC | Scope, spend ceiling and a step-by-step record of every run |
| Knowledge & Retrieval | SPEC | The business's own records indexed for grounded answers, permission-scoped at the row |

**What it owns.** `agent_runs` · `agent_steps` · `agent_scopes` · `assistant_queries` · `retrieval_index`

### The rules that matter

- **An answer carries its records.** A bare number from an assistant cannot be checked, and a figure that cannot be checked is one the business will eventually stop believing — exactly the failure Module 21 exists to prevent, arriving by a different door.
- **"I could not find it" is a valid answer; a plausible number is not.** A confident wrong figure is far more expensive than an honest blank, because nobody goes looking for it.
- **The assistant cannot see more than the person asking.** Retrieval is filtered by permission before the answer is composed, or the assistant becomes a way around every access rule in the system.
- **Money never moves, a customer is never messaged and a price is never changed without a human yes.** However confident the agent, however small the amount.
- **An agent cannot widen its own scope** mid-run, however sensible the next step looks.
- **Every run is replayable step by step.** An unexplained change made by software is worse than one made by a person, because there is nobody to ask.
- **A retrieved document is data, never an instruction.** Content that arrives from outside — a supplier's PDF, a customer's message, a marketplace's note — is reported on, not obeyed.

### The rulebook — 15 rules, 2 enforced by a test that runs today

| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R22.1 | **An answer carries the records it came from** | the assistant answers a question about a figure | the rows it used are attached and each one opens to its record | giving a bare number, which cannot be checked and therefore cannot be trusted | SPECIFIED |
| R22.2 | **An unknown answer is said, never estimated** | the assistant cannot find the figure | it says so and shows what it looked at | producing a plausible number — a confident wrong figure costs far more than an honest blank | SPECIFIED |
| R22.3 | **The assistant answers only from what the asker may already see** | a question is asked by a scoped user | retrieval is filtered to that user's permissions before the answer is composed | letting the assistant become a way around permissions that every other screen enforces | SPECIFIED |
| R22.4 | **An agent cannot widen its own scope** | an agent runs | it works within the records and the spend it was given | expanding its scope mid-run, however sensible the next step would be | SPECIFIED |
| R22.5 | **Money never moves without a human yes** | an agent proposes a refund, a payment, a payout or a credit note | it stops and waits for a person | executing it, no matter how confident or how small the amount | SPECIFIED |
| R22.6 | **A customer is never messaged by an agent without approval** | an agent drafts a message to a real customer | a person approves it before it is sent | sending on the agent's own judgement | SPECIFIED |
| R22.7 | **A price is never changed by an agent alone** | an agent proposes a price change | it enters the approvals queue with the reasoning attached | writing the new price directly | SPECIFIED |
| R22.8 | **Every agent run is replayable step by step** | an agent finishes, stops or fails | what started it, what it read, what it proposed and what was approved are all recorded | keeping only the outcome — an unexplained change made by software is worse than one made by a person | SPECIFIED |

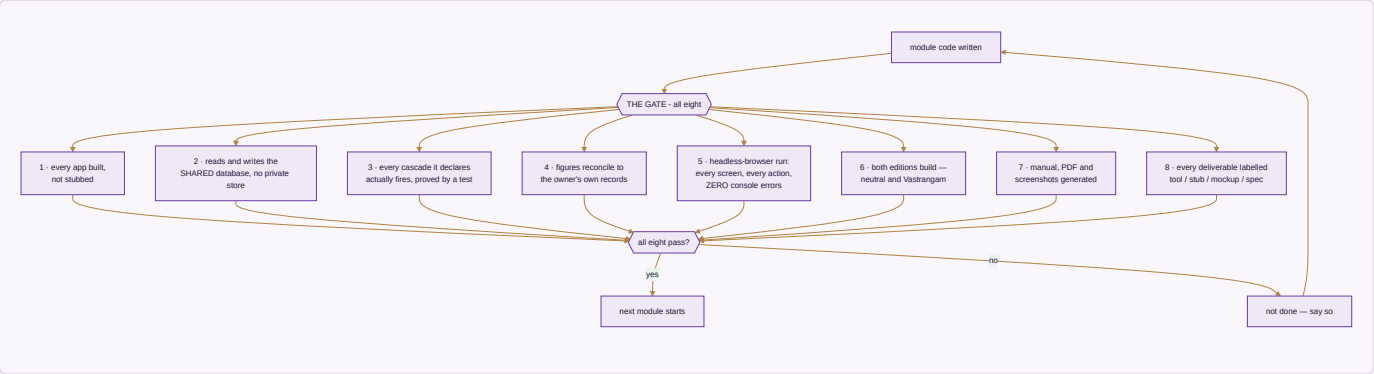
| # | The rule | When | Then | Never | State |
|---|---|---|---|---|---|
| R22.9 | **Agent spending goes through the same ceiling as everything else** | an agent calls a paid provider | it is routed through the Provider Router and refused past the ceiling | giving an agent its own unmetered budget | **ENFORCED** · brand/suite/router.js · the third call would break the ceiling and is refused |
| R22.10 | **The chatbot hands over rather than guessing about money** | a customer asks about a refund, a charge or a complaint | it hands to a person with the whole conversation attached | answering from a general idea of the policy | SPECIFIED |
| R22.11 | **The chatbot never asks a customer for a credential** | a customer is identified in a chat | identity is established through the order and the contact already on file | asking for a card number, a bank detail or a password — the promise made everywhere else does not get a chatbot-shaped exception | SPECIFIED |
| R22.12 | **A handover lands in the existing queue** | a conversation is passed to a person | it enters the Module 04 Helpdesk queue with its history | creating a second inbox that someone has to remember to watch | SPECIFIED |
| R22.13 | **An agent is not a hidden actor in the audit trail** | an agent changes anything | the change is attributed to the agent, its run, and the person who approved it | recording it under a service account, which makes an automated change indistinguishable from a human one | **ENFORCED** · core/tests/core.test.js · an audited insert leaves a before/after trail |
| R22.14 | **A retrieved document does not become an instruction** | the assistant reads a document, a review or a message while answering | that content is treated as data to report on | following instructions found inside retrieved content, which is how a supplier's PDF ends up steering the system | SPECIFIED |
| R22.15 | **An assistant answer is reproducible from the records it cites** | the assistant states a figure | re-running the same query over the same records gives the same figure | an answer that cannot be reproduced, which is a guess with citations attached | SPECIFIED |

**Reads** ← every module · **Writes** → Projects & Collaboration · CRM · Marketing

**Done when.** A question about last week's payouts returns the figure the books return, with the settlement rows attached; and an agent asked to move money stops and waits, with the refusal recorded.

# PART III — WHEN IS A MODULE DONE

A module is not finished because its code is written. It is finished when all eight of these hold.




**Rule 8 is the one that matters most,** because it is the one that is easiest to quietly skip. Nothing is called finished that is not. A mockup is labelled a mockup even when a working version would be more impressive to show.

---

# PART IV — EXECUTION

Everything up to here says what the system is and what it refuses to do. This part says how it gets built: the stack, the database, the interfaces, the order of work, the migration off the current system, and the numbers that decide whether each step is finished. A plan that stops at the module list leaves the hardest decisions to whoever opens the editor first.

## E1 · THE STACK, AND WHY IT IS NOT A DEPENDENCY

The stack below is pinned. It is also, deliberately, a list of **adapters** rather than a list of dependencies — §A.3.1 of the master spec forbids calling a provider SDK from business logic, and Module 01's Provider Router is what turns that from a wish into something the tests fail over. Both statements are true at once: this is what the system runs on, and none of it is load-bearing.

| Layer | Chosen | Behind which interface | Swappable for |
|---|---|---|---|
| Frontend | Next.js 15, TypeScript, App Router | — | any renderer; the engines are DOM-free |
| UI | Tailwind + shadcn/ui | — | — |
| Database + auth | Supabase (PostgreSQL 16, RLS, Storage) | `DatabaseService` | Neon + Clerk, self-hosted Postgres |
| Automations | n8n, self-hosted | `AutomationService` | Node-RED, Windmill, cron |
| WhatsApp | Interakt | `WhatsAppService` | Wati, AiSensy, Gallabox |
| AI | Anthropic API | `AIService` | any provider, or Ollama locally |
| Payments (domestic) | Razorpay | `PaymentService` | Cashfree, PayU |
| Payments (international) | PayPal | `PaymentService` | Stripe, Wise |
| Shipping | Shiprocket (aggregator) | `ShippingService` | Delhivery, Blue Dart, or type the AWB |
| Storage | Supabase Storage | `StorageService` | S3, MinIO, Nextcloud, this device |
| Hosting | Vercel (app), VPS (n8n) | — | any Node host |
| Mobile | PWA, Capacitor 6 shell on demand | — | same codebase, no fork |

**The test of that claim.** Every capability above appears in `brand/suite/providers.js` with its alternatives, and `brand/suite/router.js` proves at every run that each one has a fallback list ending in an option that needs nothing connected. Switching Interakt for Wati is an adapter change, not a project.

**What genuinely needs paid access to go live:** WhatsApp Business messaging, marketplace APIs, the payment gateways, courier APIs, and AI inference beyond what runs locally. Everything else runs on a laptop with the network off.

## E2 · THE DATABASE

Two schema files, and neither is a draft of the other:

- **`core/schema.sql`** — SQLite via `node:sqlite`. Loads in every test run; every table in it is exercised by `core/tests/core.test.js`. This is what runs today.
- **`core/schema.postgres.sql`** — 113 tables, PostgreSQL 16 for Supabase, organised in build- phase order so Phase 1 can be run without reading the rest. uuid keys, jsonb, the audit columns on every table, and a `company_isolation` RLS policy per business table.

`core/tests/schema.test.js` is the gate between them. It asserts that the two never disagree about a shared table, that every business table carries `company_id` and can be soft-deleted or is append-only by design, that no

money column is a float in either file, and that every company-scoped table has an RLS policy. Discovering that drift at cutover — sixty days into a parallel run — is the failure this exists to prevent.

Tables are marked `[LIVE]` where an engine already uses them and `[PHASE n]` where they are structural. That distinction is enforced: the test fails if a table marked live is absent from the schema that actually runs.

**The SKU model.** Brand → design → colour → size. A design is a photoshoot unit; its SKUs are the colour × size rows underneath it, generated when the admin picks which variants exist, never typed one at a time. Opening stock is entered at SKU level, never at design level. The old system's item code is preserved in `legacy_busy_code` forever, so a migrated voucher can still be traced to what it was.

**Row-level security.** Company isolation is enforced by the database and checked again in API middleware. A filter in a screen can be removed; a policy cannot. The policy carries both `USING` and `WITH CHECK`, so a write cannot cross companies either — reading another company's data and writing into it are two different holes.

## E3 · INTERFACES

**Resource pattern.** `/api/v1/{resource}` with the standard verbs, every request scoped to the active company from the session claim, every mutation audited. Domain routes beyond CRUD are the ones that carry business meaning: advance a production stage, assign a karigar, import a settlement file, raise a dispute, generate a payroll run, post or void a journal entry, generate a GST return, run a listing across platforms.

**Five inbound webhooks** — storefront orders, payment capture, courier AWB and COD remittance, WhatsApp messages, international payment. Every one of them: signature-verified, idempotent by the sender's own external id, and logged to `integration_errors` with its payload on failure so it can be retried rather than lost. A duplicated payout or a duplicated order is indistinguishable from a real one after the fact, which is why idempotency is a rule here and not an optimisation.

**Outbound.** Stock is pushed to the storefront on a short cycle so it cannot oversell. Marketplace orders are pulled per channel within a window. Both are rate-limit aware with backoff, and both fall back to CSV import — the manual path is not a lesser mode, it is the one that works on the day an API changes without notice.

## E4 · THE EIGHT PHASES

Thirty-two weeks, Phase 0 through Phase 7. The gate is absolute: **Phase N+1 does not start until Phase N's tests pass.** A phase is not done when the code is written; it is done when the stated result is reproduced.

| Phase | Weeks | What gets built | Done when |
|---|---|---|---|
| **0 · Setup** | 1–2 | Accounts, environments, RLS skeleton, CI, error and uptime monitoring | First commit deploys and a user can log in |
| **1 · Foundation** | 3–5 | Companies, roles, company switching, masters, the SKU model, `schema.postgres.sql` loaded | An admin creates a design with 5 colours × 7 sizes in under five minutes, and one company cannot see another's rows |
| **2 · HR + comms** | 6–9 | WhatsApp console, attendance, geofence, effective-dated salary, payroll, karigar earnings | A full month of payroll runs end to end with no manual touch, and a mid-month raise applies to the right months only |
| **3 · Inventory + manufacturing** | 10–14 | Stock by SKU × location × stage, the ten production stages, BOM, QC, procurement and three-way match | Three production orders complete — self, full job work, partial — and the karigar figures match the tool's |
| **4 · Sales, all channels** | 15–20 | Storefront sync, marketplace pull, settlement reconciliation, B2B, export, POS, customisation, returns | A full week runs on every channel with settlements reconciled and variances raised as claims |
| **5 · Finance + GST** | 21–25 | Double-entry, GST returns, ITC, TDS/TCS, bank reconciliation, period locks | A month closes: trial balance ties, GSTR-1 and 3B generate from vouchers, bank reconciles line by line |
| **6 · AI, marketing, CRM** | 26–29 | Content engine, listings, campaigns, Customer 360, automation studio, the assistant and agents | An assistant answer matches the books, and an agent asked to move money stops and waits |
| **7 · Cutover** | 30–32 | Opening balances, parallel run, smoke tests, go-live | The first month's GSTR-1 and 3B are filed from this system |

The phase order and the module order are the same plan seen twice: Phase 1 is Modules 01–04, Phase 2 is 16, Phase 3 is 03 and 06–10, Phase 4 is 05 and 11 and 15, Phase 5 is 12–14, Phase 6 is 17–22, and Phase 7 is the whole thing proving itself.

## E5 · MIGRATION AND CUTOVER

The current accounting system is not switched off on a date; it is switched off on a **result**.

**Before.** Export every master and voucher table, clean to CSV, load. Every voucher must resolve to a real party — a party code that does not join is a migration failure, not a row to skip. Customers, designs, and their colour × size explosion into SKUs all carry their legacy code.

**Opening balances at the cutover date** (start of a financial year): capital and reserves, bank balances, open receivables and payables, GST/TDS balances, fixed assets, work in progress at its current stage, and opening stock at SKU level. Entered once, from the closing trial balance.

**Sixty days of parallel running.** Every morning, five minutes: yesterday's sales total against the channels, bank inflow against the statement, cash position against the physical count, open invoices against the old system's

ageing, GST liability accruing correctly. A variance over ₹100 is investigated the same day. A pattern of variances gets a root cause, not a second look.

**Decommission criterion, stated in advance so it cannot be argued about later:** the old system is retired when the first month's GSTR-1 and GSTR-3B have been filed from this one. Not when the build is finished; not when everyone feels confident.

## E6 · SECURITY, COMPLIANCE, OPERATIONS

Row-level security in the database **and** permission checks in API middleware — defence in depth, because one layer is one mistake away from a cross-company read. Sessions expire and refresh on a fixed window. Transport encrypted; storage encrypted at rest; identity documents and bank details encrypted at the application layer with a key held outside the database, and never written into an export, a backup file or a repository.

Personal data can be exported and erased on request, with retention and consent tracked as the two separate clocks they are. Card data never reaches this system — the gateway's own secured field takes it, so there is no scope to protect. Electronic invoicing is wired when turnover crosses the threshold that makes it mandatory, not before.

Backups daily with a weekly off-site snapshot; automation workflows and server configuration in version control alongside the code. Recovery objectives: back up within four hours, lose no more than a day. A documented runbook, because a recovery plan nobody has read is a hope.

## E7 · PERFORMANCE — GATES, NOT HOPES

| Operation | p95 target |
|---|---|
| Page load, cached | < 1 s |
| Page load, uncached | < 3 s |
| Marketplace order pull, per channel | < 60 s |
| Settlement import, 1,000 lines | < 30 s |
| Invoice PDF | < 2 s |
| AI listing, one item across six platforms | < 20 s |
| Daily profit-and-loss refresh | < 10 s |
| Live stock query | < 500 ms |

The stock query is the one that matters most and looks least important: it runs on every screen that shows a quantity, and a system that takes two seconds to say how many are left is a system people stop asking.

## E8 · RISK REGISTER

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Marketplace API rate limits | Medium | Medium | Backoff, per-channel windows, CSV fallback |
| WhatsApp provider outage | Low | High | Adapter swap to an alternative; SMS for critical alerts |
| Database provider downtime | Low | Critical | Daily snapshot off-site, standby replica, documented migration |
| Tax portal down at filing time | High | Medium | Generate five days early, retry queue, manual filing path |
| Karigar resistance to digital reporting | Medium | Medium | WhatsApp is already familiar; voice notes accepted; weekly review with the supervisor |
| Power loss at the production unit | Medium | Low | Offline-capable attendance and reporting, syncing on reconnect |
| Settlement variance volume overwhelms the first 60 days | High | Medium | Auto-categorise, bulk actions, alert only above a threshold |
| Bank statement format changes | Medium | Low | CSV import with a column-mapping screen |

## E9 · SUCCESS METRICS

Twelve months after go-live, measured against how the business runs today:

| Measure | Today | Target |
|---|---|---|
| Marketplace order received → handed to courier | ~24 h | 6 h |
| Settlement reconciliation lag | 30+ days | < 7 days |
| Disputed money actually recovered | roughly half is lost | 80%+ recovered |
| Stock count variance | 5–10% annually | < 1% quarterly |
| Karigar earnings disputes | 5–10 a month | < 1 a month |
| Time to close a month | days | hours |

The dispute number is the one that pays for the build. Everything else is time; that one is money that is currently leaving.

## E10 · RUNBOOKS

**Daily** — the five-minute reconciliation during the parallel run, then the dispatch queue, the QC and alteration queue, incoming returns, and low stock. **Weekly** — channel profitability, design winners and losers, next week's content locked, marketplace ratings. **Monthly** — physical stock count, bank reconciliation per account per company, GST return generation and filing, payroll run and disbursement, karigar payout, profit and loss per company and for the group. **Quarterly** — channel mix and pricing review, vendor scorecards, dead-stock

clearance, and whether the AI spend earned its ceiling. **Annually** — year-end close and opening balances, annual returns, strategy and targets.

## E11 · THE SHELL

One application. A left sidebar grouping the modules; a top bar carrying the company switcher, the financial year, global search and quick-create; the module in the canvas. On a phone the sidebar becomes a drawer and the common actions become bottom tabs — **every key action within two taps of home**, because the shop floor is holding a phone, not sitting at a desk.

Five dashboards, because five kinds of people need five different first screens: the owner sees cash, revenue and exceptions across all companies; the manager sees today's dispatch, QC and attendance; staff see their own tasks, hours and earnings; a karigar sees today's assignment, pieces submitted and this month's running total; a customer sees their orders, wishlist and returns. Nobody is shown a figure they may not see — that is enforced by the same row-level policy as everything else, not by hiding a widget.

Tables collapse to cards on narrow screens. Every grid exports to a branded workbook in the same style as the on-screen report. Interface strings carry keys so the surfaces staff and karigars use can be read in the language they actually speak.

**The document tree.** Generated documents file themselves into a per-company folder structure — HR, production, inventory, sales, finance, marketing, vendor, customer — with a consolidated group folder alongside. A document is found by the record it belongs to; the folder is where it happens to sit, not how it is retrieved.

**Design tokens.** Deep purple `#4A2D82`, lavender `#7B5EA7`, gold `#C4963A`, near-black `#12091C`; Cormorant Garamond for display, DM Sans for body. Consolidated and total rows are visually distinct — dark ground, white bold — everywhere they appear, so a total can never be mistaken for another row of detail.

## E12 · ACCEPTANCE TARGETS

Figures from the business's own records. A module that reprocesses this data and returns something else has a bug — not a new answer.

| What | The owner's figure | Status here |
|---|---|---|
| Offline sales, three stores | 124 items · 2,601 pieces | not yet wired — the source file is not in the verification set |
| E-commerce sale / return | 59 items · sale 9,048 · return 3,995 · net 5,053 · wrong return 78 · inventory 4,975 | not yet wired — see below |
| Karigar production and cost | 143 designs · 29 units · 25,307 sets · 59,110 pieces · ₹26,90,062 | reproduces the FY2025-26 half; the cause of the rest is named in Module 08 |

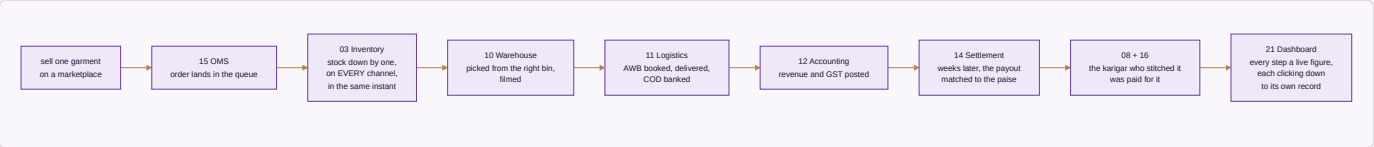
**Why two of the three say "not yet wired", plainly.** The verification suite runs against the workbooks currently in the shared folder, and those are not the same files these figures were taken from — the e-commerce figures come from a combined workbook covering a different span from the one the suite reads. Wiring a target against a file that is not the file it came from would produce a number that agrees with nothing. The targets are recorded here so they can be wired the moment the matching source is available, and marked honestly until then. The karigar

row is the worked example of doing this properly: it reproduces for the year the engine can read, and Module 08 states exactly why the other year cannot be read at all.

---

# PART V — THE PROOF

The test of whether this is one system or twenty-two programs sharing a login is a single garment followed end to end. This is re-run at every module boundary.




**One transaction. Eight modules. One database.**

Everything in this document exists to make that single sentence true. The honest state today is that of the eight modules named in that sentence, sixteen apps are working and the shared database beneath them is written and tested but not yet carrying them — which is the first job of Module 01, and the reason the build order starts where it does.

---

*Every figure in this plan traces to a source: the module and app counts are read directly from the canonical module list the website and every generated document also read, so they cannot drift; the built-versus-spec marks correspond to sixteen apps that pass their own self-tests and a full click-through audit in both editions; and the acceptance figures in Module 08 and the gate in Module 14 are the owner's own numbers, not targets invented here.*

---